



ACADEMIC REGULATIONS & SYLLABUS

Faculty of Computer Science & Applications

BCA Programme



CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Education Campus – Changa, (ECC), hitherto a conglomerate of institutes of professional education in Engineering, Pharmacy, Computer Applications, Management, Applied Sciences, Physiotherapy and Nursing, is one of the choicest destinations by students. It has been transformed into **Charotar University of Science and Technology (CHARUSAT)** through an Act by Government of Gujarat. CHARUSAT is permitted to grant degrees under Section-22 of UGC- Govt. of India.

The journey of CHARUSAT started in the year 2000, with only 240 Students, 4 Programmes, one Institute and an investment of about Rs.3 Crores (INR 30 million). At present there are seven different institutes falling under ambit of six different faculties. The programmes offered by these faculties range from undergraduate (UG) to Ph.D. degrees. These faculties, in all offer 64 different programmes. A quick glimpse in as under:

Faculty	Institute	Programmes Offered
Faculty of Technology & Engineering	Charotar Institute of Technology	B. Tech M. Tech MTM Ph. D
Faculty of Pharmacy	Ramanbhai Patel College of Pharmacy	B. Pharm M. Pharm MPM PGDCT/ PGDPT Ph. D

Faculty of Management Studies	Indukakal pcowala Institute of Management	M.B.A PGDM Dual Degree BBA+MBA Ph.D
Faculty of Computer Science &Applications	Smt. Chandaben Mohanbhai Patel Institute of Computer Applications	MCA/MCAL M.Sc. (IT) BCA B.Sc.(IT) Ph. D.
Faculty of Applied Sciences	P.D.Patel Institute of Applied Sciences	M.Sc Dual Degree B.Sc + M.Sc Ph.D

Faculty of Medical Sciences	Ashok and Rita Institute of Physiotherapy ManikakaTopawala Institute of Nursing Charotar Institute of Paramedical Sciences	B.PT M.PT Ph.D B.Sc (Nursing) M.Sc PGDHA PGDMLT GNM Ph.D
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The development and growth of the institutes have already led to an investment of over Rs.125 Crores (INR 1250 Million). The future outlay is planned with an estimate of Rs.250 Crores (INR 2500 Million).

The University is characterized by state-of-the-art infrastructural facilities, innovative teaching methods and highly learned faculty members. The University Campus sprawls over 105 acres of land and is Wi-Fi enabled. It is also recognized as the Greenest Campus of Gujarat.

CHARUSAT is privileged to have 360 core faculty members, educated and trained in IITs, IIMs and leading Indian Universities, and with long exposure to industry. It is also proud of its past students who are employed in prestigious national and multinational corporations.

From one college to the level of a forward-looking University, **CHARUSAT** has the vision of entering the club of premier Universities initially in the country and then globally. **High Moral Values like Honesty, Integrity and Transparency** which has been the foundation of ECC continues to anchor the functioning of **CHARUSAT**. Banking on the world class infrastructure and highly qualified and competent faculty, the University is expected to be catapulted into top 20 Universities in the coming five years. In order to align with the global requirements, the

University has collaborated with internationally reputed organizations like Pennsylvania State University – USA, University at Alabama at Birmingham – USA, Northwick Park Institute –UK, ISRO, BARC, etc.

CHARUSAT has designed curricula for all its programmes in line with the current international practices and emerging requirements. Industrial Visits, Study Tours, Expert Lectures and Interactive IT enabled Teaching Practice form an integral part of the unique **CHARUSAT** pedagogy. The programmes are credit-based and have continuous evaluation as an important feature. The pedagogy is student-centred, augurs well for self-learning and motivation for enquiry and research, and contains innumerable unique features like:

- Participatory and interactive discussion-based classes.
- Sessions by visiting faculty members drawn from leading academic institutions and industry.
- Regular weekly seminars.
- Distinguished lecture series.
- Practical, field-based projects and assignments.
- Summer training in leading organizations under faculty supervision in relevant programmes.
- Industrial tours and visits.
- Extensive use of technology for learning.
- Final Placement through campus interviews.

Exploration in the field of knowledge through research and development and comprehensive industrial linkages will be a hallmark of the University, which will mould the students for global assignments through technology-based knowledge and critical skills.

The evaluation of the student is based on grading system. A student has to pursue his/her programme with diligence for scoring a good Cumulative Grade Point Average (CGPA) and for succeeding in the chosen profession and life.

CHARUSAT welcomes you for a Bright Future



CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Computer Science & Applications

ACADEMIC REGULATIONS

BCA Programme

Charotar University of Science and Technology (CHARUSAT)

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Year – 2021-2022

CHARUSAT

FACULTY OF COMPUTER SCIENCE AND APPLICATIONS

ACADEMIC RULES

Bachelor of Computer Applications (B.C.A.) Programme

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. Duration of Programme

Undergraduate programme Bachelor of Computer Applications(BCA)

Minimum	6 semesters (3 academic years)
Maximum	10 semesters (5 academic years)

3. Eligibility & Mode of admissions

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

4. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

5. Attendance

5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

5.2 Student attendance in a course should be 80%.

6. Course Evaluation

6.1 The performance of every student in each course will be evaluated as follows:

6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

6.2 University Examination

6.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

6.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.

6.3 Performance at Internal & University Examination will be done on the relative grading system.

7. Grading

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

Grading Scheme

Range of Marks (%)	≥80	≥75 <80	≥70 <75	≥65 <70	≥60 <65	≥55 <60	≥50 <55	<50
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

$$\text{SGPA} = \frac{\sum C_i G_i}{\sum C_i} \quad \text{where } C_i \text{ is the number of credits of course } i$$

G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester

$$\text{CGPA} = \frac{\sum C_i G_i}{\sum C_i} \quad \text{where } C_i \text{ is the number of credits of course } i$$

G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

8. Detention Rule

8.1 No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year..

8.2 A Student will not be allowed to move to third year if he/she has not cleared all the courses of first year.

8.3 A student will not be allowed to move to fourth year if he/she has not cleared all the courses of first and second year.

9. Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

9.1.2 He should have cleared all evaluation components in every course; and

9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for the award of degree. Only latest grade will be considered.

10 Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction: CGPA ≥ 7.5 & ≤ 10

First class: CGPA ≥ 6.0 & < 7.5

Second Class: CGPA ≥ 5.0 & < 6.0

Pass Class: CGPA < 5.0

11 Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

FACULTY OF COMPUTER SCIENCE & APPLICATIONS

CHOICE BASED CREDIT SYSTEM

FOR

BACHELOR OF COMPUTER APPLICATIONS

CHOICE BASED CREDIT SYSTEM

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses:

Foundation Courses, Core Courses and Elective Courses.

Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core Courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates

proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

Vision

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

The graduates will

PEO1: Be able to understand the requirement of computing problem and implement an effective solution.

PEO2: Be able to successfully take up various available career options.

PEO3: Be able to continuously learn in their preferred domains.

PEO4: Be able to acquit themselves in ethical and professional manner while demonstrating effective skills.

PROGRAM OUTCOMES (POs)

The Graduate of Computer Science and Applications will be able to :

PO1: Computational Knowledge: Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge to provide effective solution in the area of computing.

PO2: Problem Analysis: Identify, formulate, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

PO3: Design /Development of Solutions: Design and evaluate solutions for computing problems, and design and evaluate systems, components, or processes that meet specified needs by considering several societal aspects.

PO4: Modern Tool Usage: Create, select, adapt and apply appropriate techniques, resources, and modern computing tools for any computing activities, with an understanding of the limitations.

PO5: Professional Ethics: Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

PO6: Life-long Learning: Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

PO7: Project management: Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO8: Communication Efficacy: Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.

PO9: Societal and Environmental Concern: Understand and assess societal and environmental issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.

PO10: Individual and Team Work: Function effectively as an individual and as a member or a leader in diverse teams and in multidisciplinary environments.

PO11: Innovation and Entrepreneurship: Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the programme, the student should be able to

PSO1: Analyze, evaluate and formulate workable computing solutions to real world problems using computing methods.

PSO2: Apply proficiencies of communication, teamwork and management skills with spirit of entrepreneurship.

**TEACHING SCHEME
FOR
BCA PROGRAMME
(1ST, 2ND & 3RD YEAR)

EFFECTIVE FROM
ACADEMIC YEAR 2021-2022**

Teaching and Examination Scheme (BCA Programme)

Effective from Year 2021-2022

Semester-I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA109-110	Elective-I	3	-	3	3	10	20	70	-	-	-	100
CA111	Fundamental Concepts Of Programming Language	4	3	7	7	10	20	70	15	15	70	200
CA112	Introduction to Computers	4	3	7	7	10	20	70	15	15	70	200
CA113	Programming the Internet-I	-	3	3	3	-	-	-	15	15	70	100
CA114	Foundation of Mathematics	3	-	3	3	10	20	70	-	-	-	100
HS101.02 C	Communicative English	-	2	2	2	-		-	30		70	100
FS102C	Foundation Course on Chemistry and Biology	-	2	2	2	-		-	30		70	100
		14	13	27	27	400			500			900

Elective-I

1. CA109 –Introduction to Information Technology
2. CA110 – Foundation of Computer Organization

Teaching and Examination Scheme (BCA Programme)

Effective from Year 2021-2022

Semester-II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA115-116	Elective-II	3	-	3	3	10	20	70	-	-	-	100
CA117	Operating System Concepts	4	3	7	7	10	20	70	15	15	70	200
CA118	Advanced Programming	4	3	7	7	10	20	70	15	15	70	200
CA119	Database Fundamentals	4	3	7	7	10	20	70	15	15	70	200
CA120	Fundamentals of Commerce & Business Process	3	-	3	3	10	20	70	-	-	-	100
HS201.02 C HS202.02 C HS205.02 C HS209.02 C HS210.02 C	Painting Photography Media and Graphic Design Dramatics Contemporary Dance	-	2	2	2	-		-	30		70	100
FS101C	Foundation Course On Mathematics And Physics	-	2	2	2	-		-	30		70	100
		18	13	31	31	500			500			1000

Elective-II

1. CA115–Management Information System
2. CA116–Numerical & Statistical Method

Teaching and Examination Scheme (BCA Programme)

Effective from Year 2021-2022

Semester-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		Exter nal	
						Case Study	Tests		Term work	Tests		
CA209-210	Elective - III	3	-	3	3	10	20	70	-	-	-	100
CA211-212	Elective – IV	-	3	3	3	-	-	-	15	15	70	100
CA213	Programming the Internet-II	4	3	7	7	10	20	70	15	15	70	200
CA214	Object Oriented Programming Using C++	4	3	7	7	10	20	70	15	15	70	200
CA215	System Analysis and Design	3	-	3	3	10	20	70	-		-	100
HS121.02 C	Creativity, Problem Solving and Innovation	2	-	2	2	30		70	-		-	100
CL144.01 C	Environmental Sciences	-	2	2	2	-		-	30		70	100
	University Elective – I **	-	2	2	2	-	-	-	30		70	100
		16	13	29	29	500			500			1000

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer

CA224 Introduction to Web Designing course for others.

Elective-III

1. CA209–Fundamentals of Electronic Commerce
2. CA210–Introduction to Economics

Elective-IV

1. CA211–User Interface Design
2. CA212 - Visual Basic Applications

University Elective - I

No	Course Code	Course Name	Department/Faculty
1	CE281.01	Art of Programming	Engineering
2	CL281.01	Environmental Sustainability & Climate Change	Engineering
3	IT281.01	ICT Resources & Multimedia	Engineering
4	ME281.01	Engineering Drawing	Engineering
5	PH233.01	Fundamentals of Packaging	Pharmacy
6	PD260.01	Basic Laboratory Techniques	Applied Science
7	NR251.01	First Aid & Life Support	Nursing
8	PT191.01	Health Promotion & Fitness	Physiotherapy
9	BM231	Banking & Insurance	Management
10	EC281.01	Introduction to MATLAB Programming	Engineering
11	PD261	Astrophysics, Space and Cosmos-1	Applied Science
12	CL283	SDG Handprint Laboratory	Engineering
13	EE284	Python Programming	Engineering

Teaching and Examination Scheme (BCA Programme)

Effective from Year 2021-2022

Semester-IV

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA216 – 220	Elective - V	4	3	7	7	10	20	70	15	15	70	200
CA221	Data Structures & Algorithms	4	3	7	7	10	20	70	15	15	70	200
CA222	Advanced Database System	-	3	3	3	-	-	-	15	15	70	100
CA223	Fundamentals of Computer Networks	3	-	3	3	10	20	70	-		-	100
HS111.02 C	Human Values and Professional Ethics	-	2	2	2	-		-	30		70	100
	University Elective – II **	-	2	2	2	-	-	-	30		70	100
		11	13	24	24	300			500			800

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer

CA225 Programming the Internet course for others.

Elective-V

1. CA216 -Computer Graphics
2. CA217- Multi Paradigm Programming Language
3. CA218- System Programming
4. CA219- Scripting Language
5. CA220- Basics of Web Designing

University Elective – II

No	Course Code	Course Name	Department/Faculty
1	EC282.01	Prototyping Electronics with Arduino	Engineering
2	CE282.01	Web Designing	Engineering
3	CL282.01	Basics of Environmental Impact Assessment	Engineering
4	IT282.01	Internet Technology & Web Design	Engineering
5	ME282.01	Material Science	Engineering
6	PH238.01	Cosmetics in Daily Life	Pharmacy
7	NR261.01	Life Style Diseases & Management	Nursing
8	PT192.01	Occupational Health & Ergonomics	Physiotherapy
9	BM241	Health Care Management	Management
10	PD262	Astrophysics, Space and Cosmos-2	Applied Science
11	EE287	MATLAB Programming	Engineering
12	EE288	Maintenance of Household Apparatus	Engineering

Teaching and Examination Scheme (BCA Programme)
Effective from Year 2021-2022
Semester-V

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA308–313	Elective-VI	4	3	7	7	10	20	70	15	15	70	200
CA314	Object Oriented Programming Through JAVA	4	3	7	7	10	20	70	15	15	70	200
CA315	Application development using .NET Framework	4	3	7	7	10	20	70	15	15	70	200
CA316	Software Engineering	-	3	3	3	-	-	-	15	15	70	100
HS131.02 C	Communication and Soft Skills	-	2	2	2				30		70	100
		12	14	26	26	300			500			800

Elective-VI

1. CA308 –Introduction to Open Source Technology
2. CA309 –Advanced Computer Networks
3. CA310 –Operation Research
4. CA311–Embedded Programming
5. CA312-Multi Media
6. CA313-Ruby on Rail Architecture

Teaching and Examination Scheme (BCA Programme)

Effective from Year 2021-2022

Semester-VI

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theo ry	Pract	Tot al		Internal		Exter nal	Internal		Exter nal	
						Case Study	Tests		Term work	Tests		
CA317-319	Elective-VII	4	-	4	4	10	20	70	-	-	-	100
CA320 – 322	Elective-VIII	-	3	3	3	-	-	-	15	15	70	100
CA323	Minor Project	-	15	15	15	-	-	-	100		300	400
HS132.02 C	Contributory Personality Development	-	2	2	2	-	-	-	30		70	100
		04	20	24	24	100			600			700

Elective- VII

1. CA317 –Artificial Intelligence
2. CA318–Cryptography and Network Security
3. CA319–Data Compression Techniques

Elective- VIII

1. CA320–Software Project Management
2. CA321–Basics of Mobile Applications
3. CA322–Computer Graphics-II

Teaching and Examination Scheme (BCA Programme)

Effective from Year 2021-2022

Semester-I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA109-110	Elective-I	3	-	3	3	10	20	70	-	-	-	100
CA111	Fundamental Concepts Of Programming Language	4	3	7	7	10	20	70	15	15	70	200
CA112	Introduction to Computers	4	3	7	7	10	20	70	15	15	70	200
CA113	Programming the Internet-I	-	3	3	3	-	-	-	15	15	70	100
CA114	Foundation of Mathematics	3	-	3	3	10	20	70	-	-	-	100
HS101.02 C	Communicative English	-	2	2	2	-		-	30		70	100
FS102C	Foundation Course on Chemistry and Biology	-	2	2	2	-		-	30		70	100
		14	13	27	27	400			500			900

Elective-I

1. CA109 –Introduction to Information Technology
2. CA110 – Foundation of Computer Organization

CA109 : Introduction to Information Technology

(Marks 100)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures foundations of information technology related concepts will be introduced to students. Emphasis will be given on acquisition, storage and processing of data to generate meaningful information. Students will be made familiar with applications related to information technology. Emerging trends and societal impacts of information technology will be discussed to students. Students will give practical exposure by demonstrating real information technology system.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours
		Theory
1	Basics of Information Technology	04
2	Acquisition of various types of data	07
3	Data Storage and organizations	08
4	Processing of Various types of data	08
5	Application related to Information Technology	04
6	Emerging trends and Societal Impact of IT	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Basics of Information Technology Hours: 04

Introduction, Types of Data, Simple Model of a Computer, Data Processing Using A Computer, Desktop Computer, Interrupt Controller

Unit – II: Acquisition of Various Types of data Hours: 07

Introduction to Number and Textual Data, Image Data, Audio and Video Data Numbers and Textual Data: Input Units, Internal Representation of Numerical Data, Representation of Characters in Computer, Error-detecting Codes

Image Data: Acquisition of Textual Data and Pictures, Storage Formats for Pictures, Image Compression Fundamentals, Image Acquisition with a Digital Camera,

Audio: Basics of Audio Signals, Acquiring and storing Audio Signals, Compression of Audio Signals

Video: Capturing a moving scene with a video camera, Compression of Video Data, MPEG Compression Standards

Unit – III: Data Storage and Organizations Hours: 08

Storage Cell, Physical Devices Used as a Storage Cell, Random Access Memory, Read Only Memory, Secondary Storage, Hard Disk, Compact Disc Read Only Memory, Archival Store and calculation

Unit – IV: Processing of Various Types of data Hours: 08

Introduction to Numerical Data, Textual and Multimedia,

Numerical Data: Use of Spreadsheets and numerical computational example

Textual Data: Word Processor, Desktop Publishing, Page Description Language, Markup Languages

Multimedia Data: Graphics Processing and Audio Signal Processing,

Unit – V: Application Related to Information Technology Hours: 04

Business Applications, On-Line Applications, Real-Time Applications

Unit – VI: Emerging Trends and Societal Impact of IT Hours: 05

Introduction to Emerging Trends in ICT, Ecommerce, Electronic Data Interchange, Mobile Communication, Bluetooth, Global Positioning System, Infrared Communication System, Smart Card, Imminent Technology

Introduction to Societal Impact, Social Use of World Wide Web, Privacy, Security and Integration of Information, Disaster Recovery, Intellectual Property Right, Career in IT

Core Books:

1. V Rajaraman: Introduction to Information Technology, 2nd Edition, PHI Learning Private Limited, 2013.
2. ITL Education Solution Limited private Limited: Introduction to Information Technology, 5th Impression, 2008.

Reference Books:

1. Turban, Rainer, Potter: Introduction to Information Technology, 2nd Edition, Wiley India, 2005.

2. PelinAksoy,LauraDenardis : Information Technology in Theory, Thomson Course Technology,2008.

Web References:

1. http://sct.emu.edu.tr/courses/it/index.php?id=itec103&page_type=file_directory&element_id=2 [Information Technology fundamentals]
2. <http://technology.ku.edu/software> [Information Technology related applications]
3. <http://www.managementstudyguide.com/emerging-trends-in-information-technology.htm>
[Emerging trends in Information Technology]

Upon successful completion of the course, the students will :

- C01 : Understand the data, information and tool to process them.
- C02 : Understand how to acquire, store, organize and process different kind of data.
- C03 : Understand the concept of Information Technology and real applications of it.
- C04 : Understand the Emerging trends and Societal Impact of Information Technology.
- C05 : Learn all the concepts related to information technology through case study

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Basics of Information Technology	✓				✓
2	Acquisition of Various Types of data		✓			✓
3	Data Storage and Organizations		✓			✓
4	Processing of Various Types of data		✓			✓
5	Application Related to Information Technology	✓		✓		✓
6	Emerging Trends and Societal Impact of IT				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	2	-	-	-	-	-	-	-	3	-
CO2	3	3	2	3	1	1	-	1	-	1	1	3	1
CO3	3	3	3	2	1	1	2	1	2	-	-	3	1
CO4	2	2	3	3	1	2	2	1	2	-	-	3	2
CO5	3	3	3	3	2	2	1	1	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA110: Fundamentals of Computer Organization

(Marks 100)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures the emphasis will be given on the basic foundations required to understand the operations of the Computers. Clear understanding of the Internal Hardware of the computer and various peripherals will be given to the students.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours
		Theory
1	Computer Operation and Number System	07
2	Boolean Algebra and Gates Network	07
3	Digital Logic Circuits and Digital Components	06
4	Data Representation	05
5	Buses, Interfaces and Control Unit	06
6	Computer Organization	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Computer Operation and Number System Hours: 07

Electronic digital computers, Application of computers to problems, Basic components of a digital computer, construction of memory, instructions, programming systems, Decimal system, distable devices, counting in the binary system, binary addition and subtraction, binary multiplication and division, converting decimal numbers to binary, negative numbers, use of complements to represent negative numbers, binary number complements, binary coded decimal number representation, octal and hexadecimal number systems.

Unit – II: Boolean Algebra and Gates Network Hours: 07

Fundamental concepts of boolean algebra, logical multiplication, AND gates and OR gates, complementation and inverters, evaluation of logical expressions, basic laws of Boolean algebra, De Morgan's theorem, basic duality of Boolean algebra, derivation of Boolean expression, interconnecting gates, sum of products and product of sums, NAND and NOR gates, don't care conditions, design using NAND and NOR gates.

Unit – III: Digital Logic Circuits and Digital Components Hours: 06

Digital components, combinational circuits, flip flops, sequential circuits, integrated circuits, decoders, multiplexers, registers, shift registers, binary counters, memory unit.

Unit – IV: Data Representation Hours: 05

Data types, complements, fixed point representation, floating point representation, other binary codes, error detection codes.

Unit – V: Buses, Interfaces and Control Unit Hours: 06

Interconnecting system components, interfacing buses, a standard bus interface, construction of instruction word, instruction cycle and execution cycle, sequence of operation of control registers.

Unit – VI: Computer Organization Hours: 05

Instruction word formats, representation of instructions, addressing techniques, direct addressing, immediate addressing, relative addressing, indirect addressing, indexed addressing

Core Books:

1. Thomas C. Barteo: Digital Computer Fundamentals, Sixth Edition, Tata McGraw Hill Publishing, 2012.
2. M. Morris Mano: Computer System Architecture, Third Edition, Pearson Education, 2011.

Reference Books:

1. Andrew S. Tanenbaum: Structured Computer Organization, Fourth Edition, Pearson Education, 2005.
2. Albert Paul Malvino and Jerald A. Brown: Digital Computer Electronics, Third Edition, Tata McGraw Hill Publishing, 2008.

Web References:

1. <http://www.eecs.wsu.edu/~ee314/handouts/numsys.pdf>
2. <http://www.uotechnology.edu.iq/dep-eee/lectures/1st/Digital%20techniques/part2.pdf>
3. http://ece.gmu.edu/~clorie/Spring11/ECE-301/Lectures/Lecture_16.pdf
4. <http://pages.cpsc.ucalgary.ca/~jacob/Courses/Fall00/CPSC231/Slides/04BitsAndArithmetic.pdf>

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : To understand Operation of Computer, Devices of Computer, Number System and Binary arithmetic operations
- C02 : Getting deep knowledge of digital electronic component and Boolean algebra and logical expressions.
- C03 : To understand internal circuits for various operation of computer and internal circuit for memory unit.
- C04 : To Learn various Data representation in computer and binary and error detection code.
- C05 : To understand interconnection between units of computer with bus interfaces and instruction execution cycle for instruction.
- C06: To understand Format of different instructions with addressing techniques to store instructions.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Computer Operation and Number System	✓					
2	Boolean algebra and Gates Network		✓	✓			
3	Digital Logic Circuits		✓	✓			
4	Data Representation				✓		
5	Buses, Interfaces and Control Unit					✓	✓
6	Computer Organization					✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	2	-	-	-	-	-	-	-	3	-
CO2	3	3	2	3	1	1	-	1	-	1	1	3	1
CO3	3	3	3	2	1	1	2	1	2	-	-	3	1
CO4	2	2	3	3	1	2	2	1	2	-	-	3	2
CO5	3	3	3	3	2	2	1	1	1	2	2	3	2
CO6	3	2	2	3	2	2	1	1	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA111: Fundamental Concepts of Programming Language

(Marks 200)

Contact Hours: 07

Pre- requisite: None.

Methodology & Pedagogy: During theory lectures the emphasis will be given on theoretical concepts of flow charts and algorithms. Structure of C programming language and different elements C programming will be explained in lectures. During practical sessions the concepts learned in theory lectures will be implemented.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Programming Basics	09	36
2	Introduction to C	08	
3	Data type, constants and variables	07	
4	Operators and Expression in C	06	
5	Decision control, looping and arrays	10	
6	Introduction to Functions	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit – I: Programming Basics: 09

Introduction to algorithm, Key features of algorithm, Introduction to flow charts, Significance of flow chart, Advantages of flowchart, limitation of flow chart, Introduction to pseudo code, Introduction to programming languages

Unit – II: Programming Basics: 08

Introduction, Background, characteristics of C, Structure of C Program, Writing first C program, Files used in C program, Compiling and Executing C program

Unit – III: Data type, constants and variables:07

Basic data types in C, Variables in C, Declaring and initializing variables in C, Constants, Declaring Constants, Input / Output statements in C

Unit – IV: Operators and Expression in C: 06

Operators - arithmetic, relational, logical, assignment, increment-decrement, conditional, Bitwise and special, comma operator, sizeof operator, operator precedence chart

Arithmetic expressions, evaluation of expressions, type conversions in expressions, operator precedence and associativity, mathematical functions. Type conversion and casting

Unit – V: Decision control, looping and arrays: 10

Introduction to decision control statements, Conditional and branching statements, Iterative statements and looping, Nested loops, Break, continue and goto statement.

Arrays: One-dimensional, two-dimensional,

Handling of Character Strings: Declaring and initializing string variables, reading string from terminal, writing string to screen, putting string together, comparison of two strings, string handling functions, table of strings.

Unit – VI: Introduction to Functions: 08

Need for user-defined functions, the form of C function, return values and their types, calling a function, category of functions, handling of non-integer functions, nesting of functions, recursion, functions with arrays, storage class.

Core Books:

1. Reema Thareja: Computer Fundamentals and Programming in C, 2nd Edition, Oxford University press, 2012
2. E. Balagurusamy: Programming In C, 6th Edition, 2012

Reference Books:

1. Herbert Schildt: The complete reference C, Fourth edition, Tata McGraw Hill, 2000

2. YashwantKanetkar: Let us C, 13th Edition, BPB publication, 2013.
3. PradipDey, Manas Ghosh: Programming in C, 4th Edition, Oxford University press , 2012.
4. Brian W. Kerighan and Dennis M. Ritchie : The C Programming Language,Prentice Hall.

Web References:

1. <http://cprogrampracticals.blogspot.in/p/special-c-programs.html>
[For basic c programing practical]
2. <http://cprogramsblog.blogspot.in/> [For operators and expressions in C]
3. <https://www.fayewilliams.com/> [For functions]
4. <http://www.tutorialspoint.com/cprogramming/>[For operators and expressions in C]
5. <http://www.programiz.com/c-programming> [For basic c programing practical]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Understand about algorithm, flowchart and Pseudocode with logical understanding
- C02 : Learn about the structure of C program with demo and flow of compilation and execution
- C03 : Understanding data types, variables and constants with all the operators and expressions in C
- C04 : Gain the knowledge of different conditional and iterative statements with Arrays of numbers and characters as well as manipulating with them.
- C05 : Learn and exploring User defined functions.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Programing Basics	✓				
2	Introduction to C		✓			
3	Data types, Constants and Variables			✓		
4	Operators and Expression in C			✓		
5	Decision Control, Looping and Arrays				✓	
6	Introduction to Functions					✓

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	1	2	2	3	2	-	2	2	3	2
CO2	3	1	3	2	2	3	2	-	-	2	2	3	2
CO3	2	3	2	2	2	3	1	-	-	1	2	3	2
CO4	3	3	3	1	1	3	2	-	-	2	1	3	2
CO5	3	2	3	2	1	2	1	-	-	1	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA112: Introduction to Computers

(200 Marks)

Contact Hours: 07

Pre-requisite: None.

Methodology & Pedagogy : During theory lectures students will be introduced to the fundamentals of computers, functionalities of various office automation software, Internet and Email system. During Practical sessions students will be made familiar with the working of various software packages like MS Word, MS Excel and MS PowerPoint.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Computer Fundamentals	09	36
2	The Computer System Architecture	07	
3	Elements of Word Processing	09	
4	Working with Spread Sheet	09	
5	Making Small Presentations	08	
6	Computer Communication and Internet	06	

Total Hours (Theory): 48

Total Hours (Practical):36

Total: 84

Detailed Syllabus:

Unit - I: Computer Fundamentals Hours:09

Introduction to computers, History of computers, Characteristics of computers, Block diagram of computer, Types of computers and features, Types of programming languages, Data organization, Types of memory, I/O devices, Number systems.

Unit - II: The Computer System Architecture Hours:07

Introduction, Central Processing Unit, Memory Unit, Microprocessor, Interconnecting the Units of a Computer, Performance of a Computer, Inside a Computer Cabinet, Introduction to Emerging Technologies.

Unit - III: Elements of Word Processing Hours:09

Introduction to Word Processing and MS word: Creating, saving and opening word documents, Menus, Shortcuts, Toolbars, Ruler Menus, Scroll bar, Editing a document, Formatting page and Setting Margins, Editing text documents, Using help.

Formatting Documents: Setting Font Styles, Paragraph styles and Page styles, Setting document styles, Creating tables, Drawing and formatting pictures, Spell check, Macros, Mail merge, Security, Printing documents.

Unit - IV: Working with Spread Sheet Hours:09

Features of excel, Concept of worksheet, Excel window, Navigating worksheet, entering editing data into cells, Insert/delete/hide/show rows/columns, Change column width, row height. Formatting data, Formulas & operators Range of cells, moving – copying data, Spell checking, Various types of addressing, Protecting & hiding data, sorting data, Searching & replacing data, Multiple worksheets & operations on them, Built-in functions, Look up tables, Pivot table, Data organization- what-if analysis, Charts, pictures, file operations, Macros, Circular reference, Goal seek.

Unit - V: Making Small Presentations Hours:08

Basics, Creation of Presentation, Preparation of Slides, Providing aesthetics and Animation, Slide Manipulation and Slide Show, Speaker notes, handouts, an outline, Presentation of the Slides.

Unit - VI: Computer Communication and Internet Hours:06

Basic of Computer Networks, Internet, Services on InterNet, WWW and Web Browsers, Basics of Email, Sending and receiving email messages, Email netiquette, Abbreviations and emoticons, Signature and attachments to emails, Filtering emails, Securing emails by using digital certificates, Forms of chats and conferencing.

Core Books:

1. Pradeep Sinha, Priti Sinha : Computer Fundamentals, 6th Edition, BPB Publications, 2003.
2. Joyce Cox, Curtis Frye, Dow Lambert III, Steve Lambert, John Pierce, Joan Preppernau : Step by Step 2007 Microsoft Office System, 2nd Edition, Microsoft Office Press, 2007.
3. Margaret Levine Young: Internet: The Complete Reference, Millennium Edition, Tata McGraw Hill, 1999.

Reference Books:

1. Katherine Murray, Mary Millhollon, Beth Melton : Microsoft Office Word 2007 Inside Out, Microsoft Press Publication, 2007.

2. Mark Dodge, Craig Stinson : Microsoft Office Excel 2007 Inside Out, Microsoft Press Publication, 2007.
3. Cliff Atkinson : Beyond Bullet Points: Using Microsoft Office PowerPoint 2007 to Create Presentations That Inform, Motivate and Inspire, Microsoft Press Publication, 2007.

Web References:

1. <http://office.microsoft.com/en-us/training/> [For Microsoft Office Concepts]
2. <http://www.baycongroup.com/> [For Microsoft Office Concepts]
3. <http://www.gcflernfree.org/office> [For Microsoft Office Concepts]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : To understand fundamental concept of computer hardware & software and functionality of various unit of computer, knowledge of Interconnecting the Units of a Computer and how computer works.
- C02 : To work with the basic features of MS Word, to compose, format and edit a word document with advanced formatting features like table of content, table and mail merge.
- C03 : To work with the basic features of MS Excel, to compose, format and edit an excel workbook with advanced features like filtering, pivot table, chart, in-built functions and data analysis & organization.
- C04 : To learn the creation of professional MS PowerPoint presentation by Providing aesthetics, animation& transition and multi-media.
- C05 : To familiarize about Internet, viewing information on Internet (the web), sending mails and basics of computer network.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Computer Fundamentals	✓				
2	The Computer System Architecture	✓				
3	Elements of Word Processing		✓			
4	Working with Spread Sheet			✓		
5	Making Small Presentations				✓	
6	Computer Communication and Internet					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	-	-	1	-	2	1	-	-	-	-	-	-
CO2	3	1	3	2	1	2	1	2	-	-	-	2	1
CO3	3	1	3	3	1	2	1	2	-	-	-	2	1
CO4	3	1	3	3	2	2	1	2	-	-	-	2	1
CO5	1	-	-	3	-	3	-	1	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA113: Programming the Internet - I
(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: This course focuses on providing hands-on experience to students for design and develop entire web sites using several web designing tools and HTML scripting language.

Outline of the Course:

Week No	Practical	Description
1 - 2	Structure of HTML Document, Using different tags likes Basic Tags, Paragraph Tags, List Tags, Horizontal Rule Tag, Heading Tags, Block quote Tags, Address Tags, FRONT Tag, PRE Tag, DIV tags, SPAN tag and other formatting tags demonstrate uses of those tags.	Student can learn structure of basic HTML page and also implements different HTML tags so that they became familiar with different tags and also known which tags use for specific formatting of web page.
3-4	Show how to use URL for in href tag so we make multiple web pages and make connected those pages using href tag. Also using href tag we can create email link.	Student will use HREF tag and then using it make inter link different web page of website. Also using this HREF tag student are able to create email link so they can redirect email sending outlook interface.
5-6	Introduce image tag and also implement inline & floating image inside content of webpage. Also demonstrate image mapping using area and map tags.	Using these concepts students are able to integrate images in content of webpage. Also they are able to make linkable image with different area inside the image like rectangle, circle etc.

7-8	Demonstrate Table and related tags like table, tbody, tr, td, th, caption. Also introduce the concepts like rowspan, colspan, cellspacing, cellpadding.	For displaying tabular information inside webpage table tags are used by students and show the content in well format.
9-10	Overview of Frames, Frame tags and attributes, Frame targeting, Floating Frames	Student are able to use frame inside their webpages.
11-12	Introduction to Forms, Form Tags and attributes, Form controls: Text Field, Password Field, Multiple Line Text area, Drop Down list, Check Box, Radio Buttons, Scrolled List, Reset Button, Submit Button, File field	For making interactive form inside HTML webpage student can use form and take inputs from the users.

Core Books:

1. Thomas a Powell: The Complete reference HTML, 3rd Edition, McGraw Hill,2001.
2. A. Whyte: Basic HTML, 2nd Edition, Payne-Gallway, Oxford, 2003.
3. Peter Morville& Louis Rosenfeld, Information Architecture for WWW , 3rd Edition, O'Reilly Publication, 2006.
4. Farrar : HTML Example book, BPB,2007.

Reference Books:

1. Karl Barksdale, Shane Turner : HTML and Java Scripts Basics, 4th Edition, Course Technology,2006.
2. Shelly Woods: HTML introductory concepts and techniques, 5th Edition, Course Technology,2009.
3. Jon Duket: Beginning Web Programming with HTML,XHTML and CSS,Wrox Publication.

Web References:

1. www.w3schools.com/html/ [For HTML tutorials]
2. html.net/tutorials/html/ [For HTML tutorials]
3. www.htmlgoodies.com/ [For HTML resources]
4. <https://developer.mozilla.org/en-US/learn/html/> [For HTML best practices]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Able to understand the structure of HTML document and started using basic tags.
- C02 : Able to understand how Text and Images work in HTML.
- C03 : Able to create table in HTML page.

C04 : Use the concept of frames and frameset in webpage.

C05 : Able to create webpages with form controls.

Course Outcomes Mapping:

Week No	Content	Course Outcomes				
		C01	C02	C03	C04	C05
1-4	Structure of HTML Document, Using different formatting tags, URL integration.	✓				
5-6	Insert image into HTML file	✓	✓			
7-8	Creating table in webpage	✓		✓		
9-10	Frame tags and attributes	✓			✓	
11-12	Creating form in HTML	✓				✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	1	-	3	-	-	-	-	1	-	1	-	1	-
C02	1	-	3	-	-	-	-	1	-	1	-	1	-
C03	1	-	3	3	-	-	-	1	-	1	-	1	-
C04	1	-	3	-	-	-	-	1	-	1	-	1	-
C05	1	-	3	3	-	-	-	1	-	1	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA114: Foundation of Mathematics
(Marks 100)

Contact Hours: 03

Pre-requisite : None

Methodology & Pedagogy: The emphasis during Lecture sessions will be on Understanding of Concepts rather than on complexities of Computational Techniques. Stress is also given on Visual perception through Mathematical Software. Relevant Illustrations will be provided from the Real World processes. Sufficient home assignments will be given to the students which will test their fundamentals and ability to relate Mathematical concepts with reality.

Outline of the course:

Unit No.	Title of the Unit	Minimum Numbers of Hours
		Theory
1	Set Theory	05
2	Relations & Functions	05
3	Matrices and Determinants	06
4	Limit & Continuity	06
5	Differentiation & its applications	07
6	Integration & its applications	07

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit - I: Set Theory Hours:05

Introduction to set theory, Methods of representation of a set, Operations on set, Properties of set with logical proofs, Venn Diagram, Cartesian product of sets

Unit- II: Relations & Functions Hours:05

Definition of relations, Types of relations,

Function: Definition of Function, Classification of function, Domain and Range of function, Types of functions

Unit- III: Matrices and Determinants Hours:06

Matrices, Matrix Operations, Properties of Matrix Operations, Determinants (up to order 3) and their properties, Cofactor expansion, The Inverse of a Matrix, The Rank of a Matrix and Applications.

Unit- IV: Limit& Continuity Hours:06

Limits: Definition of Limit, Evaluation Techniques

Continuity: Definition of Continuity, Conditions for the function to be Continuous, discontinuity

Unit- V: Differentiation & its applications Hours:07

Definition of differentiation, Rules of differentiation, Evaluation Techniques, Derivatives of Algebraic, trigonometric, Logarithmic, Explicit/Implicit function, Second order Derivative with example

Unit- VI: Integration & its application Hours:07

Definition of Integrations, Primitives of Standard Functions, Methods of Integration, Integration by parts

Applications of Integration: Area Bounded by the Curve

Core Books:

1. B.S.ShahPrakashan: Business Mathematics
2. S.Chand& Sons Publications
3. Thomas G. B. and R. L. Finney: Calculus and Analytical Geometry, 9th Ed., Addison Wesley, 1996.
4. Erwin Kreyszig: Advanced Engineering Mathematics, 8th Ed., John Wiley & Sons, India, 1999

Reference Books:

1. Larsen & Marx: An Introduction to Mathematical Statistics and Its Applications, Third Ed., Prentice Hall, NJ, USA, 2001.
2. Wylie & Barrett: Advanced Engineering Mathematics, Mc graw Hill pub.
3. Greenberg M D: Advanced Engineering Mathematics, 2nd ed., Pearson Education

Web References:

1. <http://www.mathtutor.ac.uk/differentiation> [For Differentiation & Integration]
2. <http://www.maths.manchester.ac.uk/~mdc/old/1K1/DiscreteMaths.html> [For Set Theory, Functions and Matrices]

Course Outcomes: Upon successful completion of the course, the students will be able to:

C01 :	Formulate problems in the language of sets and perform various set operations. Draw and use Venn diagrams to solve practical problems.
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C02 :	Understand basics of relations and functions, types of relations and functions and classification of functions.
C03 :	Understand basics of Matrix, performing various operations on matrix, types of matrices, applications of matrix (to find determinant and linear equation solutions)
C04 :	Compute basic limits of functions. Understand the importance of limits to the process of differentiation and be able to compute the derivative of a simple functions. Compare and contrast the ideas of continuity and differentiability.
C05 :	Use the basic integration rules to find indefinite integrals, Use the Fundamental Theorem of Calculus to evaluate definite integral and its application as to find the area under the curve.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Set Theory	2				
2	Relations & Functions	2	2			
3	Matrices and Determinants			2		
4	Limit & Continuity				2	
5	Differentiation & its applications				2	2
6	Integration & its applications					2

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	1	-	1	1	1	-	1	-	2	1
CO2	3	3	2	1	-	1	1	1	-	1	-	2	1
CO3	3	3	2	1	-	1	1	1	-	1	-	2	1
CO4	3	3	2	1	-	1	1	1	-	1	-	2	1
CO5	3	3	2	1	-	1	1	1	-	1	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS 101.02 C: Communicative English

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	2
Marks	--	100	--	100	

Pre-requisite courses:

- Grammar and Punctuation (Free Online Course for Beginners)
<https://www.coursera.org/learn/grammar-punctuation>

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Communicative English	03
2.	Communication Functions	06
3.	Basic Communication Skills I – Listening	03
4.	Basic Communication Skills II – Reading	03
5.	Basic Communication Skills III – Speaking	06
6.	Basic Communication Skills IV – Writing	06
7.	Developing Vocabulary	03
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

- Introduction to Communicative English 03 Hours 10%

English as a Window Language; Varieties of English: British English, American English, Indian English; Language Variations; Importance of English for Academic and Professional Development; Strategies for Language Acquisition; Formal VS Informal English

- | | | | |
|----|-------------------------|---------|-----|
| 2. | Communication Functions | 6 Hours | 20% |
|----|-------------------------|---------|-----|

Greeting and Introducing; Making Requests and Asking for Information; Expressing Likes and Dislikes; Seeking Permission; Giving and Taking Advice; Describing People, Place, Things; Retelling Past Events; Comparing and Contrasting; Persuading; Describing Cause and Effect

- | | | | |
|----|--|---------|-----|
| 3. | Basic Communication Skills I – Listening | 3 Hours | 10% |
|----|--|---------|-----|

Importance of Listening as a Language Skill; Basic Listening Skills; Types of Listening; Barriers to Listening; Strategies for Effective Listening; Listening Practice

- | | | | |
|----|---|---------|-----|
| 4. | Basic Communication Skills II – Reading | 3 Hours | 10% |
|----|---|---------|-----|

Importance of Reading as a Language Skill, Reading Strategies: Skimming, Scanning, Intensive Reading, Extensive Reading, Strategies for Effective Reading Comprehension, Reading Practice

- | | | | |
|----|---|---------|-----|
| 5. | Basic Communication Skills III – Speaking | 6 Hours | 20% |
|----|---|---------|-----|

Importance of Speaking as a Language Skill, Basic Speaking Skills; Paralanguage for Effective Speaking; Strategies for Oral Communication; Extempore and Public Speaking

- | | | | |
|----|---|---------|-----|
| 6. | Basic Communication Skills IV – Writing | 6 Hours | 20% |
|----|---|---------|-----|

Importance of Writing as a Language Skill; Process of writing: Prewriting, Drafting, Revision, Editing, Publication; Seven C's of Writing; Sentence Construction – Complex, Compound, Paragraph Development; Letter Writing (Academic Context)

- | | | | |
|----|-----------------------|---------|-----|
| 7. | Developing Vocabulary | 3 Hours | 10% |
|----|-----------------------|---------|-----|

High Frequency Vocabulary (Everyday and Academic use); Words Often Confused and Misused; Useful Phrasal Verbs, Idioms and Proverbs, Homonyms and Homographs; Lexical Range, Word Games

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Communicate with people using English language functions including greetings, introductions, making and responding to requests, suggestions, invitations and apologies, conducting simple transactions in shops and offices, asking for and giving directions, etc
CO2	Go through a text and identify specific and global information.
CO3	Become more knowledgeable about speaking strategies and speak effectively using appropriate words, expression, tone and pronunciation.
CO4	Be aware about various reading strategies and read and comprehend academic and non-academic prose (text).
CO5	Write systematically using nuances of writing
CO6	Express their opinion and likes and dislikes, advice and convince others in a more polite and accepted way.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	3
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	2	-	-	3	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Text book:

1. Sanjay Kumar and PushpLata (Second Edition, 2015), Communication Skills, Oxford University Press, New Delhi

2. M V Rodrigues (2013), Effective Business Communication, Concept Publishing Company (P) Ltd., New Delhi
3. Krishna Mohan and Meera Banerji (2010), Developing Communication Skills, Macmillan Publications India Ltd., New Delhi

❖ **Reference book:**

1. Mohan and Meenakshi Raman (2006), Effective English Communication, McGraw-Hill Publishing Company Limited, New Delhi
2. Geoffrey Leech & Jan Swartvik (1994), A Communicative Grammar of English, Longman Publications, New York
3. Jones Leo (1979), Functions of English, Cambridge University Press, UK
4. European Journal of Language and Literature Studies Vol.1 Nr, 1 April 2015
5. English for Academic Purpose: A Tool for Enhancing Students' Proficiency in English Language Skills

❖ **Web material:**

1. <https://www.futurelearn.com/courses/language-assessment>
2. <https://www.coursera.org/learn/importance-of-listening?#syllabus>
3. <https://www.futurelearn.com/courses/english-academic-study>

Teaching and Examination Scheme (BCA Programme)

Effective from Year 2021-2022

Semester-II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA115-116	Elective-II	3	-	3	3	10	20	70	-	-	-	100
CA117	Operating System Concepts	4	3	7	7	10	20	70	15	15	70	200
CA118	Advanced Programming	4	3	7	7	10	20	70	15	15	70	200
CA119	Database Fundamentals	4	3	7	7	10	20	70	15	15	70	200
CA120	Fundamentals of Commerce & Business Process	3	-	3	3	10	20	70	-	-	-	100
HS201.02 C HS202.02 C HS205.02 C HS209.02 C HS210.02 C	Painting Photography Media and Graphic Design Dramatics Contemporary Dance	-	2	2	2	-		-	30		70	100
FS101C	Foundation Course On Mathematics And Physics	-	2	2	2	-		-	30		70	100
		18	13	31	31	500			500			1000

Elective-II

1. CA115–Management Information System
2. CA116–Numerical & Statistical Method

CA115 : Management Information System

(100 Marks)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During the Theory sessions, illustrations of working real time information system will be demonstrated. Industrial visits will be arranged in order to demonstrate actual functioning of such systems. Students will be required to carry out a case study of such system.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours
		Theory
1.	Introduction to Information System	05
2.	Transaction Processing Systems	07
3.	Management Information System	06
4.	Decision Support System	06
5.	Knowledge Management & Executive Support Systems	07
6.	Introduction To ERP and Technology	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit - I: Introduction to Information System Hours:05

Data & Information, What is a system, Types of Systems, Types of information, Quality of Information, What is an Information System, Functions of Information System, classification of an Information System, Major Types of Information Systems, Interrelationship of Information Systems.

Unit - II: Transaction Processing Systems Hours:07

Introduction of Transaction Processing System (TPS), Transaction Processing System (TPS) Architecture,

Transaction Processing System (TPS) Life cycle, Methods of Transaction Processing System (TPS), Information System Availability Control, Illustrations and live demonstration of TPS such as Banking System, Railway reservation system etc.

Unit - III: Management Information System Hours:06

Introduction to Management: Approaches of Management, Function of Management, System from a Functional Perspective, Reports of Management Information System, A business perspective of MIS, Dimensions of MIS, contemporary Approaches to Information System.

Unit - IV: Decision Support System Hours:06

Business value of Improved Decision making, Types of Decision, Decision making Process, The difference between MIS and DSS, Components of DSS, System for Decision Support Group Decision Support System, Business value of GDSS.

Unit - V: Knowledge Management & Executive Support Systems Hours:07

Important Dimensions of knowledge, Organizational learning and Knowledge Management, The Knowledge Management value chain, Overview of different types of Knowledge Management Systems, Characteristics of ESS, The Role of ESS in the Firm, Business value of ESS, Monitoring Corporate Performance, Working Examples of ESS.

Unit - VI: Introduction To ERP and Technology Hours:05

An overview of ERP, Basic ERP Concepts, Risk and Benefits of ERP, ERP and Related Technologies, Business Intelligence

Core Books:

1. K. C. Laudon and J. P. Laudon: Management Information Systems, 12th Edition, Pearson Education.
2. Alexis Leon : ERP Demystified, Second Edition, Tata McGraw-Hill.

Reference Books:

1. W.S. Jawadekar: Management Information Systems, 2nd Edition, TataMcGraw-Hill.

Web References:

1. <http://www.slideshare.net/NorazilaMat1/laudon-mis12-ppt01-16595885>[Function of Information system, A business perspective of MIS, Dimensions of MIS, Contemporary Approaches to Information System]
2. <http://www.uh.edu/~mrana/try.htm>[Types and Functions of Information system]
3. <http://bisom.uncc.edu/courses/info2130/Topics/istypes.htm>[Types of Information system]
4. <http://kalyan-city.blogspot.com/2011/05/levels-of-management-top-middleand.html>[classification of information system]
5. http://my.safaribooksonline.com/book/management/9780470916803/operationalplanningand-control-systems/management_levels_comma_functions_comm[classification of information system]
6. <http://ambarwati.dosen.narotama.ac.id/files/2011/05/FIS-2011-w2.pdf>[Transaction Processing System]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : To understand basic of data & information, types of information system, function of information system and its quality.
- C02 : To understand transaction processing concepts and its working.
- C03 : Learning Management Information system, its types and its implementation.
- C04 : To gain understanding of Decision Support Systems.
- C05 : To learn Knowledge Management, Executive Support Systems , ERP and its technology.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Information System	✓				
2	Transaction Processing Systems		✓			
3	Management Information System	✓	✓	✓		
4	Decision Support System		✓		✓	
5	Knowledge Management & Executive Support Systems		✓			✓
6	Introduction To ERP and Technology		✓			✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	3	2	2	1	1	3	1	-	-	1	-	2	1
C02	3	2	2	2	2	2	2	-	-	1	-	2	2
C03	3	3	2	2	3	2	3	-	-	2	-	3	2
C04	3	3	2	3	3	3	2	-	-	3	2	3	3
C05	3	3	2	2	3	3	3	-	-	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA116: Numerical and Statistical Methods

(100 Marks)

Contact Hours: 03

Pre-requisite: Foundation of Mathematics

Methodology & Pedagogy: During Theory lectures illustrations of certain real world problems requiring Numerical computations will be explained. Numerical and Statistical methods will be introduced as tools for solving such problems also be emphasized

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours
		Theory
1	Basic Statistics Methods	06
2	Basic of Probability	05
3	Probability Distribution	05
4	Solutions of Non-linear Equation	06
5	Solution of Simultaneous Linear Equation	07
6	Interpolation and Numerical Integration	07

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit-I: Basic Statistics Methods Hours: 06

Basic vocabulary of statistics. Measure of central tendency: Mean Median and Mode for group and ungroup data, Measure of dispersion: Range, Mean Deviation, Variance and Standard Deviation for group and ungroup data.

Unit-II: Basic of Probability Hours: 05

Basic concepts of probability, Random Experiment, Events, Sample Space, Types of Events, Mathematical and Statistical definition of probability, Properties of Probability Laws

Unit-III: Probability Distribution Hours: 05

Random Variable, Discrete and continuous random variable, Standard Deviation and Variance of Random Variable, Binomial Distribution, Poisson Distribution, Normal Probability Distribution.

Unit-IV: Solutions of Non-linear Equation Hours: 06

Error in Numerical Computation: Absolute, Relative and Percentage Errors, Root of Equation: Bisection, False Position Method, Newton-Raphson Method, Self-Iterative Method.

Unit-V: Solution of Simultaneous Linear Equation Hours: 07

Basic concepts of system of linear equation, Gauss-Elimination Method, Gauss-Jordan Method, Gauss-Seidel Method, Jacobi's Iterative Method.

Unit-VI: Interpolation and Numerical Integration Hours: 07

Interpolation: Lagrange Interpolation Formulae, Newton Divided Difference Interpolation Formulae. Newton Forward Difference and Backward Difference Interpolation Formulae, Error estimates for these Formulae. Numerical Integration: Trapezoidal Rule, Simpson's 1/3 and 3/8 rules.

Core Books:

1. Levine, Krehbiel, Berenson, Viswanathan: Business Statistics, 5th Edition, Pearson.
2. B.S.Grewal: Numerical Methods in Engineering & Science, Khanna Publication
3. V. Rajaraman Computer Oriented Numerical Methods, 3rd Edition, Prentice-Hall of India Pvt. Ltd

Reference Books:

1. Hogg & Tannis: Probability and Statistical Inference, Sixth Ed., Prentice Hall, NJ, USA, 2000.
2. Larsen & Marx: An Introduction to Mathematical Statistics and Its Applications, Third Ed., Prentice Hall, NJ, USA, 2001.
3. D. N. Datta: Computer Oriented Numerical Methods, Vikas Pub. House, 2008.
4. S.S. Sastry : Introductory Methods of Numerical Analysis, Fourth Edition, PHI.

Course Outcomes: Upon successful completion of the course, the students will:

- C01: Able to calculate the mean, median, mode, and range for a set of data they collected. Students will be able to represent and statistically analyze data both graphically and numerically.
- C02: Apply problem-solving techniques to solving real-world events. Develop problem-solving techniques needed to accurately calculate probabilities and can identify types of events and their probability to occur.
- C03: Apply selected probability distributions to solve problems by various Distribution techniques (Binomial, Poisson, and Normal).
- C04: Understand practical use of numerical methods in the field of computer science and applications and applying iterative methods to find out approximate root of nonlinear equation.
- C05: Solve systems of linear equations using various methods including Gaussian and Gauss-Jordan elimination and inverse matrices. Able to identify when to apply direct and indirect method to solve linear equation.

C06: Able to estimate the missing data through various interpolation methods. Identification of errors in the experimental data and familiarize with numerical integration and finding approximate solution of a definite integral.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Basic Statistics Methods	✓					
2	Basic of Probability		✓	✓			
3	Probability Distribution		✓	✓			
4	Solutions of Non-linear Equation				✓		
5	Solution of Simultaneous Linear Equation					✓	
6	Interpolation and Numerical Integration						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	3	3	-	-	-	2	-	-	-	-	-	1	-
C02	3	3	1	1	-	2	-	-	-	-	-	2	-
C03	3	3	-	-	-	2	-	-	-	-	-	2	-
C04	3	3	-	-	-	2	-	-	-	-	-	2	-
C05	3	3	-	-	-	2	-	-	-	-	-	-	-
C06	3	3	-	1	-	2	-	-	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA117: Operating Systems Concepts

(200 Marks)

Contact Hours: 07

Pre-requisite: None.

Methodology & Pedagogy: In order to achieve the course objectives, students will first be introduced to the basic operating system concepts and basic functions. During practical sessions, the students will be required to use DOS and UNIX commands to understand the system properties and to write script files.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1.	Basic Concepts and Usage of OS	06	36
2.	Process Management	08	
3.	CPU Scheduling	08	
4.	Memory management and I/O Management	10	
5.	Operating System Commands, Editors and Filters	08	
6.	Shell Programming	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit - I: Basic Concepts and Usage of OS Hours:06

Basic Elements, Operating System Overview: OS objectives and functions (OS as User/Computer interface, OS as Resource Manager), Evolution of OS, Operating System services

Unit - II: Process Management Hours:08

Process Control Block, Process States- A two state process model, A five state process model, A seven state process model, Process Operations- creation, termination and suspension, Process Vs. Thread, Inter process communication- independent and cooperative process.

Unit - III: CPU Scheduling Hours:08

Types of scheduling - long, medium, short, I/O, Scheduling Criteria, Scheduling Algorithms- FIFO, Round Robin, Priority Scheduling, SPN, SRT, Feedback

Unit - IV: Memory management and I/O Management Hours:10

Memory Management: Memory Management Requirements, Memory Partitioning, Paging, Segmentation, I/O management: I/O Devices, Organization of the I/O Function, OS Design Issues, I/O Buffering

Unit - V: Operating System Commands, Editors and Filters Hours:08

Basic DOS commands, Batch file, History and Features of Unix OS, Directory structure, Basic commands – Metacharacter, Shell Variable, Command Substitution, Editors: vi, emacs, pico, Simple Filters and pipes, standard Input, Standard output and Standard Error, Advance Filter – Grep.

Unit - VI: Shell Programming Hours:08

Shell Script, Shell environment, Making Script interactive, Command Line Argument, Evaluating Expression, Operator, Control Statement, Looping statement.

Core Books:

1. William Stallings: Operating Systems Internals and Design Principles, 5th Edition, PHI, 2005.
2. Sumitabha Das: Unix concepts & application, 4th Edition, Tata McGraw Hill, 2010.
3. Kenneth Rosen, Douglas Host, James Farber and Richard Rosinski: The Complete Reference, Tata McGraw Hill, 1999.

Reference Books:

1. Silberschatz: Operating System Concepts, 5th Edition, John Wiley & Sons (ASIA) Ltd., 2008.
2. Mark G. Sobell: A Practical Guide to Linux, Pearson Education, 1997.
3. K.J.George, Operating System Concepts and Principles, Sroff Publishers, 2003

Web References:

1. <http://www.nondot.org/sabre/os/articles> [A useful collection of documents and papers on a wide range of OS topics]
2. <http://www.ugu.com/sui/ugu/warp.ugu> [Excellent source of UNIX information]
3. <http://www.tutorialspoint.com/unix/index.htm> [Unix Tutorials]
4. <http://williamstallings.com/OS/Animation/Animations.html> [Animations of Operating Systems Concepts]
5. <http://placement.freshersworld.com/power-preparation/technical-interviewpreparation/osinterview-questions-23351> [Operating System questions and answers]
6. <http://www.ics.uci.edu/~ics143/lectures.html> [Lecture notes of OS]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Understand operating system overview, Functions and Services
- C02 : Learn about various states of process models with reasons for process creation and termination.

- C03 : Recognizing different CPU Scheduling algorithms with types of scheduling
 C04 : Learn how memory is managed internally with Partitioning, Paging and Segmentation
 C05 : Hands on with Shell Scripts by implementing programming elements.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Basic Concept and Usage of OS	✓				
2	Process Management		✓			
3	CPU Scheduling			✓		
4	Memory Management and I/O Management				✓	
5	Operating System Commands, Editors and Filters					✓
6	Shell Programming					✓

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	2	2	1	1	1	1	-	1	1	3	2
CO2	3	2	2	3	1	1	1	1	-	1	1	3	2
CO3	3	2	2	3	1	1	1	1	-	1	1	3	2
CO4	3	2	2	3	1	1	1	1	-	1	1	3	2
CO5	2	1	2	3	1	1	1	1	-	1	1	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA118: Advanced Programming

(200 Marks)

Contact Hours: 07

Pre-requisite: Fundamental Concepts of Programming Language

Methodology & Pedagogy: During theory lectures the emphasis will be given on theoretical concepts of structure, pointers and file management. During practical sessions the concepts learned in theory lectures will be implemented in depth by considering real life systems by using file management.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Pointer in C	10	36
2	Structure, Union, and Enumerated Data Types	10	
3	Files	8	
4	Preprocessor Directives	6	
5	Stacks and Queues	6	
6	Linked Lists	8	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit - I: Pointers in C Hours: 10

Understanding the Computer's Memory, Introduction to Pointers, Declaring Pointer Variables, Pointer Expressions and Pointer Arithmetic, Null Pointers, Generic Pointers, passing arguments to Function using Pointers, Pointers and Arrays, Passing an Array to a Function, Difference between Array name and Pointer, Pointers and Strings, Array of Pointers, Pointers and 2D Arrays, Function Pointers, Array of Function Pointers, Pointers to Pointers, Memory allocation in C programs, Memory usage, Dynamic Memory Allocation, Drawbacks of Pointers

Unit - II: Structure, Union and Enumerated data types Hours: 10

Introduction, Nested Structures, Arrays of Structures, Structures and Functions, SelfReferential Structures, Union, Arrays of Union Variables, Unions Inside Structures, Enumerated Data Types

Unit - III: Files Hours: 08

Introduction to Files, Using Files in C, Read Data from Files, Writing Data to Files, Detecting the End-of File, Error Handling During File Operations, Accepting Command Line Arguments, Functions for Selecting a Record Randomly, remove(), Renaming the File, Creating a Temporary File

Unit - IV: Preprocessor Directives Hours: 06

Introduction, Types of Preprocessor Directives, #define, #include, #undef, #ifdef, #ifndef, #if, #else, #elif, #endif, #line, #pragma Directives, Conditional Directives, #error Directive, Predefined Macro Names

Unit - V: Stacks and Queues Hours: 06

Stacks, Array Representation of Stacks, Operations on a Stack, Infix, Postfix, and Prefix Notations, Evaluation of an Infix Expression, Convert Infix Expression to Prefix Expression, Applications of Stack, Queues, Operations on a Queue

Unit - VI: Linked Lists Hours: 08

Introduction, Linked Lists Versus Arrays, Memory Allocation and Deallocation for a Linked List, Different Types of Linked Lists, Singly Linked List, Circular Linked List, Doubly Linked List, Circular Doubly Linked List, Header Linked List

Core Books:

1. Reema Thareja. "Introduction to C Programming", Oxford University Press.

Reference Books:

1. Reema Thareja. "Computer fundamentals and Programming in C", Oxford University Press.
2. E. Balagurusamy. "Programming in ANSI C", Sixth Edition, McGraw Hill Education.
3. Pradip Dey and Manas Ghosh. "Programming in C", Second Edition, Oxford University Press.

Web References:

1. <https://www.codingunit.com/c-tutorial-how-to-use-pointers> [For Pointers]
2. http://www.tutorialspoint.com/cprogramming/c_pointers.htm [For Pointers]
3. http://www.tutorialspoint.com/cprogramming/c_structures.htm [For Structures]
4. http://www.tutorialspoint.com/data_structures_algorithms/dsa_queue.htm [For stack & queue]
5. <http://www.studytonight.com/data-structures/queue-data-structure> [For data structures]
6. <http://fresh2refresh.com/c-programming/> [For entire C Programming]

Course Outcomes: Upon successful completion of the course, the students will :

- C01 : To understand basic of Pointer and operation on Pointers. Understand the concept of Dynamic memory management. Learn Programming using Array with Pointer and Structure.
- C02 : Learn operation on File. Read, write and Update file in C.

C03 : Understand basic data structures such as arrays, linked lists, stacks and queues.

C04 : To gain understanding of various Application implemented in C Language.

C05 : To learn various preprocessor directives.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Pointer in C	✓		✓		
2	Structure, Union, and Enumerated Data Types	✓	✓	✓	✓	
3	Files	✓	✓			
4	Preprocessor Directives					✓
5	Linked Lists			✓	✓	
6	Stacks and Queues			✓	✓	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	2	3	2	2	1	1	1	1	-	1	1	3	2
CO2	2	-	2	3	1	1	1	1	-	1	1	3	2
CO3	3	3	2	2	1	1	1	1	-	1	1	3	2
CO4	3	2	2	3	1	1	1	1	-	1	1	3	2
CO5	3	1	-	-	1	1	1	1	-	1	1	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA119: Database Fundamentals

(200 Marks)

Contact Hours: 07

Pre-requisite: Basic knowledge of working with computer.

Methodology & Pedagogy: During the theory lectures, concepts for data storage will be discussed. Also DBMS and RDBMS concepts will be discussed in depth. The database applications used in the real world will be discussed with necessary examples. During the laboratory hours students will implement the concepts that are discussed during lecture.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Database System	04	36
2	Data Modeling using Entity Relationship Model	08	
3	Relational Data Modeling	08	
4	Database Design Methodology	08	
5	Schema Definition, Constraints, Queries	10	
6	Advanced Query Processing	10	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit - I: Introduction to Database System Hours: 04

Basic Concepts: data, database, database systems, database management system Purpose and advantages of Database management system (over file systems)

Various data models – ER Model and Relational Model, Three level architecture, Structure of DBMS, Database actors and workers

Unit - II: Data Modeling using Entity Relationship Model Hours: 08

Entity Types, Entity Sets, Attributes, Keys, Relationship Types, Relationship Sets, Roles and Structural Constraints, Weak Entity Types, ER Diagram – Notations and fundamentals, Concept of Generalization and Specialization, Associative Entity, Conversion from ER Model to Relational Model

Unit - III: Relational Data Modeling Hours: 08

Relational Model Concepts, Relational Model Constraints, Relational Model Schemas, Update Operation, Transactions, Dealing with Constraint Violations, Relational Algebra and Calculus

Unit - IV: Database Design Methodology Hours: 08

Functional Dependency and Normalization for Database – Informal Design Guidelines for Relational Schemas, Functional Dependencies, Normal Forms (1NF, 2NF, 3NF, BCNF, 4NF, 5NF), Codd Rules

Unit - V: Schema Definition, Constraints, Queries Hours: 10

Table Fundamentals: Basic Data types, Create Table Command, Viewing Data in the tables, sorting, Insert, Delete and Update Statements in SQL, Modifying the structure of tables, renaming table, truncating table, destroying table.

Data Constraints and Functions: - Pseudo columns, Null values, TAB table, DUAL table Operators, Data constraints, Type of data constraints, Modifying constraints, working with data dictionary and use of USER_CONSTRAINTS

Functions – introduction, merits and demerits, types of functions

Numeric functions, Character functions, Date functions, Conversion functions, Aggregate functions

Unit - VI: Advanced Query Processing Hours: 10

Union, Intersect and Except

Nested Queries – Introduction, Co-related Nested Queries, Set Comparison Operators Aggregate Operators, Group By clause, having clause, Order by clause Null Values – Comparison using Null Values, Logical Connectivity – AND, OR and NOT, Impact on SQL constructs, Joins (Inner Join, Outer Join, Self Join, Equi Join, Cross Join) Complex Integrity Constraints in SQL – constraints over single table, domain constraints and distinct types Creation and manipulation of database objects – indexes, views, sequences and synonym .

Core Books:

1. RamezElmasri, Shamkant B. Navathe : Fundamentals of Database Systems , 5th Edition, , Pearson Publication.
2. Ivan Bayross : SQL, PL/SQL – The programming Language Oracle.
3. Ramkrishnan, Gehrke : Database Management Systems, 3rd Edition, McGrawHill Publication.

Reference Books:

1. Silberschatz, Korth, Sudarshan : Database System Concepts, 5th Edition, McGraw Hill.
2. C.J.Date, a Kannan, S Swaminathan : An Introduction to Database Systems, 8th Edition, Pearson Education,(Equivalent Reading).
3. Scoot Urban : Oracle 9i, PL/SQL Programming, Oracle Press.
4. S. K. Singh : Database Systems: Concepts, Design and Applications, Pearson Education
5. Peter Rob, Carlos Coronel: Database Systems: Design, Implementation and Management, 7th Edition, Cengage Learning, 2007.
6. Anjali, Amisha, Roopal and Nirav : Practice book on SQL and PL/SQL.
7. Leon and Leon : Database management Systems, Vikas Publication.

Web References:

1. http://www.microsoftvirtualacademy.com/trainingcourses/databasefundamentals#?fbid=tbZ92pOp_Tt [For overall subject]
2. http://www.ntu.edu.sg/home/ehchua/programming/sql/Relational_Database_Design.html [for relational database design]
3. http://docs.oracle.com/cd/A97335_02/apps.102/a81358/05_dev1.htm [For ER Diagram]
4. <http://plsql-tutorial.com/> [For PL/SQL]

Course Outcomes: Upon successful completion of the course, the students will :

- C01 : Understand basic concepts regarding data, database systems and various data models.
- C02 : Understand various data modeling techniques and entity relationship models.
- C03 : Learn about relational data model and related concepts.
- C04 : Gain insights into database design and normalization concepts.
- C05 : Understand various database terminologies and map it in application.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to database systems	✓				
2	Data modeling using entity relationship models		✓			
3	Relational data modeling			✓		
4	Database design methodology				✓	
5	Schema definition, Constraints and queries					✓
6	Advanced query processing					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	1	1	-	3	-	-	-	-	-	2	-
CO2	1	3	3	3	-	3	1	-	-	-	-	1	5
CO3	3	3	3	3	-	3	2	-	-	2	-	3	2
CO4	-	3	3	3	-	3	3	-	-	3	3	-	3
CO5	2	3	3	3	2	3	3	-	-	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA120: Fundamentals of Commerce and Business Processes

(100 Marks)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During theory sessions, the basic terminology along with concrete Illustrations will be discussed. The fundamentals in relation with different industries will be taught and demonstrated so that students can visualize the various business entities and commercial processes.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours Theory
1	Basics of Accounts	06
2	Cost Accounting	06
3	Financial Management	07
4	Business entities	05
5	Banking	06
6	Human Resource & Marketing Management	06

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit - I: Basics of Accounts Hours: 06

Introducing the concepts of Accounting principles and convictions, objectives of Financial Accounting, Double Entry Book-keeping System, recording financial transaction, preparation of Statutory Financial Statements, Profit & Loss account:

(Balance sheet, Cash flow Statement), analysis of Financial Statement: (Ratio Analysis, Common size B/S, Comparative Financial Statement).

Unit - II: Cost Accounting Hours: 06

Meaning of Cost, Costing and Cost Accounting, Methods of Costing, Marginal Costing, Breakeven Analysis, Make or buy decision.

Unit - III: Financial Management Hours: 07

Meaning and functions of Financial Management, Sourcing of funds and cost of funds, Optimum utilizations of funds, Time value of money, Investment decision, Dividend decision, Working Capital Management, Cash flow, Taxation.

Unit - IV: Business Organization Hours: 05

Forms of business organization, sole proprietorship firm, Partnership firm, Joint stock Company, Co-operative Society.

Unit - V: Banking Hours: 06

Introduction to Banking, Deposits Management, Types and Schemes, Management of Advances, Types of Management, Banking Services, Banking Products.

Unit - VI: Human Resource & Marketing Management Hours: 06

Human Resource: HR planning in context of software industries, Selection, Development and Training, Welfare and working conditions.

Marketing Management: Selling and Marketing comparison, Core concepts of marketing, Pricing Decisions, Advertising and Sales promotion.

Core Books:

1. Dr. S.N. Maheshwari: Financial Accounting, Sultan Chand Publication.
2. I.M.Pande: Financial Management.
3. V.S.P. Rao: Human Resource Management.

Reference Books:

1. Dr. S.N. Maheshwari: Advanced Accounting, Sultan Chand Publication.
2. M.N. Arora: Advanced Cost Accounting, Himalaya Publishing House
3. B.S.Mathur: Banking Law & Practice.

Web References:

1. http://iws.collin.edu/ost/pdfs/ACNT1303/ACNT1303lecture_notes.pdf [Introduction of Accounting]
2. http://www.ucosbdc.org/Websites/ucosbdc/Images/accounting_basics.pdf [Accounting Equation, Balance Sheet]
3. <http://wiki.svtuition.org/2009/03/basic-accounting-notes.html> [Area of Accounting, Accounting Elements]
4. Training and Development Practices In Indian Hotel Industry: An Empirical Investigation International Journal of Marketing and Human Resource Management (IJMHRM), Volume 1 * Issue 1 * May 2010, <http://www.iaeme.com/MasterAdmin/UploadFolder/paper4.pdf> [Human Resources]

5. Making Better Marketing Decisions Faster with Accounting Data
<http://www.omicsgroup.org/journals/making-better-marketing-decisions-faster-with-accounting-data-2168-9601.1000e116.pdf> [Marketing Management]
6. How small businesses master the art of competition through superior competitive advantage, <http://www.aabri.com/manuscripts/121156.pdf> [Marketing Management]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : The present course builds up on the knowledge of Basic of accounts some credit and Debit rules and entries.
- C02 : Student can able to understand the costing method of any product and services.
- C03 : Various financial sources where civilians can invest and take loans.
- C04 : Understanding the functionality and services of Banking system.
- C05 : Getting some fundamental of HR and Marketing principal on which some small and large scale venture can be built.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Basics of Accounts	✓				
2	Cost Accounting	✓				
3	Financial Management		✓			
4	Business Organization			✓		
5	Banking				✓	
6	Human Resource & Marketing Management					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
C01	3	2	3	1	2	1	1	-	-	2	2	3	2
C02	3	2	3	2	1	3	2	-	-	2	2	3	2
C03	3	3	2	2	2	3	2	-	-	1	2	3	2
C04	2	3	3	1	1	2	3	-	-	2	1	3	2
C05	3	2	2	3	1	2	1	-	-	1	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS201.02 C: PAINTING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	100	-	100	

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Painting	02
2	Drawing from Nature and Object	04
3	Colour Design and Colour Value	06
4	Composition and Perspective	06
5	Figure Drawing and Proportion	04
6	Sketching	04
7	Contemporary Issues in Painting	04

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

Detailed Syllabus:

1.	An Introduction to Painting	02 Hours	06%
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	• <i>An Introduction to Painting</i> • <i>Principles of Composition</i> • <i>Medium and Techniques of Painting</i> • <i>History of Painting: Folk Indian Painting / Western Painting</i> • <i>2D and 3D Painting</i>		
2.	Drawing from Nature and Object	04 Hours	12%
	• <i>Objects of Drawing: Nature and Manmade / Artificial Objects</i> • <i>Drawing Still / Live Objects</i> • <i>Drawing from Memory</i> • <i>Drawing from Life</i>		
3.	Light and Shade	06 Hours	18%
	• Color Theory: <i>Color wheel (primary/secondary, complementary), transparency/opacity, hue, value (intensity, brightness), chroma (saturation, purity) & temperature (warm/cold)</i> • Color Contrast & Attributes: <i>Interaction, harmony, psychology/mood, culture & expression</i> • Media Characteristics & Surfaces: <i>Acrylic, oil, paper, wood & canvas (primed/unprimed)</i>		
4.	Composition and Perspective	06 Hours	18%
	• Composition: <i>Space, movement, balance, asymmetry, rhythm, shapes, proportion & lighting</i> • <i>Perspective: An approximate reproduction</i> • Types of Perspectives: • <i>Linear Perspective, One-point Perspective, Two-point Perspective, Three-point Perspective, Four-Point Perspective</i>		
5.	Figure Drawing and Proportion	04 Hours	12%
	• <i>Proportions of the Human Body</i> • <i>Three views – Anterior (front), Lateral (side) and Posterior (back)</i> • <i>Fundamental Proportion – The Big Three</i>		
6.	Sketching	04 Hours	12%
	• <i>Sketching and Freehand</i> • <i>Sketching Techniques</i> • <i>Sketch and Drawing Medium</i>		
7.	Contemporary Issues in Painting	04 Hours	12%
	• <i>Contemporary Indian Art</i> • <i>Pioneers of Contemporary Indian Art</i> • <i>Contemporary Issues in Painting</i>		

Instructional Methods and Pedagogy:

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art history, art criticism and aesthetics.

CO3	be able to recognize the elements of arts in painting.
CO4	have better cognizance and association of meaning of colors, shapes, and composition.
CO5	be able to acknowledge the principles of painting as in design and colors, concept, medium and formats.
CO6	have instantaneous painting experience about designing, lights, shades and colors and such other important aspects.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	2	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	-	-	3	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

Reference Books

1. A.N. Hodge (2007) *The History of Art: The Essential Guide to Painting Through the Ages* Arcturus Foulsham, London
2. David Lewis (1984) *Pencil Drawing Techniques* Watson-Guption Publications, New York
3. John C Van Dyke L H D (1895) *A Text Book of the History of Painting* Longmans Green and Co., London
4. Sarah Parks (2014) *Drawing Secrets Revealed - Basics How to Draw Anything*-North Light Books, Ohio

Web Material

1. <https://conceptartempire.com/what-are-the-fundamentals/>
2. <https://www.creativebloq.com/art/painting-techniques-artists-31619638>
3. <https://www.thesprucecrafts.com/perspective-in-paintings-2578098>
4. <https://www.thehansindia.com/posts/index/Hans/2016-05-19/Origin-of-painting-in-India/229138>

HS202.02 C: PHOTOGRAPHY

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	100	-	100	

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Photography	03
2	Camera and Operating System	05
3	Light and Shade	10
4	Composition	09
5	Contemporary Issues in Photography	03

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

Detailed Syllabus:

1.	An Introduction to Photography	03 Hours	10%
	• Art, Design and Visualization • Basics of Photography and Various Types of Photography • Basics of Post Production		

	• <i>A Brief History of Photography:</i> • <i>Early Experiments and Later Developments</i>		
2.	Camera and Operating System	05 Hours	16%
	• <i>Role of Camera in the Photography</i> • <i>Types of Camera</i> <i>Pin-hole, box, folding, large and medium format cameras, single lens reflex (SLR) and twin lens reflex (TLR), miniature, subminiature and instant camera</i> • <i>Principal Parts of Photographic Camera</i> <i>Lens, Aperture, Shutters, various types and their functions, focal plane shutter and in-between the lens shutter, shutter synchronization, self-timer</i> • <i>Types of Lenses</i> <i>Single (meniscus), achromatic, symmetrical and unsymmetrical lenses, telephoto, zoom, macro, supplementary and fish-eye lenses</i> • <i>Different Models of Camera, their Features and Operating Systems</i> • <i>Camera and Size of the Image, Speed and Power of Lens</i>		
3.	Light and Shade	10 Hours	34%
	• <i>Reflection and refraction of light</i> • <i>Dispersion of light through a glass prism, lenses</i> • <i>Colour Filters:</i> <i>Different kinds, Red, yellow, green, neutral density, half filters, filter factor, colour correction filter</i> • <i>Photographic Light Sources:</i> <i>Natural source, the Sun, nature and intensity of the sunlight at different times of the day, different weather conditions</i> • <i>Artificial light sources:</i> <i>Nature, intensity of different types of light sources used in photography namely; (i) Photo flood lamp, (ii) Spot light, (iii) Halogen lamp, Barn doors and snoot, lighting stands</i> <i>Flash unit: Bulb flash and Electronic flash, main components, electronic flash units, studio flash, slave unit, multiple flash, computer flash, x-contact, exposure table</i>		
4.	Composition	09 Hours	30%
	• <i>Different kinds of image formations</i> • <i>Principal focus and focal length of the lens</i> • <i>Depth of field, angle of view and perspective</i> • <i>Perspective and composition</i> • <i>Rules of composition</i>		
5.	Contemporary Issues in Photography	03 Hours	10%
	• <i>Present Day Photography</i> • <i>Contemporary Photographers and their Contributions</i> • <i>Major Issues in Contemporary Photography</i>		

Instructional Methods and Pedagogy:

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Understand, appreciate and demonstrate innovative approach, beauty and acute acumen in the area of photography
CO2	Develop photography skills and become familiar with the functions and importance of the visual elements of nature and artificial objects

CO3	Become independent thinkers who will contribute inventively and critically to culture through the making of art photography
CO4	Have thorough understanding and acute sense of light and shade, composition, and presentation of a piece of an art
CO5	Experiment and Represent the cultivated sense and skills in Photography to the mass
CO6	Prepare an impressive portfolio encompassing holistic approach to art and other the areas of study

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	2	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	2	-	-	-
CO4	-	-	-	-	-	2	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	2	-	-	3	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

Reference Books

1. Maria Short, Sri-Kartini Leet and Elisavet Kalpaxi (2020) *Context and Narrative in Photography*-Routledge, New York.
2. Kevin Los (2010) *Improve Your Photography: 50 Essential Digital Photography Tips & Techniques* Digital Photography Series, New York.

Web Material

1. <https://photographylife.com/photography-basics>
2. <https://artofvisuals.com/the-basics-of-photography-introduction-to-photography-tutorials/>
3. <https://www.photographytalk.com/beginner-photography-tips/camera-basics-for-beginners>
4. <https://www.makeuseof.com/tag/5-photography-tips-for-beginners/>
5. <https://annamcnaught.com/blog/basics-of-photography>

HS205.02 C: MEDIA AND GRAPHIC DESIGN

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	100	-	100	

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Media and Graphic Design	03
2	Layout and Design	07
3	Form and Space	06
4	Computer Graphics	04
5	Fonts	04
6	Basic Print Media	03
7	Contemporary Issues in Graphic Design	03

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

Detailed Syllabus:

1.	An Introduction to Media and Graphic Design	03 Hours	10%
	<i>Creating Art, Art in Context and Art as Inquiry</i> <i>History of Graphic Design</i> <i>Constructional, Representational, and Simplification Drawing</i>		

2.	Layout and Design	07 Hours	23%
	• <i>Layout, Design and Aesthetics</i> • <i>Elements of Design</i> • <i>Principles of Design:</i> <i>Harmony, Balance, Rhythm, Perspective, Emphasis, Orientation, Repetition and Proportion</i> • <i>Impact/function of Design</i> • <i>Indigenous design practices</i> • <i>Role of design in the changing social scenario</i>		
3.	Form and Space	06 Hours	20%
	• <i>Types of Forms: Man-made, Nature</i> • <i>Types of Space: Negative and Positive</i> • <i>Composition of Form and Space to create Layout</i> <i>Exploring Creativity</i>		
4.	Computer Graphics	04 Hours	12%
	• <i>An Introduction to Graphic Software</i> • <i>Flash, Coreldraw, Illustrator and Photoshop</i> • <i>Pre-press Process</i>		
5.	Fonts	04 Hours	12%
	• <i>Construction of Type</i> • <i>Anatomy of Type</i> • <i>Visual Language</i> • <i>Creating Logo and Symbol</i>		
6.	Basic Print Media	03 Hours	10%
	• <i>An Introduction to Press and its Development Phases</i> • <i>Types of Press</i> • <i>Types of Printing Technologies</i> • <i>Post-press Processes</i>		
7.	Contemporary Issues in Graphic Design	03 Hours	10%
	• <i>Present Day Graphic Designs</i> • <i>Contemporary Designers and their Contribution</i> • <i>Major Contemporary Issues in Graphic Design</i>		

Instructional Methods and Pedagogy:

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art and designs, art criticism and aesthetics.
CO3	be able to recognize the elements of arts in graphic design.
CO4	have better cognizance and association with the meaning of designs, shapes, colors, print and medium.
CO5	be able to design graphics using computer softwares like Photoshop, CorelDraw, and Illustrator.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	3	-	-	2	-	-	-	-
CO4	-	-	3	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	3	-	-	-	-	2	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

Recommended Study Material:

Reference Books

1. Ellen Lupton and Jennifer Cole Philips (2008) *Graphic Design: The New Basics*, Princeton Architectural Press, New York
2. Ryan Hembree (2011) *The Complete Graphic Designer: A Guide to Understanding Graphics and Visual Communication* Rockport Publishers, Beverly

Web Material

1. <https://99designs.com/blog/tips/graphic-design-basics/>
2. <https://edu.gcfglobal.org/en/beginning-graphic-design/fundamentals-of-design/1/>
3. <https://ultragraphicsmt.com/10-basic-concepts-to-improve-your-graphic-design/>
4. <https://www.jcsocialmedia.com/graphic-design/>

HS209.02 C DRAMATICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	100	-	100	

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Drama	06
2	History of Drama and Contemporary Theatre	06
3	Theatre Design and Techniques	06
4	Technicalities of Stage Performance	08
5	Contemporary Trends in Drama	04

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

Detailed Syllabus:

1.	Introduction to Drama	06 Hours	20%
	• Introduction to performing arts • Drama - An art, a socializing activity, & a way of learning • Form of Drama • Elements of Drama • Types of Drama		
2.	History of Drama and Contemporary Theatre	06 Hours	20%

	• Important world dramatists & drama—from Greek to modern • Evolution of contemporary theatre in the context of developments in Indian theatre • Major Movements in Drama		
3.	Theatre Design and Techniques	06 Hours	20%
	• Theatre Architecture • Stage craft: Set, light, costume, make up, sound, props • Theatre techniques: from selection of script to final performance		
4.	Technicalities of Stage Performance	08 Hours	26%
	• Selection of plot and character • Improvisation • Movement • Voice, Speech, Imagination • Character Development • Scene Enactment		
5.	Contemporary Trends in Drama	04 Hours	14%
	• New Tendencies in theatre • Drama and Society • Using drama for Social Change and Education		

Instructional Methods and Pedagogy:

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05

3	Project	-	15	15
4	Attendance	-	05	05
	Total			30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
	Total			70

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	be aware about the concept of performing art and its nuances.
CO2	display a working knowledge of historic of drama, its development and current trends in dramatics.
CO3	demonstrate skills in the technical/design preparation and execution of a theatre performance.
CO4	demonstrate the ability to work collaboratively.
CO5	develop essential transferable skills in various relevant areas of the theatre.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	2	-	-	2	-	-	-	-
CO3	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	3	-	-	-
CO5	-	-	-	-	-	-	-	-	2	-	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

Reference Books

1. Bridget Panet (2009) Essential Acting: A Practical Handbook for Actors, Teachers and Directors Routledge, New York.
2. Tom Bancroft (2012) Character Mentor: Learn by Example to Use Expressions, Poses, and Staging to Bring Your Characters to Life Focal Press, New York

Web Material

1. <https://entertainism.com/elements-of-drama>
2. <https://www.rcboe.org/cms/lib/GA01903614/Centricity/Domain/5069/the%20elements%20of%20drama.pdf>
3. <https://marrzipandrama.co.nz/the-6-elements-of-drama/>
4. <https://www.backstage.com/magazine/article/theater-trends-broadway-2019-66726/>

HS210.02 C || CONTEMPORARY DANCE

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	100	-	100	

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to dance	04
2	Types of Dance	06
3	Basic Elements of Dance	04
4	Technical Skills in Professional Contemporary Dance	06
5	Contemporary Trends in Dance	10

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

Detailed Syllabus:

1.	Introduction to dance	04 Hours	14%
	- Dance as a Performing Art - Dance as a Medium of Expression - History and Development of Dance		
2.	Types of Dance	06 Hours	20%
	- Western dance and classical dance - Salsa, rumba, hip hop, tap dance, belly dance, etc. - Indian Classical Dance forms: Odissi, Bharatanatyama, Kathak, Kathakali, Kuchipudi etc. - Other Regional dance forms in India		
3.	Basic Elements of Dance	04 Hours	14%

	• Movements of different parts of a body for Expression • Concepts of: Nritya, Laya and Taal		
4.	Technical Skills in Professional Contemporary Dance	06 Hours	20%
	• Dance technique: alignment, balance, co-ordination, flexibility and control • Expressive / presentation skills: Dynamic energy, physical engagement with the given material and stage, etc. • Skills and processes of rehearsal and production: • physical energy, stamina and athleticism		
5.	Contemporary Trends in Dance	10 Hours	32%
	• Prevalent trends and techniques in contemporary dance • Future trends in contemporary dance form • On Stage Performance		

Instructional Methods and Pedagogy:

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
	Total			30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	be able to develop ability to express through the form of dance.
CO2	have enhanced aesthetic sensitivity.
CO3	have improved concentration, mental alertness, quick reflex action, and physical agility.
CO4	be able to express a natural way human feelings and expressions by creating harmony.
CO5	be able to deliver contemporary dance performance.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	3	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	-	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

Reference Books

1. Helene Scheff, Marty Sprague and Susan Mc Greeve. (1939) *Experiencing dance from student to dance artist*. Human kinetics, New York

2. Robin Grove, Catherine J. Stevens, Shirley McKenzie (2005) Thinking in Four Dimensions: Creativity and Cognition in Contemporary Dance. Melbourne university press, Melbourne

Web Material

1. <https://www.liveabout.com/what-is-contemporary-dance-1007423#:~:text=Contemporary%20dance%20is%20a%20style,body%20through%20fluid%20dance%20movements.>
2. <https://dancemagazine.com.au/2014/01/whats-contemporary-dance-days/>
3. <https://www.frontiersin.org/articles/10.3389/fpsyg.2019.00071/full>
4. <https://www.thehindu.com/features/friday-review/dance/what-is-contemporary-dance/article5096022.ece>

Teaching and Examination Scheme (BCA Programme)

Effective from Year 2021-2022

Semester-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		Exter nal	
						Case Study	Tests		Term work	Tests		
CA209-210	Elective - III	3	-	3	3	10	20	70	-	-	-	100
CA211-212	Elective – IV	-	3	3	3	-	-	-	15	15	70	100
CA213	Programming the Internet-II	4	3	7	7	10	20	70	15	15	70	200
CA214	Object Oriented Programming Using C++	4	3	7	7	10	20	70	15	15	70	200
CA215	System Analysis and Design	3	-	3	3	10	20	70	-		-	100
HS121.02 C	Creativity, Problem Solving and Innovation	2	-	2	2	30		70	-		-	100
CL144.01 C	Environmental Sciences	-	2	2	2	-		-	30		70	100
	University Elective – I **	-	2	2	2	-	-	-	30		70	100
		16	13	29	29	500			500			1000

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CA224**

Introduction to Web Designing course for others.

Elective-III

1. CA209–Fundamentals of Electronic Commerce
2. CA210–Introduction to Economics

Elective-IV

1. CA211–User Interface Design
2. CA212 - Visual Basic Applications

University Elective - I

No	Course Code	Course Name	Department/Faculty
1	CE281.01	Art of Programming	Engineering
2	CL281.01	Environmental Sustainability & Climate Change	Engineering
3	IT281.01	ICT Resources & Multimedia	Engineering
4	ME281.01	Engineering Drawing	Engineering
5	PH233.01	Fundamentals of Packaging	Pharmacy
6	PD260.01	Basic Laboratory Techniques	Applied Science
7	NR251.01	First Aid & Life Support	Nursing
8	PT191.01	Health Promotion & Fitness	Physiotherapy
9	BM231	Banking & Insurance	Management
10	EC281.01	Introduction to MATLAB Programming	Engineering
11	PD261	Astrophysics, Space and Cosmos-1	Applied Science
12	CL283	SDG Handprint Laboratory	Engineering
13	EE284	Python Programming	Engineering

CA209: Fundamentals of Electronic Commerce

(100 Marks)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures, the fundamental concepts regarding Ecommerce, E-Payment, E-Security, E-branding and E-Data Interchange will be discussed. Students will be required to carry out case studies using the concepts and techniques they have learnt during theory sessions.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours
		Theory
1	Introduction to E-commerce	07
2	E- Marketing	05
3	E-Security and E-Payment	06
4	Internet Bandwidth and Technology Issue	05
5	Mobile Commerce	07
6	E-Commerce Solutions	06

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit - I: Introduction to E-commerce Hours: 07

Emergence and use of Internet, origin of web and world wide web, advantages and disadvantages of E-commerce, Features of E-Commerce, Application of E-Commerce, Electronic Commerce over the Internet, Types of E-commerce.

Unit - II: E- Marketing: 05

Traditional Marketing, Online Marketing, E- advertising, Internet Marketing Trends and strategies, E-branding.

Unit - III: E-Security and E-Payment: 06.

Security on the Internet, E-Business risk management issues, Digital token based E-Payment System, Properties of Electronic Cash, Digital Signature.

Unit - IV: Mobile Commerce: 05

Definition of M-Commerce, Features of M-Commerce, Advantages and Disadvantages of MCommerce, Areas of M-Commerce Applications, Payment Systems and Models in MCommerce.

Unit V: E- CRM and ERP Hours: 07

Customer Relationship Management, CRM capabilities and Customer life cycle, Introduction to ERP, Reasons for the growth of the ERP Market, Advantages and Disadvantages of ERP.

Unit VI: E-Commerce Solutions: 06

Categories of E-Commerce Solutions, Solutions in Detail: Magento, Prestashop, Drupal Commerce, Demandware, Intershop, OXID eShop, RBS Change and Hybris.

Core Books:

1. P.T.Joseph, S.J.: E-Commerce- An Indian Perspective, 3rd Edition, PHI learning Private Limited.
2. Kamlesh K. Bajaj, Debjani Nag: E-Commerce, The Cutting Edge of Business, 2nd Edition, McGraw-Hill Education.

Reference Books:

1. Anita Rosen: The e-commerce Question and Answer Book: A Survival Guide for Business Managers, 2nd Edition, Amacom.
2. Janice Reynolds: The Complete E-Commerce Book: Design, Build & Maintain a Successful Web-based Business, 2nd Edition, CRC Press.
3. By Philippe Humeau&Matthieu Jung : The White Book of Ecommerce Solutions, NBS Systems
4. IshitaLahiri and Sujit Kumar Ghose : Principals of Marketing and E-Commerce

Web References:

1. <http://www.ecommercetutorial.net/> [For Tutorial]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : To understand basic of Electronic Commerce and its Application

C02 : Getting knowledge of E-marketing and its Strategies.

- C03 : Learn E-security and E-Payment in Detail and its term. Also, learn Encryption and decryption in basic.
- C04 : To gain understanding of Mobile Commerce and its Application through real life examples.
- C05 : To learn concepts of Customer Relationship Management and various solutions of E-Commerce (Software and Framework).

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to E-commerce	✓				
2	E- Marketing	✓	✓			
3	E-Security and E-Payment		✓	✓		
4	Mobile Commerce		✓	✓		
5	E- CRM and ERP				✓	
6	E-Commerce Solutions					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	3	3	2	2	3	1	2	3	2	3	3	2
CO2	2	3	3	2	3	3	2	2	3	3	3	3	2
CO3	2	3	3	2	3	3	3	3	3	2	3	3	1
CO4	2	3	3	2	2	3	2	3	3	3	3	3	2
CO5	2	3	3	2	2	3	2	2	3	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA210: Introduction to Economics

(100 Marks)

Contact Hours: 03

Prerequisite : None

Methodology & Pedagogy: During theory sessions, concepts of economics such as demand, supply, production, cost function, market structure, factor pricing and macroeconomics will be discussed with real time examples. Appropriate case studies will be assigned to students in order to give practical exposure of theoretical concepts of economics.

Unit No.	Title of unit	Minimum number of hours Theory
1.	Overview of Economics	06
2.	Demand and Supply	07
3.	Production and Cost function	08
4.	Market Structure	04
5.	Factor Pricing	05
6.	Macroeconomics	06

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit I: Overview of Economics Hours: 06

What is economy? Revolution of economics, The scope and method of economics, Basic Competitive model, prices, property rights and profits; incentives and information

Unit II: Demand and Supply Hours: 07

Concept of demand, determinants of individual and market demand, functions, elasticity of demand – price, income and cross elasticities, concept of supply, determinants of individual and market supply functions, elasticity of supply, Equilibrium price.

Unit III: Production and Cost function Hours: 08

Production function in short run – law of variable proportion, production function in the long run – isoquants, isocosts, ridge lines, returns to scale; producer's equilibrium -optimum combination of inputs to (i) maximize output - given cost and (ii) minimize cost -given output (least - cost combination of inputs)

Cost function – short run costs, long run average cost, long run marginal cost

Unit IV: Market Structure Hours: 04

Price and output determination under perfect competition and monopoly. Comparison between perfect competition and monopoly with respect to Efficiency

Unit V: Factor Pricing Hours:05

Demand and supply of factors of production, pricing of a single variable, factor under perfect competition and monopoly; modern theory of rent, quasi-rent

Unit VI: Macroeconomics Hours:06

Meaning of macroeconomics, Keynesian theory of determination of income, and employment in the three sector economy, multiplier analysis, IS-LM model of equilibrium, income and interest rate. Meaning and objectives of fiscal and monetary policies.

Core Books:

1. H. L. Ahuja: Modern Economics: An Analytical study, 20th Edition, S. Chand Publishing.
2. Paul A. Samuelson & William D. Nordhaus: Economics: 18th Edition, Tata McGraw-Hill Publishing

Reference Books:

1. H. L. Ahuja: Macro Economics: Theory and Policy, 20th Edition, Sultan Chand & Company.
2. Robert E. Hall & Mark Lieberman: Macroeconomics- Principles and Applications, 5th Edition, South-Western Cengage Learning.

Web References:

1. <http://www.investopedia.com/university/economics/economics3.asp>[Basics of Economics]
2. <https://www.youtube.com/watch?v=LwLh6ax0zTE>[Video Lecture of Supply and Demand]
3. <http://www.tutor2u.net/economics/reference/key-summary-on-market-structures>
4. <https://www.khanacademy.org/economics-finance-domain/macroeconomics>

Upon successful completion of the course, the students will :

C01 : Understand the concepts of basic economic theory.

C02 : Be able to apply the concepts of demand, supply, production ,cost function, market structure and factor pricing.

C03 : Be able to explain the differences between macroeconomics and microeconomics. C04:
Learn all the concepts related to Economics through case study

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes			
		C01	C02	C03	C04
1	Overview of Economics	✓			✓
2	Demand and Supply		✓		✓
3	Production and Cost function		✓		✓
4	Market Structure		✓		✓
5	Factor Pricing		✓		✓
6	Macroeconomics			✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	-	-	2	-	-	-	-	1	2	1
CO2	2	3	2	-	-	2	-	-	-	-	1	2	1
CO3	2	2	2	-	-	2	-	-	-	-	1	2	1
CO4	3	3	2	-	-	2	-	-	-	-	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA211: User Interface Design
(Marks 100)

Contact Hours:03

Pre- requisite: Basic Concepts of HTML

Methodology & Pedagogy: During Practical sessions, students will be required to work with image editing using Adobe Photoshop and able to develop webpages using macromedia Dreamweaver.

Outline of the Course:

Sr. No	Practical	Description
Week 1	Getting Started with Photoshop and Working with layers	Introduction Photoshop Interface, Image size and resolution , creating new images , Opening, importing & placing images , Introduction to Tool Box & Tool Option bar , Overview of layers and Blending Modes.
Week 2	Selections Tool & Transformation	Introduction, Selecting images, Cropping images ,Transforming Objects.
Week 3	Retouching Tool	Retouching the Image, Healing Brush Tool, Clone Stamp Tool, Pattern Stamp Tool, Eraser Tool.
Week 4	Painting Tool and Working with text	Painting tools available in Photoshop , Brush Presets, type tool
Week 5	Creating Shapes	Painting ,Drawing ,Pen tool ,Work path
Week 6,7	Masking, Adjustment and other Tools, Filters and Animation in PHOTOSHOP	Introduction to layer masks and clipping masks, adjustment and fill layer effects. Hands on session to implement animation on objects.
Week 8	Introduction to Web Design Macromedia Dreamweaver Tool. Create HTML 5 web page	Create website and set up project, its properties in Macromedia Dreamweaver. Explanation of Structure web pages. Work with

		HTML layout, Create HTML5-compliant pages, Visually modify HTML5 Web Page structure
Week 9	Implementation of Validation and advance features, Add Header and Footer	Advanced code completion and validation on web page, Work with header and footer of web page
Week 10	Style text content, Add images and links	Work with text, Create bulleted and numbered lists, Work with Edge Web Fonts, Working with internal and external hyperlink, Insert image and image link
Week 11	Working with tables , forms and controls	Create table, Create web form, use different controls, work with web page properties
Week 12	Create website and explanation of Testing and publishing of website	Create website using all concepts and work with Testing and publishing of website

CORE BOOKS:

1. Lesa Snider, Photoshop CS5 - The missing manual, O'reilly Media, First Edition, May2010.
2. Lisa DaNaeDayley and Brad Dayley, Photoshop CS5 - Adobe Photoshop CS5 Bible,Wiley India Pvt. Ltd., June 2010.
3. David Sawyer McFarland, Dreamweaver CS3: The Missing Manual, O'Reilly Media,fifthedition, 2007

REFERENCE BOOKS:

1. Michael Toot, Sherry Kinkoph, Master Adobe Photoshop, Illustrator, Premiere and after Effects Visually, First Edition 2002
2. Adobe Creative Team,Adobe Dreamweaver CC Classroom, Adobe; Pap/Psc edition, 2013

WEB REFERENCES:

1. <https://helpx.adobe.com/photoshop/tutorials.html>[Photoshop Tutorial]
2. https://www.photoshoptutorials.ws/?doing_wp_cron=1471335186.3795449733734130859375[Photoshop Tutorial]
3. <http://www.photoshop.com/tutorials>[PhotoshopTutorial]
4. <https://helpx.adobe.com/dreamweaver/tutorials.html>[Adobe Dreamweaver

Tutorial]

5. <http://www.tutorialspoint.com/listtutorials/dreamweaver/1>[Adobe Dreamweaver Tutorial]
6. <https://www.youtube.com/watch?v=08CshKXqzoA>[Adobe Dreamweaver Tutorial]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Able to get familiar with the Photoshop tool, workspace and it's working.
- C02 : Able to work with the selection, resize, crop, navigate, color modes, text for editing and creating new images as well as shape.
- C03 : Able to understand and implement retouching and repairing techniques to correct images
- C04 : Use basic Photoshop skills and concepts to develop effective graphics for both web and print media.
- C05 : Able to create simple animation.
- C06: Able to design HTML template in Photoshop and convert the template in HTML,CSS

Course Outcomes Mapping:

Week No.	Content	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Introduction to Photoshop	✓					
2-4	Selection tools, Retouching tool , Painting tool, Text and transformation		✓	✓			
4-6	Creating Shapes, Masking, Adjustment and other Tools, Filters and Animation in PHOTOSHOP		✓		✓	✓	
8-12	Introduction to Web Design using Macromedia Dreamweaver Tool		✓		✓		✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	3	3	2	2	3	1	2	3	2	3	3	2
CO2	2	3	3	2	3	3	2	2	3	3	3	3	2
CO3	2	3	3	2	3	3	3	3	3	2	3	3	1
CO4	2	3	3	2	2	3	2	3	3	3	3	3	2
CO5	2	3	3	2	2	3	2	2	3	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA212: Visual Basic Applications
(Marks 100)

Contact Hours: 03

Pre- requisite:CA111 Fundamental Concepts of Programming Language, CA112 Introduction to Computers

Methodology & Pedagogy: The prime focuses of this course is on integrating programming of visual basic in Microsoft Office Products. It emphasizes on basic programming constructs of visual basic language.

Outline of the Course:

Sr No	Practical	Description
Week 1	Overview of VBA and its applications	This topic will provide introduction to Visual Basic for Applications and its real use.
Week 2, Week 3	Excel VBA terminologies and macros	The overview of excel VBA terminologies with some example will be discussed in this topic.
Week 4	Comments, Message box and Input box	This topic provides introduction to comments, message and input box available in VBA
Week 5 Week 6	Variables, Constants, Operators, Decisions, Loops	This topic provides basic understanding of various programming terms such as variables, constants, operator etc.
Week 7	String, Date and Time	This topic covers concepts of data types such as String and Date and Time
Week 8 Week 9	Array, Function, Sub procedures, Events	Student can know very well about arrays, functions, sub procedures and event handling with suitable example.
Week 10	Error handling, Excel objects	Advance topics such as handling the errors and accessing excel objects using VBA

Week 11	Working with files, programming charts and user forms	The implementation of real applications will be demonstrated using various options such as files, charts etc.
Week 12	Case Study discussion	The concepts learned during previous weeks will be illustrated.

Core Books:

1. John Walkenbach: Excel 2013 Power Programming with VBA, Wiley publication 1st Edition, 2013
2. John Walkenbach: Excel VBA Programming For Dummies, Wiley publication, 3rd Edition, 2013

Reference Books:

1. Paul Lomax : VB & VBA in a Nutshell: The Language (In a Nutshell) 1st Edition, O'Reilly publication , 1998
2. MaayanPoleg: Excel VBA: for Non-Programmers (Programming in Everyday Language) (Volume 1), CreateSpace Independent Publishing Platform, 3rd Edition, 2013

Web References:

1. <http://www.tutorialspoint.com/vba/> [All concepts of Visual Basic Applications]
2. [https://msdn.microsoft.com/enus/library/office/ee814735\(v=office.14\).aspx](https://msdn.microsoft.com/enus/library/office/ee814735(v=office.14).aspx)[For MS Office and Visual Basic Applications]

Course Outcomes: Upon successful completion of the course, the students will :

- C01 : Learn about VBA applications, terminologies and macros
 C02 : Learn to use different programming constructs of VBA
 C03 : Understand and use different inbuilt functions, sub procedures and events.
 C04 :Learn and use concepts about error handling and working with different excel objects.
 C05 : Learn to make use of files and charts with user forms in VBA

Course Outcomes Mapping:

Week No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Overview of VBA and its applications	✓				
2	Excel VBA terminologies and macros	✓				

3	Excel VBA terminologies and macros	✓				
4	Comments, Message box and Input box		✓			
5	Variables, Constants, Operators and Decisions, Loops		✓			
6	Variables, Constants, Operators and Decisions, Loops		✓			
7	String, Date and Time		✓			
8	Array, Function, Sub Procedures, Events			✓		
9	Array, Function, Sub Procedures, Events			✓		
10	Error handling, Excel Objects				✓	
11	Working with files, programming charts and user forms					✓
12	Case Study Discussion	✓	✓	✓	✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	1	1	1	-	-	2	-	-	-	3	-	1	1
CO2	-	2	2	-		2	-	-	-	3	-	2	2
CO3	-	-	3	-	-	2	-	-	-	3	2	2	1
CO4	-	-	2	-	-	2	-	-	-	3	1	1	1
CO5	-	2	2	-	-	2	-	-	-	3	1	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA213: Programming the Internet - II

(Marks 200)

Contact Hours: 07

Pre - requisite: CA113: Programming the Internet - I

Methodology & Pedagogy: This course focuses on providing hands-on experience to students to design and develop entire web sites using several web designing tools, HTML5 tags, CSS and Java Script.

Outline of the Course:

Unit No.	Unit Title	Minimum number of hours	
		Theory	Practical
1	Introduction to Internet Programming	07	
2	Web Design using Hypertext Markup Language	09	36 Hours
3	Cascading Style Sheets	09	
4	Java Script Programming - I	07	
5	Java Script Programming - II	09	
6	Introduction to Content Management System & Search Engine Optimization	07	

Total Hours (Theory): 48

Total Hours (Practical): 36

Total: 84

Detailed Contents:

Unit I: Introduction to Internet Programming Hours:07

Overview of Internet and WWW, Basic elements of the Internet, Internet services, Internet Browsers and Servers, Hardware and Software requirements to connect to the internet, Internet Service Provider (ISP), Internet Addressing: standard Internet Address, Domain Name Server (DNS), Pseudo Internet Address, Introduction to Internet Protocols: TCP/IP, UDP, FTP, SMTP, POP3, MIME, SOAP, IPv4 and IPv6

Unit II: Web Design using Hypertext Markup Language Hours:09

HTML5: Introduction to Lists, Images, Tables, Linking documents, New Elements, Semantics,

Forms: Input Types, Form Elements, Form Attributes, Media: Audio and Video, HTML Canvas Reference, Inline SVG, Image Maps

Unit III: Cascading Style Sheets Hours:09

Introduction to Cascading Style Sheet (CSS), CSS essentials, CSS selectors, CSS Box Model and Wrapper class, CSS Display, CSS Backgrounds and Borders, CSS Text Effects and Fonts, CSS Navigation, CSS Image Gallery

CSS3-2D/3D Transformations, Transitions, Animations, Multiple Column Layout, User Interface.

Unit IV : Java Script Programming - I Hours:07

Introduction to Scripting Language and Client Side Scripting, Basics of Java Script, Java Script Statements, Comments, Data Types, Objects, Operators and Expressions, Constructs, Conditional checking, Loops, Functions, Dialog boxes.

Unit V : Java Script Programming – II Hours:09

The Java Document Object Model (DOM): Introduction to JS DOM, Objects in HTML, Object hierarchy, Event handling,

DHTML: Introduction to DHTML, DHTML CSS, DHTML Java Script, DHTML HTML DOM, DHTML Events

Unit - VI Introduction to Content Management System & Search Engine Optimization Hours:07

Introduction to Content Management System: - CMS Types, CMS Working and CMS Applicability
Web Promotion Search Engine Optimization: Using Good Keywords for Keyword Search, Creating the Title Tag and Meta Tags for Website, Optimizing Website for Search Engine - Keyword Density and Content, Submitting Site to the Search Engines, Increase Rankings by Building Link Popularity

Core Books:

1. Harley Hahn: The Internet Complete Reference, Second Edition, Tata McGraw-Hill Edition.
2. Matthew MacDonald: HTML5: The Missing Manual, O'Reilly Media, August 2011
3. Peter Gasston: The Book of CSS3: A Developer's Guide to the Future of Web Design, No Starch Press, April 2011
4. Richard York: Beginning CSS: Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2005.
5. Michael Morrison: Head First JavaScript, O'Reilly, 2014
6. Deepak Bansal: A Complete Guide to Search Engine Optimization, B.R. Publishing Corporation

Reference Books:

1. Ivan Bayross: Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP, BPB Publication, 4th revised edition, 2009

2. Adrian Farrel :The Internet and its Protocol – A comparative approach , Morgan Kaufmann Publishers, 2004
3. David Mc Farland: CSS: The Missing Manual, O'Reilly, 2006.
4. Thomas Powell: HTML & CSS: The Complete Reference, Fifth Edition Paperback, 2010

Web References:

1. http://www.w3schools.com/html/html5_intro.asp[HTML5 notes]
2. <http://www.w3schools.com/js/>[JavaScript Materials]
3. <http://www.w3schools.com/css/>[CSS notes]
4. http://www.tutorialspoint.com/internet_technologies/html.htm[HTML notes]
5. <http://james-star.com/answers/en/css3-hover-effect-transitions-transformationsandanimations/>[Animation in CSS]
6. http://www.w3schools.com/js/js_events_examples.asp[JavaScript HTML DOM Events]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Get familiar with basics of the Internet Programming.
- C02 : Gain deep understanding of the implementation of HTML tags and able to create HTML static web pages.
- C03: Able to use basic properties of CSS and CSS3 in html file
- C04 : Able to design template using HTML5 and CSS3
- C05 : Able to create dynamic web pages using JavaScript
- C06: Able to understand different types of CMS and how search engine works.

Course Outcomes Mapping:

Unit No.	Unit Name	C01	C02	C03	C04	C05	C06
1	Introduction to Internet Programming	✓					
2	Web Design using Hypertext Markup Language		✓		✓		
3	Cascading Style Sheets			✓	✓		
4	JavaScript Programming I					✓	
5	JavaScript Programming II					✓	

6	Introduction to Content Management System & Search Engine Optimization						✓
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Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	1	-	-	3	2	2	-	1	-	-	-	-	-
CO2	-	3	3	3	-	-	1	1	-	-	-	-	-
CO3	-	3	3	2	2	-	-	-	-	2	3	-	1
CO4	1	3	3	3	2	-	1	1	-	2	3	1	1
CO5	2	3	3	3	-	2	2	2	-	2	3	2	2
CO6	-	3	3	-	-	3	-	-	1	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA214: Object Oriented Programming using C++

(200 Marks)

Contact Hours: 07

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures, illustrations of certain complex real world applications, which emphasis the use of advanced concepts of programming, will be discussed. The fundamental concepts regarding development activities, various object oriented systems and other advanced issues in object oriented programming will also be discussed. In addition there may be announced or unannounced quizzes/assignments. During Practical sessions, students will be required to carry out case studies using the concepts and techniques they have learnt during theory sessions.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1.	Foundation of C++	07	36
2.	Classes and Objects	08	
3.	Operator Overloading and Type Conversion	09	
4.	Inheritance, Virtual Functions and Polymorphism	09	
5.	Exception Handling	08	
6.	Using Files of IO	07	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit - I: Foundation of C++ Hours:07

Overview of C-Expressions, Arrays, Functions, Structures, Pointers- Call by value, call by reference, Introduction to C++,Structure of C++ program, Fundamental of OOP, Characteristics of OOP, inline function, Default arguments , function overloading

Unit - II: Classes and Objects Hours:08

Define class, objects, visibility modes, static members, friend function, Constructors and destructors, Default Constructor, Copy constructors, Parameterized Constructor.

Unit - III: Operator Overloading and Type ConversionHours:09

Introduction, Operator overloading through member function, Need of Function objects- user defined Conversions.

Unit - IV: Inheritance, Virtual Functions and PolymorphismHours:09

Introduction, Base class access controls, Inheritance and protected members, Virtual base classes, Virtual functions – calling through a base class, pure virtual functions, abstract class, early vs. late binding

Unit - V: Exception Handling Hours:08

Introduction, Exception handling fundamentals, catching class types, using multiple catch statements, exception handling options, handling derived class exceptions, applying exception handling

Unit - VI: Using Files for IO Hours:07

<fstream> and the file class, opening and closing file, reading and writing text files, Unformatted and Binary IO, more get() functions, detecting EOF, Random access, I/O status

Core Books:

1. Herbert Schildt: C++ A Beginner's Guide, TMH Publications.
2. E Balagurusamy: Object Oriented Programming with C++, 2nd Edition, TMH Publications.

Reference Books:

1. Robert Lafore: Object Oriented Programming in C++, 4th Edition, Sams Publications.
2. YashavantKanetkar: Let Us C++, 8th Edition, BPB Publications.
3. Herbert Schildt: The Complete Reference C++, 3rd Edition, TMH Publications.

Web References:

1. <http://www.cplusplus.com/doc/tutorial/> [For Entire Syllabus]
2. <http://www.learncpp.com/> [For coding standards and concepts]
3. http://www.tutorialspoint.com/cplusplus/cpp_basic_input_output.htm [For Practicals]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Revise the C programming basic concepts and moved further through Introduction to C++ programming, it's structure, Object Oriented concepts, able to differentiate C with advance C++ and what's new features. Also most useful topics like inline functions, default arguments and function overloading and its need.

- C02 : Able to create basic class and its objects and access members of class using object. Understanding and Creating different types of constructors and destructors.
- C03 : Understand the need of operator overloading and using it at different scenarios. Creating Function Objects and how to do type conversions.
- C04 : Able to understand the concept of inheritance, how it is created, differentiate between base class and sub class and its access control mechanism. Use of protected members, Virtual functions and pure virtual functions.
- C05 : Able to handle the runtime errors or exceptions in C++.
- C06 : Create files in C++ for Input and Output and access it for reading, writing and updating data into files.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Foundation of C++	✓					
2	Classes and Objects		✓				
3	Operator Overloading and Type Conversion			✓			
4	Inheritance, Virtual Functions and Polymorphism				✓		
5	Exception Handling					✓	
6	Using Files of IO						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	3	3	3	3	3	3	2	2	-	-	-	3	-
CO2	3	3	3	3	3	3	2	2	-	-	-	3	-
CO3	3	3	3	3	3	3	2	2	-	-	-	3	2
CO4	3	3	3	3	3	3	2	2	-	-	-	3	2
CO5	3	3	3	3	3	3	2	2	-	-	-	-	2
CO6	3	3	3	3	3	3	2	2	-	1	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA215: System Analysis and Design

(100 Marks)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures emphasis will be given on understanding how Information Systems are designed and constructed and how major modifications are made to existing systems, analyze business needs for information and develop an appropriate strategy to solve a particular business problem, understand the issues associated with physical design, utilize some of the tools used by systems analysts in the design of Information Systems, a significant portion of the course will deal with computer aided system engineering (CASE) tools and techniques in the context of a case study.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours
		Theory
1.	Introduction to System Development	05
2.	Classical Systems Development Methodology	07
3.	Structured Systems Development Methodology	07
4.	Prototype Development Methodology and Tools	05
5.	Systems Design	07
6.	Systems Implementation & Maintenance	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit - I: Introduction to System Development Hours: 05

Concept of system, Basic components, Information systems categories, Business Information Systems, Examples of real-life systems, System analyst – meaning, role and need, Users, Need of information system development, Application development portfolio, Integration of portfolio, Controlling and monitoring committees.

Unit - II: Classical Systems Development Methodology Hours: 07

Classical Systems Development Life Cycle (SDLC) method, Requirement determination, Fact finding techniques, Modern methods for determining systems requirements, Joint Application Development Program, Tools for documenting procedures and decisions, Tools for systems development, understanding the format of system requirement specification .

Unit - III: Structured Systems Development Methodology Hours: 07

Structured Analysis Development Strategy, Data Flow Diagrams (DFDs) for analysis and design, Physical and logical DFDs, Data Dictionary.

Unit - IV: Prototype Development Methodology and Tools Hours: 05

Purpose of Application Prototyping, Steps in prototype method, Use of prototypes, Tools for prototyping, Strategies for prototype development, misconceptions about prototyping, Role of tools, Categories and use of automated tools, using a CASE tool

Unit - V: Systems Design Hours: 07

Analysis to design transition, Design of output system, Design of input system, Design of security and control, Design of user interface and Design of Database Interaction.

Unit - VI: Systems Implementation & Maintenance Hours: 05

Designing reliable and maintainable systems, design of software, software design and documentation tools, managing quality assurance, user training, training methods and conversion.

Core Books:

1. James A Senn : Analysis and Design of Information System, McGraw Hill International, 2003.
2. Kendall and Kendall : Systems analysis and Design, 5th Edition, Prentice-Hall of India Private Limited, 2003.

Reference Books:

1. V. Rajaraman : Analysis and Design of Information Systems, Prentice-Hall of India Private Limited, 2003.
2. Jeffrey L. Whitten, Lonnie D. Bentely and Kevin C. Dittman : Systems Analysis and Design Methods, Tata McGraw Hill Publishing Co. Ltd., 2001.
3. Tuthill and Leavy : Knowledge Based Systems : Mangers Perspectives : Tab professional and Reference Books, 1991.

Web references:

1. <http://lecture-notes-forstudents.blogspot.in/2010/04/system-analysis-and-design-sad.html> [For referring SDLC]
2. <http://www.slideshare.net/aroravinay/1-introduction-to-ado-infosys>[For Introduction of System analysis and design]
3. <http://bcastuff.blogspot.in/p/sad-notes.html#.U6uW-ZzWleo>[For brief notes on System analysis]

4. <http://www.eis.mdx.ac.uk/staffpages/geetha/bis2030/DFD.html>[For DFD notes and examples]
5. <http://faculty.washington.edu/ytan/is460/notes/LN11.pdf>[For DFD notes and examples]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Understand the basics of system, type of information systems, system analyst- meaning, role and need, portfolio development of the system, the methods or ways to review and select the project.
- C02 : Understand the concept of SDLC, the fact finding techniques, detailed study of modern methods for collecting the requirements of system to be developed, Decision and procedure documentation and a brief introduction to the SRS document.
- C03 : Learn the concept of structured analysis, detail study of DFD and its types and the study of data dictionary.
- C04 : Gain understanding of prototypes, its uses and tools for development, clarification of the misconception about the prototype and categories of tools including the use of automated tools like case tools.
- C05 : Be able to learn objective of collecting data, methods to collect data for designing the input and output interface, the concept and objective of database design,
- C06 : Study the designing of reliable and maintainable systems, documentation tool, quality assurance management and need for user training.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Introduction to system development	✓					
2	Classical system development Methodology		✓				
3	Structured System Development Methodology			✓			
4	Prototype Development Methodology and Tools				✓		
5	System Design					✓	

6	Systems Implementation & Maintenance						✓
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Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	2	2	3	3	2	2	3	2	3	2
CO2	2	2	3	2	3	3	2	2	3	2	3	3	2
CO3	2	2	3	3	1	2	2	1	2	1	1	3	1
CO4	2	2	3	1	2	2	1	1	1	1	1	3	2
CO5	2	2	2	2	1	2	2	1	1	1	1	2	1
CO6	1	2	2	2	2	2	2	1	1	2	2	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS121.02 C: CREATIVITY, PROBLEM SOLVING AND INNOVATION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	2
Marks	--	100	--	100	

Pre-requisite courses:

Creative Problem Solving

<https://www.coursera.org/learn/creative-problem-solving>

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Creativity, Problem Solving and Innovation	06
2.	Questioning, Learning and Visualization	06
3.	Creative Thinking and Problem Solving	06
4.	Logic, Language and Reasoning	06
5.	Contemporary Issues and Practices in Creativity and Problem Solving	
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1.	Introduction to Creativity, Problem Solving and Innovation	06 Hours	20%
	Definitions of Creativity and Innovation, Need for Problem Solving and Innovation, Scope of Creativity in various Domains, Types and Styles of Thinking, Strategies to develop Creativity, Problem Solving and Innovation skills		
2.	Questioning, Learning and Visualization	6 Hours	20%
	Strategy and Methods of Questioning, Asking the Right Questions, Strategy of Learning and its Importance, Sources and Methods of Learning, Purpose and Value of Creativity Education in real life, Visualization strategies - Making thoughts Visible, Mind Mapping and Visualizing Thinking		
3.	Creative Thinking and Problem Solving	6 Hours	20%
	Creative Thinking and its need, Strategy of Thinking Fluency, Generating all Possibilities, SCAMPER Technique, Divergent Vs Convergent Thinking, Lateral Vs Vertical Thinking, Fusion of Ideas for Problem Solving, Applying strategies for Problem Solving		
4.	Logic, Language and Reasoning	6 Hours	20%
	Basic Concepts of Logic, Statement Vs Sentence, Premises Vs Conclusion, Concept of an Argument, Functions of Language: Informative, Expressive and Directive, Inductive Vs Deductive Reasoning, Critical Thinking & Creativity, Moral Reasoning		
5.	Contemporary Issues and Practices in Creativity and Problem Solving	6 Hours	20%
	Cognitive Research Trust Thinking for Creatively Solving Problems, Case Study on Contemporary Issues and Practices in Creativity and Problem Solving		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Demonstrate creativity in their day to day activities and academic output.
CO2	Solve personal, social and professional problems with a positive and an objective mindset.
CO3	Think creatively and work towards problem solving in a strategic way.
CO4	Initiate new and innovative practices in their chosen field of profession.
CO5	Give logical ideas, opinions, and solutions to problems.
CO6	Think critically over the situation and drawing conclusion.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	2
CO2	-	-	3	-	-	-	-	-	3	-	-	-	2
CO3	-	-	-	-	-	-	-	-	3	-	-	-	2
CO4	-	-	-	-	3	-	-	-	3	-	-	-	2
CO5	-	-	-	-	-	-	-	-	3	-	-	-	2
CO6	-	-	-	-	-	-	-	-	3	-	-	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ **Text book:**

4. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity, Jossy-Bass Publication 2013
5. Michael Michalko, Crackling Creativity, The Secrets of Creative Genius, Ten Speed Press 2001

❖ **Reference book:**

6. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
7. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
8. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
9. Tony Buzan, How to Mind Map, Thorsons Publication 2002
10. Scott Berkum, The Myths of Innovation, Expanded and revised edition, Berkun Publication 2010
11. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
12. Ira Flatow, The all Laughed, Harper Publication 1992
13. Paul Sloane, Des MacHale & M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

❖ **Additional Readings**

1. Keith Sawyer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
2. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
3. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
4. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
5. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
6. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
7. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
8. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994

9. Scott G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
10. Donald J Treffinger, Scott G Isaksen, K Brain Stead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
11. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
12. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
13. Howard Gardner, Creating minds, Basic Books Publication 1993
14. Mihaly Csikszentmihalyi, Creativity—Flow and Psychology of Discovery and Invention, Harper Publication 1996
15. Martin Gerdner, W. H., Aha! Insight, Freeman Publication 1978
16. Paul Slovic, Test Your Lateral Thinking IQ, Sterling Publication 1994
17. Paul Slovic & Des Macale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

❖ **Web material:**

Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

Teaching and Examination Scheme (BCA Programme)

Effective from Year 2021-2022

Semester-IV

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA216 – 220	Elective - V	4	3	7	7	10	20	70	15	15	70	200
CA221	Data Structures & Algorithms	4	3	7	7	10	20	70	15	15	70	200
CA222	Advanced Database System	-	3	3	3	-	-	-	15	15	70	100
CA223	Fundamentals of Computer Networks	3	-	3	3	10	20	70	-		-	100
HS111.02 C	Human Values and Professional Ethics	-	2	2	2	-		-	30		70	100
	University Elective – II **	-	2	2	2	-	-	-	30		70	100
		11	13	24	24	300			500			800

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CA225**

Programming the Internet course for others.

Elective-V

1. CA216 -Computer Graphics
2. CA217- Multi Paradigm Programming Language
3. CA218- System Programming
4. CA219- Scripting Language
5. CA220- Basics of Web Designing

University Elective – II

No	Course Code	Course Name	Department/Faculty
1	EC282.01	Prototyping Electronics with Arduino	Engineering
2	CE282.01	Web Designing	Engineering
3	CL282.01	Basics of Environmental Impact Assessment	Engineering
4	IT282.01	Internet Technology & Web Design	Engineering
5	ME282.01	Material Science	Engineering
6	PH238.01	Cosmetics in Daily Life	Pharmacy
7	NR261.01	Life Style Diseases & Management	Nursing
8	PT192.01	Occupational Health & Ergonomics	Physiotherapy
9	BM241	Health Care Management	Management
10	PD262	Astrophysics, Space and Cosmos-2	Applied Science
11	EE287	MATLAB Programming	Engineering
12	EE288	Maintenance of Household Apparatus	Engineering

CA216: Computer Graphics

(200 Marks)

Contact Hours: 07

Pre-requisite:CA111: Fundamental Concepts of Programming Language, CA214: Object Oriented Programming Using C++

Methodology & Pedagogy: During theory sessions, the emphasis would be given on theoretical aspects of computer graphics & its elements. During practical sessions, the students will learn graphics programming using C, C++.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Basic of Computer Graphics	06	36
2	Introduction to Graphics Systems	08	
3	Output Primitives - I	08	
4	Output Primitives – II	09	
5	Two Dimensional (2D) Transformations	09	
6	Two Dimensional (2D) Viewing	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit – I Basic of Computer Graphics Hours:06

Basic of Computer Graphics, Applications of computer graphics, Image Processing, Visualization, Graphical User Interfaces

Unit – II Introduction to Graphics SystemsHours:08

Display devices, Random and Raster scan systems, Graphics input devices, Graphics software and standards

Unit – III Output Primitives – I Hours:08

Points, Lines, Circles, Line Drawing Algorithms: Digital Differential Analyzer (DDA) and Bresenham, Circle & Ellipse generating algorithm: Midpoint Circle & Midpoint Ellipse Algorithm

Unit – IV Output Primitives – II Hours:09

Fill area primitives including scan-line polygon filling, inside-outside test, boundary and floodfill, character generation, line attributes, area-fill attributes, character attributes

Unit – V Two Dimensional (2D) Transformations Hours:09

2-D Geometric Transformations: Translation, Rotation, Scaling, Reflection & Shear, Matrix representation & homogeneous coordinates, composite transformations

Unit – VI Two Dimensional (2D) Viewing Hours:08

Viewing Pipeline, Window-to-Viewport transformation, Point Clipping Line clipping, Cohen Sutherland line clipping algorithm Polygon Clipping, Text clipping

Text Books:

1. Computer Graphics - C Version: Donald Hearn, M. Pauline Baker, 2nd Edition, Pearson Education, 2008.
2. Computer Graphics with OpenGL: Hearn and Baker, 3rd Edition, Pearson Education.

Reference Books:

1. Computer Graphics: Sinha & Udai, Tata McGraw-Hill Education.
2. Computer Graphics: Foley and van Dam, 3rd Edition, Person Education.

Web Resources:

1. <http://www.idav.ucdavis.edu/education/GraphicsNotes/BresenhamsAlgorithm.pdf>[algorithm]
2. http://web.cs.wpi.edu/~emmanuel/courses/cs543/slides/lecture09_p1.pdf[computer graphics]
3. <https://users.soe.ucsc.edu/~pang/160/f12/slides/dda2.pdf>[dda algorithm]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be able to identify the basic concepts of computer graphics.
- C02: Be able to understand about how graphics system works
- C03 : Be able to work with different types of line, circle and fill algorithms as a output primitives.
- C04 : Able to create and apply 2D Geometric Transformations
- C05 : Able to create and apply 2D viewing techniques.

Course Outcomes Mapping:

Unit No.	Content	Course Outcomes				
		C01	C02	C03	C04	C05
1	Basic of Computer Graphics	✓				
2	Introduction to Graphics Systems		✓			
3	Output Primitives – I			✓		✓
4	Output Primitives – II			✓		✓
5	Two Dimensional (2D) Transformations		✓		✓	
6	Two Dimensional (2D) Viewing		✓			✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	2	2	3	-	-	2	-	-	-	-	-	2	1
CO2	2	2	3	3	2	2	-	-	-	1	-	2	1
CO3	2	2	3	2	-	3	-	-	2	2	1	2	1
CO4	2	2	-	3	3	1	-	-	2	2	2	2	1
CO5	2	2	-	3	3	1	-	-	2	2	2	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA217: Multi Paradigm Programming Language

(200 Marks)

Contact Hours: 07

Pre-requisites: CA111: Fundamentals Concepts of Programming Language, CA214 : Object Oriented Programming Using C++

Methodology & Pedagogy: During the theory lectures, concepts required to design any application using programming language Python in functional, object oriented and web based aspects are discussed in detail. During laboratory hours students will implement the concepts that are discussed during theory lectures. Using case study approach student will develop any real application using any paradigm by learning some advance concepts of Python programming language.

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Multi paradigm approach using Python	06	36
2	Objects, Numbers and Sequences in Python	09	
3	Conditionals ,loops and files	08	
4	Functional Programming	09	
5	Object Oriented Programming	08	
6	Web Programming	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit I: Introduction to Multi paradigm approach using Python Hours: 06

Overview of programming paradigms-Imperative, Functional, logic and object oriented
Introduction to Python programming Language- Features, Downloading and Installing, Running

Python, Python Documentation, Python Basics- The print statement, comment, statements and syntax, variable assignments ,identifier

Unit II: Objects, Numbers and Sequences in Python Hours: 09

Python Objects- Standard types, Other built-in types, Internal types, Numbers- Integer, Double precision floating point, complex numbers, operators, Sequences- String, List and Tuples

Unit III: Conditional, loops and files Hours: 08

If, else, elif, conditional expressions, while, for, break , continue, pass, File objects, File built in functions, standards files, command line arguments

Unit IV: Functional programming Hours: 09

Creating Functions, Passing Functions, Functional Programming, Scope of variables, Introduction to Modules, Modules and Files, Importing Modules, Module Built-in functions, packages, Other features of modules

Unit V:Object Oriented Programming Hours: 08

Classes, Classes attributes, Instances, Instance attributes, Binding and Method Invocation, Static and Class methods, Inheritance, Built-in functions for classes, instances and other objects

Unit VI: Web Programming Hours: 08

Introduction, Creating Simple Web Client, Advanced Web Clients, CGI, Building CGI Applications

Core Books:

1. Wesley J. Chun : Core Python Programming, 2nd edition, Pentice Hall,2006.
2. Megnus Lie Hetland : Beginning Python from novice to professional, 2nd edition, Apress,2009.

Reference Books:

1. Mark Lutz : Programming Python, 4thEdition,O'reilly, 2011 .
2. Dusty Philips: Python 3 Object oriented Programming , PACKT publishing, 2010.
3. Steve Holden: Python Web Programming, 1st edition,2002.

Web References:

1. http://people.cs.aau.dk/~normark/prog3-03/html/notes/paradigms_themesparadigmoverview-section.html[Overview of Programming Paradigms]
2. www.tutorialspoint.com/python[For tutorials of Python]
3. <https://docs.python.org/>[Python Documentation]
4. <https://www.python.org/about/gettingstarted/>[For beginners of Python]
5. <http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-189-agentleintroduction-to-programming-using-python-january-iap-2008/> [Python Materials]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Able to understanding Python scripting language.

- C02 : Clear Understanding of the built-in objects of Python and object-oriented concepts
 C03 : Express different Decision Making statements and Functions
 C04 : Be exposed to advanced applications such as Web applications,

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes			
		C01	C02	C03	C04
1	Introduction to Multi paradigm approach using Python	✓			
2	Objects, Numbers and Sequences in Python	✓	✓		
3	Conditionals ,loops and files			✓	
4	Functional Programming			✓	
5	Object Oriented Programming		✓		
6	Web Programming				✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	3	2	3	3	2	3	1	2	1	3	3	3	3
CO2	3	2	3	3	2	3	2	2	1	3	2	3	2
CO3	3	3	3	3	3	3	1	2	2	3	2	3	1
CO4	3	3	3	3	2	3	2	3	2	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA218: System Programming
(200 Marks)

Contact Hours: 07

Pre-requisite: CA117: Operating System concepts.

Methodology & Pedagogy: During theory sessions topics related to concepts of system level programming shall be covered. During Practical sessions, students will develop programs using the assembly language and understand the working of the program.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Overview of System Software	07	36
2	Overview of Language Processors	07	
3	Assemblers	07	
4	Macro and Macro Processors	09	
5	Linkers and Loaders	09	
6	Compilers, Interpreters & Debuggers	09	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit – I: Overview of System Software Hours:07

Introduction, Software, Software Hierarchy, Systems Programming, Machine Structure, Interfaces, Address Space, Computer Languages, Tools, Life Cycle of a Source Program, Different Views on the Meaning of a Program, System Software Development, Recent Trends in Software Development, Levels of System Software.

Unit – II: Overview of Language Processors Hours:07

Programming Languages and Language Processors, Language Processing Activities, Program Execution, Fundamental of Language Processing, Symbol Tables. Data Structures for Language Processing: Search Data structures, Allocation Data Structures.

Unit – III: Assemblers Hours:07

Elements of Assembly Language Programming, Assembler, Assembler Design Criteria, Types of Assemblers, Two-Pass Assemblers, One-Pass Assemblers, Algorithm of Single Pass

Assembler, Multi-Pass Assemblers, Advanced Assembly Process, Variants of Assemblers
Design of two pass assembler.

Unit – IV: Macro and Macro Processors Hours: 09

Introduction, Macro Definition and Call, Macro Expansion, NestedMacro Calls, Advanced Macro Facilities, Design Of a Macro Pre-processor, Design of a Macro Assembler, Functions of a MacroProcessor, Basic Tasks of a Macro Processor, Design Issues of MacroProcessors, Features, Macro Processor Design Options, Two-Pass MacroProcessors, OnePass Macro Processors.

Unit – V: Linkers and Loaders Hours: 09

Introduction, Relocation of Linking Concept, Design of a Linker, Self-Relocating Programs, Linking in MSDOS, Linking of Overlay Structured Programs, Dynamic Linking, Loaders, Different Loading Schemes, Sequential and Direct Loaders, Compile-and-Go Loaders, GeneralLoader Schemes, Absolute Loaders, Relocating Loaders, Practical Relocating Loaders, Linking Loaders, Relocating Linking Loaders, Linkers v/s Loaders, Scanning and Parsing, Programming Language Grammars, Classification of Grammar, Ambiguity in Grammatic Specification, Scanning, Parsing, Top DownParsing, Bottom up Parsing, Language Processor Development Tools, LEX, YACC.

Unit – VI: Compilers, Interpreters & Debuggers Hours: 09

Causes of Large Semantic Gap, Binding and Binding Times, DataStructure used in Compiling, Scope Rules, Memory Allocation, Compilation of Expression, Compilation of Control Structure, Code Optimization ,Benefits of Interpretation, Overview of Interpretation, The Java Language Environment, Java Virtual Machine, Types of Errors, Debugging Procedures, Classification of Debuggers, Dynamic/Interactive Debugger.

Core Books:

1. D M Dhamdhare : System Programming, Tata McGraw Hill Publication, 2011.
2. Srimanta Pal: Systems Programming : Oxford University Press, 2011.

Reference Books:

1. Anand A. Godbole R.K. Maurya: System Programming And Compiler Construction (Includes Labs),Wiley, 2014
2. Leland L. Beck,DhanabalachandranManjula:System Software – An Introduction to Systems Programming,3rdEdition, Pearson Education Asia, 2012.
3. SantanuChattopadhyay:System Software, Prentice-Hall India,2007.

Web References:

1. https://en.wikipedia.org/wiki/System_programming[Introduction to system programming]

2. <http://nptel.ac.in/courses/106102067/>[Introduction to programming language]
3. <http://www.cse.psu.edu/~djp284/cmpsc311-s14/slides/01-systemsprogramming.pdf>[Introduction to system programming]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : To understand the basics of system programs like editors, compiler, assembler, linker, loader, interpreter and debugger
- C02 : Describe the various concepts of assemblers and macro-processors.
- C03 : To understand the various phases of compiler and compare its working with assembler.
- C04 : To understand how linker and loader create an executable program from an object module created by assembler and compiler
- C05 : To know various editors and debugging techniques.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Overview of System Software	✓				
2	Overview of Language Processors	✓	✓			
3	Assemblers		✓	✓		
4	Macro and Macro Processors		✓			
5	Linkers and Loaders			✓	✓	
6	Compilers, Interpreters & Debuggers				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	2	2	2	2	-	2	1	-	-	-	1	2	1
CO2	3	2	2	2	-	2	1	-	-	-	1	2	1
CO3	3	3	2	2	-	2	1	-	-	-	1	2	1
CO4	3	3	2	2	-	2	1	-	-	-	1	2	1
CO5	2	2	2	2	-	2	1	-	-	-	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA219: Scripting Language
(Marks 200)

Contact Hours: 07

Pre-requisite: Basic knowledge of programming language.

Methodology & Pedagogy: During theory lectures basics of scripting language and concepts of Perl scripting language will be introduced to students. Emphasis will be given on basic syntax of Perl language and various features of Perl like array, hashes, references, pattern matching etc. Students will be made familiar with applications related to Perl scripting language. During practical sessions the concepts learned in theory lectures will be implemented.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Basics of Scripting Language	05	36
2	Introduction to Perl Language	09	
3	Statements and Subroutine	08	
4	Basic Data Structures in Perl	08	
5	Pattern Matching	09	
6	References and Modules	09	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit – I: Basics of Scripting Language Hours: 05

Introduction, Motivation for and applications of scripting, how scripting languages differ from non-scripting languages, Types of Scripting language, Important features of Scripting Language, Advantages and Disadvantages of scripting language, uses of scripting language.

Unit – II: Introduction to Perl Language Hours: 09

What is Perl?, Perl Features, Perl and the web, Applications of Perl, Perl Philosophy, The Perl Interpreter, sample script, the life cycle of Perl Program, variables, Built-in data types, scalar values, numbers, strings, String functions, undef, formatted IO, Operators, Input operators.

Unit – III: Statements and Subroutine Hours: 08

What is truth?, Compound Selection Structures, Compound Repetition Structures, Simple Statement Modifiers, Loop Miscellaneous, switch, subroutine, subroutine scope, context.

Unit – IV: Basic Data Structures in Perl Hours: 08

Arrays, Lists, Hashes, Arrays of Arrays, Hashes of Arrays, Arrays of Hashes, Hashes of Hashes, Hashes of Functions

Unit – V: Pattern Matching Hours: 09

Regular expression, Pattern Matching Operators, Metacharacters and Metasymbols, Quantifiers, Positions, Grouping and capturing, split and join, Special variables.

Unit – VI: References and Modules Hours: 09

Introduction to references, creating references, using hard references, symbolic references, braces, brackets and quoting, loading modules, unloading modules, creating modules, overriding built-in functions, CPAN.

Core Books:

1. Tom Christiansen: Programming Perl, 4th edition, Shroff Publishers and Distributors O'RELLY, Oct. 2013
2. Simon Cozens: Beginning Perl, WROX Press Ltd, June 2000

Reference Books:

1. Martin C. Brown: The complete reference Perl, second edition, Osborne/McGraw-Hill, Jan. 2001

Web References:

1. <http://userpages.umbc.edu/~dhood2/courses/cmsc433/fall2005/?section=Notes&topic=Perl>
2. https://docs.google.com/viewer?url=http%3A%2F%2Fblob.perl.org%2Fbooks%2Fbeginningperl%2F3145_Chap01.pdf
3. <https://www.perl.org/learn.html>

Upon successful completion of the course, the students will :

- C01 : Learn the importance of scripting language in real life.
- C02 : Learn fundamental concepts of Perl Scripting language and related aspects.
- C03 : Be able to extract information from text file using scripting language.
- C04: Learn to design entire application using scripting language.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes			
		C01	C02	C03	C04
1	Basics of Scripting Language	✓			✓
2	Introduction to Perl Language		✓		✓
3	Statements and Subroutine		✓		✓
4	Basic Data Structures in Perl		✓		✓
5	Pattern Matching		✓	✓	✓
6	References and Modules		✓		✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	3	3	2	2	-	2	1	-	-	-	2	3	1
CO2	3	3	2	2	-	2	1	-	-	-	2	3	1
CO3	3	3	2	2	-	2	1	-	-	-	2	2	1
CO4	3	3	2	2	-	2	1	-	-	-	2	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA220: Basics of Web Designing
(200 Marks)

Contact Hours: 07

Pre-requisite:CA113 - Programming the Internet-I andCA213 - Programming the Internet-II.

Methodology & Pedagogy: During theory lectures students will be introduced to the fundamental basics of Responsive web design, Media Query and AngularJS framework. During Practical sessions students will be required to design and develop responsive web sites using HTML5, CSS3 and AngularJS.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Foundation of Responsive Design	07	36
2	Responsive Web Design	09	
3	Responsive Layout Designing	08	
4	Introduction to JavaScript Framework – AngularJS	07	
5	Elements of AngularJS	08	
6	Form Designing and Validation using AngularJS	09	

Total Hours (Theory): 48

Total Hours (Practical):36

Total: 84

Detailed Syllabus:

Unit - I: Foundation of Responsive Design Hours:07

Basics of Responsive Design, History of Responsive designing, Responsive Content: Content Strategy, Managing Content, Developing Content, Content Parity, Content Governance, Adaptive Content.

Unit - II: Responsive Web DesignHours:09

Responsive Element and Media, Resizing Image, Video, button, Responsive Typography, creating fluid, responsive typography, Text Shadow, Text Rotating, 3D text, texture to text using CSS3.

Unit - III: Responsive Layout Designing Hours:08

Controlling Layout, Media query in CSS, Responsive width layout with Media query, Changing Image size with media query, Hiding an element with media query, Responsive Framework, Fluid 960 grid layout, Blueprint grid layout.

Unit - IV: Introduction to JavaScript Framework – AngularJS Hours:07

Basic fundamentals of AngularJS: Overview of AngularJS, Features of AngularJS, Advantages and Disadvantages of AngularJS, MVC architecture of AngularJS.

Unit - V: Elements of AngularJS Hours:08

AngularJS expressions, AngularJS Modules, AngularJS Directives, AngularJS ng-model Directive, AngularJS Data Binding, AngularJS Controllers.

Unit - VI: Form Designing and Validation using AngularJS Hours:09

AngularJS DOM, AngularJS Forms, AngularJS Tables, AngularJS Select, AngularJS Events, AngularJS Validation.

Core Books:

1. HTML5 and CSS3 Responsive Web Design Cookbook, Benjamin LaGrone, PACKT publishing.
2. Learning Responsive Web Design: A Beginner's Guide, Clarissa Peterson, O'REILLY
3. AngularJS – 1st Edition, Brad Green, ShyamSeshadri, O'REILLY
4. AngularJS Programming by Example 1st Edition -2014, AgusKurniawan,

Reference Books:

1. Pro AngularJS 2014 Edition, Adam Freeman, Apress.
2. Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics 4th Edition, Jennifer Niederst Robbins, O'REILLY

Web References:

1. <http://www.w3schools.com/angular/default.asp> [Tutorial link for AngularJS]
2. <http://www.tutorialspoint.com/angularjs/> [Tutorial link for AngularJS]
3. https://www.tutorialspoint.com/angularjs/angularjs_tutorial.pdf [E-book for AngularJS]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be able to understand need of responsive design and different layout for web page design with its difference.

- C02: be able to design responsive web page for different devices (desktops, tablets, and phones) using bootstrap and media query.
- C03 : Be familiar with JavaScript Framework - AngularJS and its applications.
- C04 : Be able to create a single page application using AngularJS.
- C05 : Be able to create web application using the MVC architecture. Also, able to maintain the two-way data binding between model and view.
- C06 : Be able to work with AngularJS DOM and validation.

Course Outcomes Mapping:

Unit No.	Content	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Foundation of Responsive Design	✓					
2	Responsive Web Design		✓				
3	Responsive Layout Designing		✓				
4	Introduction to JavaScript Framework – AngularJS			✓	✓	✓	
5	Elements of AngularJS			✓	✓	✓	
6	Form Designing and Validation using AngularJS				✓		✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	1	3	-	-	2	3	1	2	3	-	-	2	1
CO2	3	3	3	3	3	3	1	2	3	2	2	3	1
CO3	3	1	3	3	2	2	1	1	2	2	2	3	1
CO4	3	2	3	3	2	2	2	2	2	2	2	3	1
CO5	3	3	3	3	3	3	3	2	3	3	3	3	2
CO6	3	2	2	3	3	3	3	2	2	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA221: Data Structures and Algorithms

(200 Marks)

Contact Hours: 07

Pre-requisites: CA111: Fundamentals Concepts of Programming Language, CA214 : Object Oriented Programming Using C++

Methodology & Pedagogy: During theory lectures emphasis will be given on understanding fundamental concepts of data structures and algorithms and how to use and choose the most efficient data structures from a variety of available ones for different kind of applications. In practical sessions emphasizes given on designing, analyzing and implementation of all kind of data structures for solving computational problems using high level programming language.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1.	Introduction to Data Structures and Algorithms	06	36
2.	Linear Data Structures Using Array	10	
3.	Dynamic Data Structures	09	
4.	Trees Data Structure	08	
5.	Graph Data Structure	08	
6.	Sorting and Searching	07	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit - I: Introduction to Data Structures and Algorithms Hours: 06

Atomic and composite data, Data type, Data object, Data Structure, Abstract Data type, Types of Data Structures, Introduction to Algorithms, Relationship among Data, Data Structures and Algorithms, Analysis of Algorithms, space and time complexity algorithm.

Unit - II: Linear Data Structures Using Array Hours: 10

Linear Data Structures using sequential organization, Concepts of Ordered list, Representation of Stacks using sequential organization, Applications of Stack, Recursion, Application of

Recursion, Concepts of Queues, Realization of Queues using sequential organization, Multiqueue, Deque, Priority Queue, Applications of Queue.

Unit - III: Dynamic Data Structures Hours: 09

Linked list, Comparison of sequential and linked organizations, Dynamic Memory Management, Singly linked list, Doubly linked list, Applications of linked list.

Unit - IV: Trees Data Structure Hours: 08

Basic Terminologies, Definition and concepts, Representation of Binary Tree, Operations on Binary Tree, Types of Binary Trees, B Tree, B+ Tree Indexing, Trie Tree Indexing.

Unit - V: Graph Data Structure Hours: 08

Graph Terminology, Representation of Graph, Operations on Graph, Applications of Graph Structure.

Unit - VI: Sorting and Searching Hours: 07

Sorting Notations and Concepts, Sorting Techniques, Sequential Searching, Binary Searching, Search Trees, Hash Table Methods.

Core Books:

1. Jean-Paul Tremblay, Paul G. Sorenson: An introduction to data structures with applications, 2nd Edition, Tata McGraw Hill Publications, 1991.
2. Varsha h. Patil : Data Structures Using C++, 1st Edition, Oxford University Press, 2012.
3. D. Samanta: Classic Data Structures, 2nd Edition, PHI Publications, 2002.

Reference Books:

1. YashvantKanethkar : Data structures through C++, 2nd Edition, BPB Publications,2003.
2. YedidyahLangsam, Moshe J. Augenstein, Aaron M. Tenenbaum : Data structures using C and C++ , 2nd Edition, Eastern economy edition, Pearson Education,2007.
3. Mark Allen Weiss: Data Structures and Algorithm Analysis in C++, 3rd Edition, Pearson Education, 2009.
4. ReemaThareja: Data Structures using C, 1st Edition, Oxford University Press, 2012.

Web References:

1. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/ds_ToC.html [For materials of Data Structures and Algorithms]
2. <http://www.cs.usfca.edu/~galles/visualization/> [For Data Structures Visualization]
3. <https://www.cs.auckland.ac.nz/~jmor159/PLDS210/mst.html>[For Graph Data Structures]
4. <http://interactivepython.org/runestone/static/pythonds/Trees/trees.html> [For Tree Data Structures]

5. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/niemann/s_man.pdf [For Sorting and Searching Cookbook]
6. <http://www.cs.princeton.edu/~rs/AlgsDS07/10Hashing.pdf> [For Hashing Materials]
7. <http://nptel.ac.in/video.php?subjectId=106102064> [For Data Structures videos]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Analyze the asymptotic performance of algorithms
- C02 : Explore linear and nonlinear data structures
- C03 : Apply important algorithmic design paradigms and methods of analysis.
- C04 : Use efficient data structures which are a key to designing efficient algorithms.
- C05: Compare various sorting and searching techniques

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Data Structures and Algorithms	✓	✓			
2	Linear Data Structures Using Array		✓	✓		
3	Dynamic Data Structures		✓	✓		
4	Trees Data Structure		✓		✓	
5	Graph Data Structure		✓		✓	
6	Sorting and Searching					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	1	2	1	-	-	-	-	-	-	1	-	2	-
C02	2	2	-	1	-	-	1	-	-	2	2	2	1
C03	1	2	3	1	-	-	1	-	-	2	1	2	1
C04	1	2	3	1	-	-	1	-	-	1	1	2	1
C05	2	1	2	1	-	-	-	-	-	1	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA222: Advanced Database System

(100 Marks)

Contact Hours: 03

Pre-requisite: CA119 : Database Fundamentals.

Methodology & Pedagogy: During the practical session student will learn how to create table and perform operation on data and table using different oracle objects. Through procedure and function students can understand the concept of reusability. Student will learn how to handle error using exception and using trigger they can understand how to restrict the operation at particular time or given condition.

Outline of the Course:

Week No	Practical	Description
1	Create, Insert, Update, Delete and Alter Statements, Datatypes, Constraints, select statement	Student will learn how to create table with/without constraints and how to apply different operations on table and data.
2	INNER Join, SELF Join, LEFT Join, RIGHT Join Subquery, Nested query, Group by and Having Clause	Execution of JOIN and Subquery and usage of Group by and having
3-4	Datatypes of PL/SQL, Looping Structures, Conditional Statements, Operators	Basic syntax of PL/SQL Block and usage of Conditional statements and Looping structure inside the block
5-6	Implicit Cursor, Explicit Cursor, Parameterized Cursor	Usage of Cursor and its working
7-8	User defined Procedure & Function	Student will learn how to create procedure and function and its execution
9-10	Error Handling using pre-defined exception and user-defined exception	How to create exception and how to use predefined exception
11-12	Row level and Statement level Trigger, GRANT & REVOKE	Syntax of trigger and its working, Assign privileges at different level

Total Hours: 36

Core Books:

1. SQL, PL/SQL – The programming Language Oracle-by Ivan Bayross
2. Teach Yourself Database Technologies-by Ivan Bayross

Reference Books:

1. Oracle 9i, PL/SQL Programming by Scoot Urban, Oracle Press
2. Practice book on SQL and PL/SQL by Anjali, Amisha, Roopal and Nirav publications.

Web References:

1. <http://www.w3schools.com> [For SQL]
2. <http://plsql-tutorial.com/index.htm> [For PL/SQL]
3. http://www.tutorialspoint.com/plsql/plsql_collections.htm [For PL/SQL]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : To Learn basic commands for Creating Table and manipulate data in table and applying constraint to data.
- C02 : Learning basic Queries and Retrieval of data from multiple tables with grouping of data.
- C03 : Learn to create generic block of PL\SQL with condition and looping structures.
- C04 : Learn to generate data reports with multiple records and multiple tables with PL\SQL.
- C05 : Learn to create user define objects of database like Functions and Procedure.
- C06 : Learn to Handle errors during PL\SQL block executions with user define exceptions.
- C07 : Learn to create different types of Triggers and privileges assignment to user.

Course Outcomes Mapping :

Week No.	Unit Name	Course Outcomes						
		C01	C02	C03	C04	C05	C06	C07
1	Datatypes, DDL, DML and DCL statements and Constraints	✓						
2	Joins, Subquery, Nested query, Grouping with Having clause		✓		✓			
3-4	Basic of PL/SQL, condition and Looping statements, Operators			✓				
5-6	Implicit Cursor, Explicit Cursor, Parameterized Cursor				✓			
7-8	User defined Procedure & Function			✓	✓	✓		
9-10	Error Handling using pre-defined exception and user-defined exception						✓	
11-12	Row level and Statement level Trigger, GRANT & REVOKE							✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	3	2	1	2	3	-	2	1	1	3	-
CO2	3	3	3	2	2	2	3	2	2	2	2	3	2
CO3	3	3	3	2	3	2	3	2	2	2	2	3	1
CO4	3	3	3	3	1	3	3	1	2	2	3	3	2
CO5	3	3	3	3	3	3	3	1	3	2	3	3	2
CO6	3	3	3	2	3	2	3	2	3	2	3	3	1
CO7	3	3	3	1	3	3	3	2	3	2	2	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA223: Fundamentals of Computer Networks

(Marks 100)

Contact Hours:03

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures the emphasis will be given on the Introduction to the computer network basics and terminologies used in computer networking. The teacher will also discuss different reference models and the working of each layer of models. The teacher will discuss different design issues related to layers of model and networking devices and networking protocols.

Outline of the Course:

Unit No.	Title of Unit	Minimum Number of Hours
		Theory
1	Basics of Computer Network	05
2	Network Communication Concepts	07
3	Reference Models	05
4	Issues in Data Link Layer Design	08
5	Issues in Network Layer Design	06
6	Network Protocols & Devices	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit - I: Basics of Computer Network Hours: 05

Definition of Computer Network, Goals of Computer Network: For Businesses, For People, Advantages & Disadvantages of Computer Network, Applications of Computer Network, Components of Computer Network, Types of Networks, Network Topologies.

Unit - II: Network Communication Concepts Hours: 07

Parallel & Serial Communication, Synchronous & Asynchronous Communication. Modes of Communication: Simple, Half-Duplex, Full-Duplex, Multiplexing, De-Multiplexing: FDA, TDM. Communication Media: Guided Media, Unguided Media. Switching Techniques: Circuit Switching, Message Switching, Packet Switching.

Unit - III: Reference Models Hours: 05

OSI Reference Model: Architecture, Function of Each Layer, TCP/IP Reference Model: Architecture, Function of Layers. Comparison of OSI & TCP/IP Models.

Unit - IV: Issues in Data Link Layer Design Hours: 08

Services provided to the Network Layer, Framing, Error Control (error detection and correction code), Flow Control, Data Link Layer in the Internet (SLIP, PPP). MAC sub layer: CSMA/CD/CA, IEEE standards (IEEE802.3 Ethernet IEEE 802.4 Token Bus, IEEE 802.5 Token Ring).

Unit - V: Issues in Network Layer Design Hours: 06

Routing Algorithms: Optimality Principle, Shortest Path Routing, Congestion Control Policies, Concept of Internetworking.

Unit - VI: Network Protocols & Devices Hours: 05

Protocol Hierarchy, Network Protocols: SMTP, FTP, MIME, POP, Telnet, DNS, DHCP, ARP, RARP. Network Devices: Hub, Switch, Repeater, Router, Gateway, Bridge.

Core Books:

1. Andrew S. Tanenbaum: Computer networks, 4th Edition, Pearson Publication.
2. Behrouz Forouzan: Data communications and networking, 4th Edition, McGraw Hill.

Reference Books:

1. Achyut S Godbole, AtulKahate: Data Communications and Networks, 2nd Edition, McGraw Hill.
2. Honey Cutt: Using the Internet, 4th Edition, PHI Learning.

Web References:

1. <http://nptel.ac.in/> [Online Lectures].
2. http://www.tutorialspoint.com/data_communication_computer_network/index.html [C ourse Material, Notes].

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Understand basic of networks, networking technologies and concepts.
- C02 : Come to know how communication systems work.
- C03 : Learn different networking reference models
- C04 : Gain understanding of various layers' design issues and their solutions

C05 : Learn functionality of various network protocols and how devices work

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Basics of Computer Network	✓				
2	Network Communication Concepts		✓			
3	Reference Models			✓		
4	Issues in Data Link Layer Design				✓	
5	Issues in Network Layer Design				✓	
6	Network Protocols & Devices					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	2	1	-	1	1	1	-	-	1	2	-
CO2	2	3	3	2	-	1	1	1	-	-	1	3	1
CO3	3	3	3	2	-	1	1	1	1	-	1	3	3
CO4	1	3	3	2	-	1	1	1	1	-	1	3	2
CO5	3	3	3	2	-	1	1	1	1	-	1	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS111.02 C: HUMAN VALUES & PROFESSIONAL ETHICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/01	--	30/15	2
Marks	--	100	--	100	

Pre-requisite courses:

- Ethical Leadership through Giving Voice to Values
<https://www.coursera.org/learn/uva-darden-giving-voice-to-values?skipBrowseRedirect=true>

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Values and Ethics	05
2.	Elements and Principles of Values	08
3.	Applied Ethics	08
4.	Value, Ethics & Global Issues	05
5.	Contemporary Issues in Values and Ethics	04

Total hours (Theory): --
Total hours (Practical): 30
Total hours (Lab) : --
Total hours : 30

Detailed Syllabus:

1.	Introduction to Values and Ethics	5 Hours	17%
	Need, Relevance and Significance of Values General, Concept and Meaning of Values and Ethics		
2.	Elements and Principles of Values	8 Hours	26%
	Universal & Personal Values, Social, Civic & Democratic Value		
3.	Applied Ethics	8 Hours	26%
	Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics		
4.	Value, Ethics & Global Issues	5 Hours	17%
	Cross-Cultural Issues, Role of Ethics & Values in Sustainability		
5.	Contemporary Issues in Values and Ethics	4 Hours	14%
	Case Studies, Presentations, Projects		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the concepts and mechanics of values and ethics.
CO2	Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.
CO3	Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.
CO4	Develop their responsibility towards society.
CO5	Comprehend their own core values and adhere to those values at their workplace.
CO6	Practice Ethical Leadership.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	3	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	3	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	2	-	-	-
CO5	-	-	-	-	3	-	-	-	2	-	-	-	-
CO6	-	-	-	-	-	-	-	-	2	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ **Reference book:**

1. Human Values and Ethics in Workplace, United Nations Settlement Program, 2006.
(http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
2. Ethics for Everyone, Arthur Dorbin, 2009.
(<http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf>) .
3. Values and Ethics for 21st Century, BBVA. (https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)

❖ **Web material:**

- www.ethics.org

Teaching and Examination Scheme (BCA Programme)

Effective from Year 2021-2022

Semester-V

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theor y	Pract	Tot al		Internal		Exter nal	Internal		Exter nal	
						Case Study	Tests		Term work	Tests		
CA308–313	Elective-VI	4	3	7	7	10	20	70	15	15	70	200
CA314	Object Oriented Programming Through JAVA	4	3	7	7	10	20	70	15	15	70	200
CA315	Application development using .NET Framework	4	3	7	7	10	20	70	15	15	70	200
CA316	Software Engineering	-	3	3	3	-	-	-	15	15	70	100
HS131.02 C	Communication and Soft Skills	-	2	2	2				30		70	100
		12	14	26	26	300			500			800

Elective-VI

1. CA308 –Introduction to Open Source Technology
2. CA309 –Advanced Computer Networks
3. CA310 –Operation Research
4. CA311–Embedded Programming
5. CA312-Multi Media
6. CA313-Ruby on Rail Architecture

CA308: Introduction to Open Source Technology

(Marks 200)

Contact Hours: 07

Prerequisite: Working knowledge of Html

Methodology & Pedagogy : During theory lectures students will be introduced to the fundamentals of PHP, various functions available in PHP and working with form elements. During Practical sessions students will be made familiar to develop website using PHP.

Outline of the Course:

Sr .No.	Title of the Unit	Minimum No of Hours	
		Theory	Practical
1.	Introduction	07	36
2.	PHP Configurations and Basics	07	
3.	Control Structures	07	
4.	Functions in PHP	09	
5.	Working With Data	09	
6.	PHP Utilities	09	
		48	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus

Unit – I: Introduction Hours: 07

Introduction of open source software, Development philosophy of open source software, pros and cons of open source software, open source vs. close source software Introduction to Webpage and Website, Static and Dynamic Website, Client & Server Side Scripting, Introduction to other server side languages.

Unit – II: PHP Configurations and Basics Hours: 07

Introduction to PHP, Installation of Apache, MySQL and PHP, How PHP code is parsed, Embedding PHP and HTML, Executing PHP and viewing in Browser, Data types, Operators, PHP variables: static and global variables, Comments in PHP.

Unit – III: Control Structures Hours: 07

Condition statements : If...Else, Switch, Ternary operator.

Loops: While, Break Statement, Continue, Do...While, For, For each.

Exit, Die, Return. Working with array in PHP.

Unit – IV: Functions in PHP Hours: 09

Overview of built in functions of PHP, PDF functions, file handling functions, CSV file handling using PHP, User Defined Functions.

Unit – V: Working With Data Hours: 09

FORM element, INPUT elements, validating the user input, passing variables between pages: Passing variables through GET, Passing variables through POST, Passing variables through REQUEST.

Sessions and cookies: Concept of Session, Starting session, Modifying session variables, Un registering and deleting session variable, Concept of Cookies.

Unit – VI: PHP Utilities Hours: 09

File Uploading: Upload Single and Multiple file using PHP script, Understanding HTTP requests, Exploring and modifying HTTP responses, getting information from web server, Sending mails, Ajax with PHP.

Core Books:

- 1.SteverHolzner: The Complete Reference PHP, McGraw Hill
- 2.Leon Atkinson: Core PHP Programming, Pearson publishers
- 3.Matt Doyle ,Beginning PHP 5.3, Wrox : 2010.

Reference books:

1. Peter Moulding: PHP Black Book
2. SharnamShah,Vaishali Shah: LAMP programming for professionals,ShroffPublishers,December 2009
3. Jason Garner,Morgan Owens, Elizabeth Naramore,MattWarden,JeremyStolz: Professional LAMP: Linux,Apache,MySQL and PHP Web Development(Paperback) Wrox,2005.

Web References:

1. <http://www.w3schools.com/php>[Lecture Notes of php]
2. <http://www.php.net/> [New release information of php]
3. <http://www.tutorialspoint.com/> [Lecture notes of php]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Gain the skills and knowledge to install and use an integrated PHP/MySQL environment.
- C02 : To acquire knowledge and skills for creation of web site considering both client and server side programming.
- C03 : Learn the request/response cycle, including GET/POST.
- C04 : Write PHP code to produce outcomes and solve problems using various available functions and control structures. able to build dynamic web applications with MySQL database connections.
- C05 : Learn the different state management techniques.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction		✓			
2	PHP Configurations and Basics	✓	✓			
3	Control Structures	✓	✓			
4	Functions in PHP	✓	✓			
5	Working With Data			✓	✓	✓
6	PHP Utilities		✓	✓		

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	-	-	2	-	2	-	-	-	-	-	3	3
CO2	3	2	2	-	1	1	-	-	1	1	-	3	3
CO3	2	-	-	2	-	-	-	-	-	-	-	3	3
CO4	-	2	-	-	1	1	2	1	-	2	1	2	2
CO5	-	1	1	-	-	1	-	-	-	1	-	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA309: Advanced Computer Networks

(200 Marks)

Contact Hours: 07

Prerequisites : CA223 Fundamentals of Computer Networks.

Methodology & Pedagogy: During theory sessions the students shall be introduced to various internet and intranet technologies and network services. As a case study an institute level network design, devices, components and security mechanism will be demonstrated. During practical sessions students will be trained to develop network based applications using available technologies.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Addressing Techniques	07	36
2	Internet Protocol (TCP/IP)	08	
3	Routing	08	
4	UDP and Mobile IP	08	
5	Cellular Systems	08	
6	Internet Security	09	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Unit – I : Addressing Techniques Hours:07

Classful Internet Addresses, Mapping Internet Addresses (ARP), Determining An Internet Address (RARP)

Unit – II : Internet Protocol (TCP/IP) Hours:08

Internet Protocol: Connectionless Datagram Delivery, Internet Protocol: Routing IP Datagrams, Internet Protocol: Error and control Messages, ICMP

Unit – III : Routing Hours:08

Congestion, Router Discovery and solicitation, Classless and Subnet Address Extensions, Transparent routers, Proxy ARP and Subnet addressing, CIDR, Subnet routing. Static and Dynamic routing, Difference between static and dynamic routing. Introduction to dynamic routing algorithms.

Unit – IV: UDP and Mobile IP Hours:08

User Datagram Protocol, Reliable Stream Transport Service, Sliding Window, TCP, Karn's Algo, Congestion, RED Silly window syndrome, starting and closing TCP connection, Mobile IP

Unit – V: Cellular Systems Hours:08

Cellular system overview - Cellular system organization, Frequency Reuse, Increasing Capacity, Operation of cellular system, steps in a Mobile switching center(MSC) controlled call between mobile users, Mobile Radio Propagation effect, Additional Function in MSC Controlled Call, Handoff Performance metrics, Handoff Strategies Used to Determine Instant of Handoff, Power control, Traffic Engineering.

Unit – VI : Internet Security Hours: 09

Electronic mail, TCP/IP, SMTP, MIME, Applications: World Wide Web Internet Security and Firewall Design, Internet Security, Firewalls

Core Book:

1. Douglas E Comer: Internetworking with TCP/IP Volume – I, Fourth Edition, PHI.
2. Andrew S. Tanenbaum, David Wetherall: Computer Networks, PHI
3. Ajit Pal: Data Communications and Computer Networks, PHI

Reference Book:

1. Behrouz A. Forouzan: TCP / IP Protocol Suite, Third Edition
2. Natalia Olifer, Victor Olifer: Computer Networks: Principles, Technologies and Protocols for Network Design, Wiley
3. Debra Littlejohn Shinder: Computer Networking Essentials: An essential guide to understanding networking theory, implementation and interoperability, Cisco Press.

Web References:

1. <http://www.firewall.cx>[Basics of Network Protocols]
2. <http://nptel.ac.in>[Video Lectures]
3. <https://www.cs.purdue.edu/homes/comer/netbooks.html>[Teaching Material]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Understand basic of network addressing schemes and types of addresses.
- C02 : Come to know various protocol working at various layers of TCP/IP protocol suite.
- C03 : Learn different routing techniques and algorithms
- C04 : Gain understanding of connection-less protocol such as UDP and working of mobile communication
- C05 : Get overview of cellular system organization and working of it.
- C06 : Gain knowledge of firewalls and other security techniques

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Addressing Techniques	✓					
2	Internet Protocol (TCP/IP)		✓				
3	Routing			✓			
4	UDP and Mobile IP				✓		
5	Cellular Systems					✓	
6	Internet Security						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	3	-	1	1	-	-	1	1	-	3	-
CO2	3	2	3	2	1	1	-	-	-	1	-	3	1
CO3	3	3	2	2	1	2	2	1	-	2	1	2	1
CO4	3	2	3	2	1	1	2	1	-	1	2	1	1
CO5	2	3	3	2	1	2	3	1	-	2	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA310: Operation Research

(Marks 200)

Contact Hours: 07

Pre-requisite: None.

Methodology & Pedagogy: Theory sessions are responsible to make the students aware with the concepts to develop a report that describes the model and the solving techniques and analyze the results. Practical sessions will be inclined towards the real implementation of mathematical models.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Introduction to Operation Research	08	36
2	Linear Programming	09	
3	Transportation Problem	07	
4	Simulation	08	
5	Stochastic Processes	08	
6	Random Numbers	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total:84

Unit - I: Introduction to Operation Research Hours:08

Introduction, Definition and Characteristics of OR, Scope of operation Research, Models And Modelling in OR, An Overview of OR Techniques, Role Of Computers in Solving operation Research Problems, Limitation of Operation Research.

Unit - II: Linear ProgrammingHours:09

Introduction to Linear Programming, Formulation of LP Problems, Standard form of LP Problem, Graphical Solution of two variable LP Problems, Simplex Method: Computational Procedure, Special Cases: (Unbounded, Multiple Solutions, Non-feasible Solution),Degeneracy.

Unit - III: Transportation ProblemHours:07

Mathematical formulation and tabular representation of Transportation problem, Initial Basic Feasible Solution- North West Corner Rule, Least Cost Method, Vogel's Approximation

Method, Optimal Solution, Optimality Test, Stepping-Stone Method, MODI's Method, Degeneracy in Transportation Problem, Unbalanced Transportation Problem.

Unit - IV: Simulation Hours:08

Introduction to the simulation, Simulation as a tool, advantages of Simulation, Areas of application, general steps followed in simulation experiment Discrete and Continuous Systems, Discrete-Event System Simulation, The Monte Carlo Method, Comparison of Simulation and Analytical Methods.

Unit - V: Stochastic Processes Hours:08

Stochastic variables, Characteristics of Stochastic processes, Discrete and Continuous Probability functions.

Unit - VI: Random Numbers Hours:08

Uniformly Distributed Random Numbers, Properties of Random Numbers ,Generation of Pseudo-Random Numbers, Techniques for generating Random numbers(Linear Congruential Method, Combined Linear Congruential Generators),Tests for Random numbers (Frequency Tests)

Core Books:

1. J K Sharma: Operation Research Theory and Application, MacMillan.
2. V.K. Kapoor : Operation Research Techniques for Management, Sultan Chand & Sons.
3. S.D. Sharma : Operation Research, KedarNath& Co.
4. Taha : Operation Research, MacMillan, New York.

Reference Books:

1. S. Kalavathy:Operation Research, PHI.
2. Gordon G. : System Simulation, PHI.
3. NarsinghDeo: System Simulation and Digital Computer, PHI
4. Banks, Carson, Nelson, Nicol: Discrete-Event System Simulation, Pearson Education
5. Reitman, J.: Computer Simulation Application, Wiley

Web References:

1. <http://web.itu.edu.tr/topcuil/ya/OR.pdf> [Operation Research notes]
2. <http://nptel.ac.in/courses/112106134/1> [Linear Programming notes]

Upon successful completion of the course, the students will:

- C01 : Able to Identify and develop operational research models from the verbal description of the real system.

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C02 : Able to formulate a given simplified description of a suitable real-world problem as a linear programming model in general, standard and canonical forms

C03 : Recognize and formulate transportation, assignment problems and drive their optimal solution.

C04 : Understand the role of simulation for real application.

C05 : Learn the usefulness of stochastic processes and random numbers for real applications.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Operation Research	✓				
2	Linear Programming	✓	✓			
3	Transpiration Problem	✓		✓		
4	Simulation	✓			✓	
5	Stochastic Processes	✓				✓
6	Random Numbers	✓				✓

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	3	3	3	1	1	2	-	2	1	2	1	2	1
C02	2	3	3	1	1	2	-	2	1	2	1	2	1
C03	2	3	3	1	1	2	-	2	1	2	1	2	1
C04	2	1	3	1	1	2	-	2	1	2	2	2	1
C05	2	1	3	1	1	2	-	2	1	2	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA311: Embedded Programming
(200 Marks)

Contact Hours: 07

Pre-requisite: Operating System Concepts, C, C++

Methodology & Pedagogy: Theory sessions are responsible to make the students aware with the concepts of traps and interrupts which make necessary calls and alerts for the hardware interactions. Practical sessions will be inclined towards the real implementation of hardware interactions through implementation of core programming C/C++.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Introduction to Embedded systems & Hardware Components	09	36
2	Characteristics and quality attributes of embedded systems	08	
3	Programming Embedded Systems	07	
4	Embedded hardware	08	
5	Design and Development	08	
6	Introduction to VC++ for Embedded Systems	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit – I: Introduction to Embedded systems & Hardware Components Hours: 09

Systems, history, classifications, applications and purpose of embedded systems.

Core of Embedded Systems: Microprocessors and microcontrollers, RISC and CISC controllers, Big endian and Little endian processors, Application specific ICs, Programmable logic devices, COTS, sensors and actuators, communication interface, embedded firmware, other system components, PCB and passive components.

Unit – II: Characteristics and quality attributes of embedded systems Hours: 08

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Characteristics, Operational and nonoperational quality attributes, application specific embedded system - washing machine, domain specific - automotive.

Unit – III: Programming Embedded Systems Hours: 07

Structure of embedded program, infinite loop, compiling, linking and locating, downloading and debugging.

Unit – IV: Embedded hardware Hours: 08

Memory map, i/o map, interrupt map, processor family, external peripherals, memory - RAM, ROM, types of RAM and ROM, memory testing, CRC, Flash memory.

Unit – V: Design and Development Hours: 08

Control and Status Registers, Device Driver, Timer Driver-Watchdog Timers, Embedded Operating System, Real-Time Characteristics, Selection Process.

Unit – VI: Introduction to VC++ for Embedded Systems Hours: 08

Introduction to Microsoft Visual C++ environment. Floatable & Dockable windows, Visual C++ projects & files, Introduction to Microsoft Foundation Class library.

Core Books:

1. Michael Barr and Anthony Massa: Programming Embedded Systems, Second Edition with C and GNU Development Tools, O' Reilly
2. Visual C++ and MFC Programming, © FunctionX, Inc.

Reference Books:

1. K V Shibu: Introduction to Embedded Systems (Paperback), Tata McGraw Hill, 2009
2. Michael Barr: Programming Embedded Systems in C and C++, First Edition January, O' Reilly
3. Raj Kamal: Embedded Systems: Architecture, Programming and Design, Tata McGraw Hill Publications

Web References:

1. https://books.google.co.in/books?id=pWlW0H3IAC&printsec=frontcover&source=gbs_ge_summ ary_r&cad=0#v=onepage&q&f=false [For additional reading from reference book]
2. https://www.tutorialspoint.com/embedded_systems/embedded_systems_tutorial.pdf [Tutorials for embedded systems]
3. <http://www.tutorialspoint.com/listtutorials/visual-c++/1> [For VC++ tutorials]
4. https://en.wikipedia.org/wiki/Embedded_system [For fundamentals related to embedded systems]
5. https://www.google.co.in/url?sa=t&rct=j&q=&esrc=s&source=web&cd=12&cad=rja&uact=8&ved=0ahUKewjW5afcmenQAQhXDOo8KHUdrDfw4ChAWCCewAQ&url=http%3A%2F%2Fstroustrup.com%2FProgramming%2F25_embedded.ppt&usg=AFQjCNHaPQKbYzFRrMV8LMG7tSLsjcnh3g [For basics of embedded system in Power Point Presentation]

Course Outcomes: Upon successful completion of the course, the students will:

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C01 : Foster ability to understand the internal architecture and interfacing of different peripheral devices with Microcontrollers.

C02 : An ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability

C03 : Foster ability to understand the role of embedded systems in industry.

C04 : Foster ability to developed embedded systems.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes			
		C01	C02	C03	C04
1	Introduction to Embedded systems & Hardware Components	✓			
2	Characteristics and quality attributes of embedded systems	✓		✓	
3	Programming Embedded Systems				✓
4	Embedded hardware	✓	✓		
5	Design and Development				✓
6	Introduction to VC++ for Embedded Systems				✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	2	-	-	1	-	1	-	-	-	-	-	-	-
CO2	3	2	3	2	-	2	1	-	-	-	1	2	1
CO3	1	-	-	1	3	1	-	2	-	-	-	1	1
CO4	1	1	3	1	1	1	-	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA312: Multimedia
(200 Marks)

Contact Hours: 07

Pre-requisite: None.

Methodology & Pedagogy: During theory sessions topics related to common technologies and techniques used in the multimedia will be covered with suitable examples. During Practical sessions, students will be required to develop and produce animations using appropriate tool.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Introduction	08	36
2	Digital representation	09	
3	Text and Image	08	
4	Audio and Video technology	07	
5	Working with Web Animations	08	
6	Advanced Animations	08	

Total Hours (Theory): 48

Total Hours (Lab): 36 Total: 84

Detailed Syllabus:

Unit – I Introduction Hours:08

What is Multimedia? Defining the scope of multimedia. Usage of Multimedia, Stages of a Multimedia Project, the Multimedia Team.

Unit – II Digital Representation Hours:09

Introduction, Analog representation, waves, digital representation, need for digital representation, Analog to Digital conversion, Digital to Analog conversion, relation between sampling rate and bit depth, Quantization error, Fourier representation, pulse modulation. Importance and drawback of digital representation.

Unit – III Text and Image Hours:08

Introduction, Types of text, Font, insertion, compression, File formats. Types of images, color models, Using Text in Multimedia, Computers and Text, Font Editing, Hypermedia, Hypertext, Image in Multimedia: Organizing Tools, Bitmap Images, Vector Drawings, 3-D Drawing and Rendering, Color models, Image File Formats

Unit – IV Audio and Video Technology Hours:07

Sound Multimedia: Power of Sound, Digital Audio, Preparing Digital Audio Files and MIDI Files, Function of editing Digital Audio Files, Audio File Formats, MIDI versus Digital Audio, Adding Sound to Your Multimedia Project Video Multimedia: Using Video, Analog Display Standards, Digital Display Standards, Digital Video, Video Recording, Shooting and Editing Video, Optimizing Video Files.

Unit – V Working with Web Animations Hours:08

CSS3 Fundamentals, CSS3 Transforms and Transitions, CSS3 Transitions for Images, CSS3 Transitions for UI Elements

Unit – VI Advanced Animations Hours:08

CSS3 Keyframe Animations, CSS3 Keyframe Animations for Web Content, Integrating CSS3 Animations with SVG and Filters, CSS3 3D Transforms, Transitions, and Animations, Moving Between Landscapes, Tools, Technologies, and the Future of CSS Animation

Core books:

1. Principles of Multimedia by Ranjan Parekh. Tata McGraw-Hill
2. Multimedia: Making it work by Tay Vaughan - TMH, Eighth edition. 2006 3. Pro CSS3 Animation by Dudley Storey. Apress Publisher, 2012

Reference books:

1. Multimedia systems by John F. Koegel Buford-Pearson Education.
2. Multimedia Systems Design by Prabhat K. Andleigh and Kiran Thakrar-PHI publication 3. Learning CSS3 Animations and Transitions: A Hands-on Guide to Animating in CSS3 with Transforms, Transitions, Keyframes, and JavaScript by Addison-Wesley Professional, 2012

Web References:

1. https://www.w3schools.com/css/css3_animations.asp [CSS3 Animations]
2. https://www.tutorialspoint.com/css/css3_animation.htm [CSS3 Animations]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be able to identify the basic concepts of multimedia and the web.
- C02: Be able to understand analog and digital representation

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C03 : Be able to work with different file format such as text, image, audio, video using tools.

C04 : Able to create animated banner ads using CSS3

C05 : Apply end-user design skills of different application software in creative industry of digital media.

Course Outcomes Mapping:

Unit No.	Content	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction	✓				
2	Digital representation		✓			
3	Text and Image			✓		✓
4	Audio and Video technology			✓		✓
5	Working with Web Animations	✓			✓	✓
6	Advanced Animations				✓	✓

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	2	2	2	2	2	3	1	1	-	1	-	2	1
C02	2	1	1	2	2	2	-	1	1	1	-	1	1
C03	2	2	2	2	3	2	-	1	-	1	1	1	1
C04	3	1	3	3	3	3	2	2	-	1	1	1	1
C05	3	1	3	3	3	3	2	2	-	1	1	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA313: Ruby on Rail Architecture

(200 Marks)

Contact Hours: 07

Pre-requisite: None

Methodology & Pedagogy: During theory sessions topics related to common technologies and techniques used in the developing of web-based applications will be covered with suitable examples. During Practical sessions, students will be required to design and develop entire web sites using Ruby On Rails.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Getting started and basics of Ruby Language Literacy.	08	36
2	Working with standard types, Methods and Expression	08	
3	Working with Arrays, Hashes and Enumerable	08	
4	Working with Times and Dates, Exceptions, Catch and Throw	08	
5	Files, I/o and Databases	08	
6	OOP and Dynamic features in Ruby	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit – I: Getting started and basics of Ruby Language Literacy Hours: 08

What is Ruby; Ruby download and installation, An Introduction to Object Orientation, Basic Ruby Syntax and Semantics OOP in Ruby, Dynamic Aspects of Ruby.

Unit – II: Working with standard types, Methods and Expression Hours: 08

Numbers, Strings, Ranges, Regular Expressions, defining a method, Calling a method, Operator expressions, Miscellaneous expressions, Assignment, Conditional execution, Case expression, Loops, variable Scope, Loops and Blocks.

Unit – III: Working with Arrays, Hashes and Enumerable Hours: 08

Comparing arrays, Sorting array, selecting from an array by criteria, using multidimensional arrays, Finding, transforming, removing, Concatenating and reversing in arrays and hashes with key value pairs. Searching and selecting, Counting and comparing, iterating, Extracting and converting in Enumerable

Unit – IV: Working with Times and Dates, Exceptions, Catch and Throw Hours: 08

Determining and finding time and dates, conversion, parsing and formatting of date and time. Exception class, Handling and Raising exceptions, Catch and Throw.

Unit – V: Files, I/o and Databases Hours: 08

What is an IO object, Opening and closing files, Reading and writing files, Connecting to external data stores like MySQL, ORM and SQLite.

Unit – VI: OOP and Dynamic features in Ruby Hours: 08

Revision of OPP concepts, Learning some advance techniques and dynamic features and Program introspection.

Core Books:

1. Hal Fulton, Andre Arko: The ruby way, Third edition, Addison Wesley
2. Dave Thomas, Chad Fowler, Andy Hunt: Programming Ruby -The Pragmatic Programmers' Guide, Second Edition, The Pragmatic Bookshelf
3. David A. Black: The Well-Grounded Rubyist, Manning Publications Co., Greenwich

Reference Books:

1. Peter Cooper: Beginning Ruby, Second Edition, Apress
2. Chris Pine: Learn to Program, Second Edition, The Pragmatic Bookshelf

Web References:

1. <https://www.udemy.com/ruby-on-rails-for-web-development/>[Video tutorial for web development]
2. <https://www.udemy.com/learn-ruby-on-rails-from-scratch/>[For basics of Rails - video tutorial]
3. <https://www.railstutorial.org/book>[For additional reading of web development using Ruby On

Rails]

4. <https://learnrubythehardway.org/book/> [For additional reading from references]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Employ control structures, methods, procs, arrays and hashes to create Ruby programs.

C02 : Understand object-oriented programming and input/output processing and apply these concepts to develop dynamic interactive Ruby applications.

C03 : Able to handle run time exception.

C04 : Able to develop applications.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes			
		C01	C02	C03	C04
1	Getting started and basics of Ruby Language Literacy.	✓			
2	Working with standard types, Methods and Expression	✓			
3	Working with Arrays, Hashes and Enumerable	✓	✓		
4	Working with Times and Dates, Exceptions, Catch and Throw			✓	
5	Files, I/O and Databases			✓	✓
6	OOP and Dynamic features in Ruby				

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	1	1	2	1	2	1	1	-	2	2	2	1
CO2	2	1	2	2	2	2	1	1	-	2	2	2	1
CO3	2	1	1	2	1	2	1	1	-	2	2	2	1
CO4	2	1	1	2	2	2	2	2	-	2	2	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA314: Object Oriented Programming through JAVA
(Marks 200)

Contact Hours: 07

Objective: This course is intended to emphasize the importance of Object oriented programming concepts and Java not only as an Object Oriented programming language but as a platform for developing Applications. The objectives also include extensively train the students in Core Java techniques and to create simple Java applets for use in web pages.

Pre-requisite: Basic knowledge of programming language.

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for Object Oriented Programming concepts will be discussed. Emphasize will be given on Core Java concepts, various Java packages, Multithreading, and GUI using AWT and Applets. During Practical sessions, students will be required to develop Simple Java programs, develop simple to intermediate level applications on Java platform and create simple Java Applets for use in Web pages.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Object Oriented Programming and Introduction to Java	07	36
2	Fundamental programming structures in java	08	
3	Inheritance, Abstract class and Interface	09	
4	Packages	08	
5	Exception handling and Multithreading	08	
6	GUI using AWT and Applet	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit – I: Object Oriented Programming and Introduction to Java Hours: 07

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1. Ivor Horton: Beginning Java JDK, 5th Edition, Wiley Press, 2008.
2. Cay Horstmann and Gary Cornell: Core Java Volume 1, 8th Edition or later, Pearson Education, 2008.
3. Ken Arnold, James Gosling and David Holmes: The Java Programming Language, 4th Edition, Addison Wesley.

Web References:

1. <http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-092introduction-toprogramming-in-java-january-iap-2010/lecture-notes/> [Lecture Notes in PDF]
2. <http://www.javatpoint.com/java-tutorial> [Notes]

Course Outcomes: Upon successful completion of the course, the students will :

- C01 : To understand basic of Object Oriented Concepts, Java Buzz Words and application of Java, execution environment in IDEs i.e. NetBeans & Eclipse .
- C02 : To understand syntax and semantics of Java, structure of java.
- C03 : Learning Inheritance, Abstract class and Interface to understand how the java works ethically .
- C04 : To gain understanding creating packages and how to import packages , creating application using multithreading & exception handling.
- C05 : To learn how to create GUI based application using AWT and Applet..

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Object Oriented Programming and Introduction to Java	✓	✓			
2	Fundamental programming structures in java	✓	✓			
3	Inheritance, Abstract class and Interface	✓	✓	✓		
4	Packages		✓	✓	✓	
5	Exception handling and Multithreading		✓	✓		
6	GUI using AWT and Applet		✓	✓	✓	✓

Course Articulation Matrix:

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C01	2	3	2	-	-	-	-	-	-	-	-	-	3	3
C02	3	3	-	-	-	-	-	-	-	-	-	-	3	2
C03	-	-	2	2	3	-	-	-	-	-	-	-	3	2
C04	-	-	2	2	-	-	-	-	-	-	-	-	3	3
C05	-	-	2	3	3	-	-	-	-	-	-	-	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA315: Application Development using .NET Framework

(Marks 200)

Contact Hours: 07

Pre- requisite: None.

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for advanced features of .Net framework and to develop windows based application using C# language will be given. During Practical sessions, students will be required to develop Web Applications using concepts discussed during class.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction to .NET framework	07	36
2	Basics of C# language	08	
3	Object Oriented programming using C#	08	
4	Building desktop application	09	
5	Database connectivity through ADO.NET	08	
6	Building WPF application	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit – I: Introduction to .NET framework Hours: 07

Introducing .NET Framework, .NET Framework Component, .NET Framework Version Compatibility, Core of .NET Framework: Application Services, Base Class Library and CLR, Types of Applications which can be developed using MS.NET, MS.NET Base Class Library, MS.NET Namespaces, The Common Language Runtime (CLR), Managed Code, MS.NET Memory Management / Garbage Collection, Common Type System (CTS), Common Language Specification (CLS), Types of JIT Compilers.

Unit – II: Basics of C# language Hours:08

C# Fundamentals: Basic classes, declarations, conditionals, loops, arrays, strings, enumerations, structures, and Reference Type and Value Type, Datatypes & Variables Declaration, Implicit and Explicit Casting, Checked and Unchecked Blocks – Overflow Checks, Casting between other datatypes, Boxing and Unboxing, Enum and Constant, Operators, Control Statements, Working with Arrays, C# Application Basics: Command line and VS.NET compilation.

Unit – III: Object oriented programming using C# Hours: 08

Overview of Object-Oriented Programming, Principles of Object Orientation programming : Encapsulation, inheritance, polymorphism, Object, Lifecycle of an object, Building Classes, Defining Properties, Using Auto-Implemented Properties, Defining Methods, Understanding Constructors, Extending .NET Classes via Inheritance, Defining and Implementing Interfaces, Understanding the Role of Interfaces in .NET

Unit – IV: Building desktop application Hours: 09

Building Windows Forms Applications, Setting Form Properties, Understanding the Life-cycle of a Form, Using the Windows Forms Designer, MessageBox Class and .config File, Working with Windows Forms Controls: TextBox, Button, Selection, List, Container, Image, ErrorProvider, TooltipProvider, etc. Handling Events: Understanding the Event-Driven Programming Model, Writing Event Handlers, Sharing Event Handlers

Unit – V: Database Connectivity through ADO.NET Hours: 08

Evolution of ADO.NET, Understanding the ADO.NET Object Model, Connected vs. Disconnected Access, Use of Connection, Command, DataReader, DataAdapter and Dataset class.

Unit – VI: Building WPF Application Hours: 08

Overview of WPF, Features of WPF, Windows Forms vs WPF , Introduction to XAML, WPF Layout: Grid Panel, Stack Panel, Dock Panel, Wrap Panel and Canvas Panel, WPF Controls: TextBlock, PasswordBox, ListBox, Slider, Popup, Menus and Expander.

Core Books:

1. Christian Nagel, Bill Evjen, Jay Glynn, Karli Watson, Morgan Skinner: Professional C# 2012 and .NET 4.5, Wiley – India, WROX
2. .NET 4 for Enterprise Architects and Developers
3. Albahani Josheph: C# 4.0 Pocket Reference, Shroff publication, 2010

Reference Books:

1. Drayton David: C# Language Pocket Reference, Shroff publication, 2005

Web References:

1. <http://csharp.net-informations.com/> [to learn C# and .NET fundamentals]
2. <http://www.wpftutorial.net/> [to learn WPF]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Understanding .NET Framework core features and Components.
- C02 : To develop object oriented programming using C#.
- C03 : To learn how to manage files with C# language in console based applications
- C04 : To develop windows based application using core feature of .NET framework and C# language.
- C05 : Access and manipulate data in a Microsoft SQL Server database by using Microsoft ADO.NET.
- C06 : Use modern approach to build windows based application using WPF.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to .NET framework	✓				
2	Basics of C# language	✓				
3	Object Oriented programming using C#		✓			
4	Building desktop application			✓		
5	Database connectivity through ADO.NET				✓	
6	Building WPF application					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
C01	3	2	2	3	-	2	1	-	-	1	1	2	-
C02	2	3	2	2	-	2	1	-	-	1	-	1	-
C03	1	2	2	-	-	-	-	-	-	-	-	-	-
C04	2	2	3	2	-	2	1	-	-	1	-	2	-
C05	-	3	3	-	-	-	-	-	-	-	1	-	-
C06	-	2	2	3	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA316: Software Engineering

(Marks 100)

Contact Hours: 03

Pre- requisite: None.

Methodology & Pedagogy: This course focuses on providing hands-on experience in designing and developing large-scale software systems with emphasis on the use of analyzing real world application and techniques that enable large-scale software development. Students will generate concrete software engineering artifacts at all stages of the software life-cycle. Design principles and methods for largescale software system development using object oriented software engineering.

Outline of the Course:

Sr No	Practical	Description
Week 1	Introduction to Software engineering, nature of software , stake holders in software engineering and software quality.	Students will get familiarized with basics of software engineering concepts and software quality.
Week 2	Review of object orientation	Students will get knowledge about object oriented concepts.
Week 3	Assigning a problem definition and make students understand the problem domain Gathering requirements and analyzing requirements	Students will be assign real world definition to understand Students will gather the information for respective definition using various requirement gathering techniques
Week 4	Modeling with UML: Introduction to UML	Students will be introduced to UML concepts and notations.
Week 5	Modeling with Use Case Diagram	Students will create use case diagram using event table which specifies functional requirement of the problem domain.

Week 6 Week 7	Modeling with Class Diagram	Students will analyze and identify the entities in form of classes on the basis of use case diagram and activity diagrams And classes will also be implemented using any object oriented programming language .
Week 8 Week 9	Modeling with Interaction Diagrams	Students will draw sequence diagram and collaboration diagram on basis of knowledge acquire from use case and class diagram which are base for interaction diagram, it will also include the implementation using any object oriented programming language.
Week 10 Week 11	Modeling with activity diagrams	Students will create activity diagram which specifies flow of events for the problem domain.
Week 12	Review of sessions	

Core Books:

1. Timothy C. Lethbridge and Robert Laganière: Object-Oriented Software Engineering: Practical Software Development using UML and Java : Second Edition : McGraw-Hill Education: 2005
2. Bernd Bruegge : Object oriented software engineering :Second Edition, Pearson Education.
3. Roger Pressman: Software Engineering: Sixth Edition: Tata McGraw Hill.

Reference Books:

1. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modeling Language User Guide, Addison Wesley.
2. Jacobson, Booch, Rumbaugh :The Unified Software Development Process : Pearson Education 1999.
3. Stephan R. Schach : Object oriented software engineering :Tata McGraw Hill.

Web References:

1. <http://pl.cs.jhu.edu/oose/lectures>
<http://www.site.uottawa.ca/school/research/lloseng/supportMaterial/videos>
2. http://www.cs.uic.edu/~jbell/CourseNotes/OO_SoftwareEngineering

4.http://www.abssw.com/papers/UML_Overview.pdf

Course Outcomes: Upon successful completion of the course, the students will :

- C01 : Understanding Software Engineering Concept.
- C02 : Understand Object Oriented Software Engineering.
- C03 : Understand a problem, Gather Requirements and analyze functional and non-functional requirements.
- C04 : Get Familiar with UML tokens for drawing various diagrams.
- C05 : Understand and draw various Structural and Behavioral diagrams.

Course Outcomes Mapping :

Week No.	Content	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Software engineering	✓				
2-3	Review of object orientation , Assigning a problem definition and make students understand the problem domain Gathering requirements and analyzing requirements		✓	✓		
4-6	Modeling with UML: Introduction to UML , Modeling with Use Case Diagram				✓	✓
7-8	Modeling with Class Diagram				✓	✓
9-10	Modeling with Interaction Diagrams Modeling with activity diagrams				✓	✓
11-12	Review of sessions				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
C01	2	3	2	2	2	3	3	3	3	3	1	2	2
C02	2	3	2	2	2	3	3	2	1	3	2	3	2
C03	3	2	3	2	3	3	2	2	3	2	3	3	2
C04	3	2	3	3	1	2	2	1	2	1	1	3	1
C05	2	2	3	2	2	2	1	1	1	1	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS 131.02 C: COMMUNICATION AND SOFT SKILLS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	02
Marks	--	100	--	100	

Pre-requisite courses:

- Communicative English

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An Introduction to Communication	06
2.	Cross-cultural Communication and Globalization	03
3.	Communication for Career Building	10
4.	Group Dynamics and Soft Skills	05
5.	Effective Presentation Strategies	04
6.	Contemporary Issues in Communication and Soft Skills	02
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	An Introduction to Communication	06 Hours	20%
	Basics of Communication: Origin, Concept, Process, Levels, Principles and Barriers; Applications of Communication; Rhetoric in Professional Communication; Importance of Ethos, Logos, and Pathos in Communication		

2.	Cross-cultural Communication and Globalization	03 Hours	10%
	Basic Concepts: Culture, Globalization and Cross-cultural Communication; Social and People Skills; Communicating with People of Different Cultures; Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them; Persuasive Communication		
3.	Communication for Career Building	10 Hours	33%
	Cover Letters and Resume; E-mail and Report; Types of Resume; Concept and Rationale of Group Discussion Skills and Aspects assessed in Group Discussion; Concept and Rationale of Personal Interview; Types of Personal Interview; Writing Statement of Purpose		
4.	Group Dynamics and Soft Skills	05 Hours	17%
	An Introduction to Group Dynamics and Soft Skills; Groups and their Structures; Roles and Functions of Members in Groups; Conflict Management; Aptitude and Attitude; Various Intelligences; Developing an Open Mindset		
5.	Effective Presentation Strategies	04 Hours	14%
	Designing Appealing Presentation; Audience Analysis and Supporting Material; Presentation Mechanics and Presentation Process; Managing Yourself during Q and A Session; Fundamentals of Persuasion		
6.	Contemporary Issues in Communication and Soft Skills	02 Hours	06%
	Trends and Practices in Communication, Case Studies		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Gain thorough understanding and proficiency in various Professional Communication Skills.
CO2	Develop awareness and competence in cross-cultural communication in their personal, academic and professional environments.
CO3	Develop business writing and presentation skills to succeed in career.
CO4	Develop soft skills to stand out and take their career to the next level.
CO5	Develop various intelligences and open Mindset to function in multi-disciplinary and cross-cultural work environment.
CO6	Practice new trends in communication in multiple perspectives at personal, professional, and social level.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	2
CO2	-	-	-	-	-	-	-	-	3	-	-	-	2
CO3	-	-	-	-	-	-	-	-	3	-	-	-	2
CO4	-	-	-	-	-	-	-	-	3	-	-	-	2
CO5	-	-	-	-	-	-	2	-	3	-	2	-	2
CO6	-	-	-	-	-	-	-	-	3	-	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ **Text book:**

1. Koneru, A. Professional Communication, Tata McGraw Hill Education Private Limited
2. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
3. Raman, M & Singh, P. Business Communication, Oxford University Press

❖ **Reference book:**

1. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
2. Anandamurugan, A. Placement Interviews – Skills for Success, Tata McGraw Hill Education Private Limited

❖ **Web material:**

1. <https://www.coursera.org/learn/careerdevelopment>
2. <https://www.futurelearn.com/courses/writing-applications>
3. <https://www.futurelearn.com/courses/workplace-englis>

Teaching and Examination Scheme (BCA Programme)

Effective from Year 2021-2022

Semester-VI

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theo ry	Pract	Tot al		Internal		Exter nal	Internal		Exter nal	
						Case Study	Tests		Term work	Tests		
CA317-319	Elective-VII	4	-	4	4	10	20	70	-	-	-	100
CA320 – 322	Elective-VIII	-	3	3	3	-	-	-	15	15	70	100
CA323	Minor Project	-	15	15	15	-	-	-	100		300	400
HS132.02 C	Contributory Personality Development	-	2	2	2	- -		-	30		70	100
		04	20	24	24	100			600			700

Elective- VII

1. CA317 –Artificial Intelligence
2. CA318–Cryptography and Network Security
3. CA319–Data Compression Techniques

Elective- VIII

1. CA320–Software Project Management
2. CA321–Basics of Mobile Applications
3. CA322–Computer Graphics-II

CA317: Artificial Intelligence

(100 Marks)

Contact Hours: 04

Pre-requisite: None

Methodology & Pedagogy: The theory sessions will be focused on basics of Artificial Intelligence, problem solving paradigms, and search strategies. Areas of application such as Expert Systems, Software Agents, knowledge representation, natural language processing, expert systems, robotics will be explored.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours Theory
1	Fundamental issues in intelligent systems	06
2	Search and constraint satisfaction	09
3	Knowledge representation and reasoning	09
4	Expert Systems	08
5	Agents	08
6	Introduction to AI Techniques and Application Areas	08

Total Hours (Theory): 48

Detailed Syllabus:

Unit – I Fundamental issues in intelligent systems Hours:06

Concepts and definitions of Artificial Intelligence (AI) – Brief history of AI – AI and related fields – The AI problems and underlying assumptions.

Unit – II Search and constraint satisfaction Hours:09

Problem spaces; brute-force search; best-first search; two-player games; constraint satisfaction

Unit – III Knowledge representation and reasoning Hours:09

Review of propositional and predicate logic; resolution and theorem proving; non-monotonic inference; probabilistic reasoning; Bayes theorem.

Unit – IV Expert Systems Hours:08

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Introduction – Representing and using domain knowledge – Knowledge acquisition and representation – General structure of Expert Systems – Expert System Shell – Advantages and disadvantages of Expert Systems

Unit – V Agents Hours:08

Definition of agents; successful applications and state-of-the-art agent-based systems; software agents, personal assistants, and information access; multi-agent systems.

Unit – VI Introduction to AI Techniques and Application Areas Hours:08

Introduction to fuzzy logic – Introduction to various application areas of AI like Natural Language Processing (NLP), Game Playing, Robotics.

Core Books:

1. Norvig, Peter, Paradigms of AI Programming, Morgan Kauffman, 1992.
2. S. Russell and P. Norvig, Modern Approach to Artificial Intelligence, Prentice Hall of India Ltd., 2006.
3. Rich & Knight, Artificial Intelligence, 2nd edition, TMH, 1991.

ReferenceBooks:

1. R.Akerker and P. S. Sajja, Knowledge-Based Systems, Jones and Bartlett, MIT, 2010.
2. George Luger, Artificial Intelligence, 5th Edition, Addison Wesley, 2004.

Web References:

1. http://www.tutorialspoint.com/artificial_intelligence/[Basics of AI]
2. <http://intelligence.worldofcomputing.net/ai-branches/expert-systems.html>[Expert Systems]
3. <http://pages.cs.wisc.edu/~bolo/shipyard/neural/local.html>[Neural Network]

Course Outcomes: Upon successful completion of the course, the students will :

- C01 : To understand basic problems of AI and history of AI.
- C02 : Getting deep knowledge of searching algorithms and its applications in AI.
- C03 : Learning Knowledge Representation and reasoning techniques.
- C04 : To gain understanding Expert System and Agent Based Systems.
- C05 : To explore various application areas of AI.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Fundamental issues in intelligent systems	✓				
2	Search and constraint satisfaction		✓			

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3	Knowledge representation and reasoning			✓		
4	Expert Systems				✓	
5	Agents				✓	
6	Introduction to AI Techniques and Application Areas					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	-	2	2	2	1	-	-	-	-	-	-	1	2
CO2	3	2	2	3	2	2	-	3	-	-	-	2	2
CO3	-	-	3	3	2	-	-	-	-	-	-	1	-
CO4	3	2	2	-	2	2	-	3	-	-	-	1	2
CO5	2	-	-	-	2	-	3	3	-	-	-	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA318: Cryptography and Network Security
(100 Marks)

Contact Hours: 04

Prerequisites:CA309 Advanced Computer Networking.

Methodology & Pedagogy: During theory sessions the students shall be introduced to various techniques of cryptography and concepts of network security. As case study students will have to develop or study a cryptographic protocol or a network security system.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours
1	Foundation and Stream Cipher	07
2	Block Ciphers	08
3	Message Integrity	08
4	Introduction to Network Security	08
5	Threats to Communication Networks	08
6	Network Security Assessment	09

Total Hours (Theory): 48

Unit –I : Foundation and Stream Cipher Hours : 07

Introduction to cryptography: History of cryptography, Applications of cryptography. Stream Cipher: Information Theoretic Security and The One Time Pad, Stream Ciphers and Pseudo Random Generators, Attacks on Stream Ciphers and The One Time Pad, Real-World Stream Ciphers, PRG Security Definitions, Semantic Security

Unit –II : Block Ciphers Hours: 08

Block Ciphers: DES, Exhaustive Search Attacks, The AES Block Cipher, Block Ciphers from PRGs Review: PRPs and PRFs, Modes of Operation: One Time Key, Many Time Key (CBC, CTR)

Unit –III: Message Integrity Hours:08

Message Authentication Codes, MACs Based On PRFs, CBC-MAC and NMAC, MAC Padding, PMAC and the Carter-Wegman MAC, Generic Birthday Attack, The Merkle-Damgard Paradigm, Constructing Compression Functions, HMAC, Timing attacks on MAC verification, Digital Signatures

Unit –IV : Introduction to Network Security Hours:08

Introduction to Network Security, Vulnerability to Threat and Attack, Importance of Effective Network Security Strategies, Typical Network Security When Using Your Computer: Firewalls, Anti-Virus software

Unit –V : Threats to Communication Network Hours:08

Development of Network Security Measures, Data Storage, Data Transmission, Passive and Active Attacks, Potential Network Vulnerabilities, Tapping into transmission media.

Unit –VI : Network Security Assessment Hours:09

IP network scanning: ICMP Probing, TCP Port Scanning, UDP Port Scanning, Low Level IP Assessment.

Core Book:

1. Schneier Bruce: Applied Cryptography 2nd Ed., Wiley Publications
2. McNab Chris: Network Security Assessment 2nd Ed., O'Reilly Publication.
3. Jose MiggaKizza: Computer Network Security, Springer

Reference Book:

1. Stallings, William: Network Security Essentials: Applications and Standards, PHI
2. Richard Mollin: An Introduction to Cryptography, Chapman & HALL/CRC
3. Christof Parr, Jan Pelzl: Understanding Cryptography: A textbook for Students and Practitioners.

Web References:

1. <https://www.coursera.org/learn/crypto>[Video Lectures]
2. <http://williamstallings.com/NetworkSecurity/>[Materials for teachers]
3. https://onlinecourses.nptel.ac.in/noc16_cs21[Video lectures and notes]
4. <https://alison.com/courses/Fundamentals-of-Network-Security>[Network Security]

Course Outcomes: Upon successful completion of the course, the students will :

- C01 : Learn about basic concepts of cryptography and stream ciphers.
- C02 : Understand concepts of block ciphers.
- C03 : Gain knowledge about message integrity and related concepts.
- C04 : Learn about various concepts of network security.
- C05 : Know and understand various tools used for network security assessment.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Foundation and stream cipher	✓				
2	Block ciphers		✓			
3	Message integrity			✓		
4	Introduction to network security				✓	
5	Threats to communication network				✓	
6	Network security assessment					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	3	2	1	1	-	2	1	-	-	-	2	2	1
CO2	2	2	2	1	-	2	1	-	-	-	2	2	1
CO3	2	2	1	1	-	2	1	-	-	-	2	2	1
CO4	2	2	2	1	-	2	1	-	-	-	2	2	1
CO5	2	2	1	1	-	2	1	-	-	-	2	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA319: Data Compression Techniques

(100 Marks)

Contact Hours: 04

Pre-requisite: None

Methodology & Pedagogy: During theory lectures the emphasis will be given on theoretical concepts of data compression techniques, model and various coding techniques.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours
		Theory
1	Introduction to Compression Techniques, Model & Coding Hours	09
2	Huffman Coding	09
3	Arithmetic Coding	09
4	Dictionary Techniques	08
5	Video Compression	07
6	Audio Compression	06

Total Hours (Theory): 48

Detailed Syllabus:

Unit – I: Introduction to Compression Techniques, Model & Coding Hours: 09

Introduction to Compression Techniques, Lossless Compression, Lossy Compression, Measures of Performance, Models: Physical Models, Probability Models, Markov Models and Composite Source Model, Coding: Uniquely Decodable Codes & Prefix Codes

Unit – II: Huffman Coding Hours: 09

Introduction of Huffman Coding Algorithm, Minimum Variance Huffman Codes, Adaptive Huffman

Coding: Update Procedure, Encoding Procedure and Decoding Procedure, Golomb Codes, Rice Codes, Applications of Huffman Coding

Unit – III: Arithmetic Coding Hours: 09

Introduction to Arithmetic Coding, Coding a Sequence, Generating a Tag, Deciphering the Tag, Generating a Binary Code, Uniqueness and Efficiency of the Arithmetic Code, Comparison of Huffman and Arithmetic Coding, Application of Arithmetic coding

Unit – IV: Dictionary Techniques Hours: 08

Introduction to Dictionary Techniques, Static Dictionary, Diagram Coding, Adaptive Dictionary, The LZ77 Approach, The LZ78 Approach, Applications of Dictionary Techniques, File Compression—UNIX compress, Image Compression—The Graphics Interchange Format (GIF), Image Compression—Portable Network Graphics (PNG)

Unit – V: Video Compression Hours: 07

Analog Video, Composite and Components Video, Digital Video, Video Compression, MPEG and MPEG-4

Unit – VI: Audio Compression Hours: 06

Sound, Digital Audio, The human Auditory System, Speech Compression

Core Books:

1. Khalid Sayood: Introduction to Data Compression, 3rd Edition, Morgan Kaufman
2. David Salomon: Data Compression: The Complete Reference, 3rd Edition, Springer

Reference Books:

1. David Salomon, Giovanni Motta: Handbook of Data Compression, 5th Edition, Springer
2. Mark Nelson, Jean-Loup Gailly: The Data Compression Book, 2nd Edition, IDG Books Worldwide, Inc

Web References:

1. <https://www.ics.uci.edu/~dan/pubs/DataCompression.html> [For basics of data compression]
2. <http://www.axis.com/global/en/learning/web-articles/technical-guide-to-networkvideo/videocompression-guide> [For video compression]
3. <https://users.cs.cf.ac.uk/Dave.Marshall/Multimedia/node200.html> [For Video and Audio compression techniques]
4. <http://www.cprogramming.com/tutorial/computersciencetheory/huffman.html> [For Huffman coding]
5. <http://marknelson.us/2014/10/19/data-compression-with-arithmetic-coding/> [For arithmetic coding]
6. <http://www.linktionary.com/c/compression.html> [For Dictionary techniques]

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Course Outcomes: Upon successful completion of the course, the students will:

C01 : Be able to understand the concept of data compression.

C02 : Be familiar with compression techniques.

C03 : Be able to understand image compression technique.

C04 : Be able to understand video compression technique.

C05 : Be able to understand the audio compression technique.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Compression Techniques, Model & Coding Hours	✓				
2	Huffman Coding		✓			
3	Arithmetic Coding		✓			
4	Dictionary Techniques		✓	✓		
5	Video Compression		✓		✓	
6	Audio Compression		✓			✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	2	2	-	2	1	-	-	-	1	2	1
CO2	3	2	2	2	-	2	1	-	-	-	1	2	1
CO3	3	3	2	2	-	2	1	-	-	-	1	2	1
CO4	3	3	2	2	-	2	1	-	-	-	1	2	1
CO5	2	2	2	2	-	2	1	-	-	-	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA320: Software Project Management
(100 Marks)

Contact Hours : 03

Pre-requisite: System Analysis and Design.

Methodology & Pedagogy: In order to achieve the course objective, tools related to Software Project Management will be used by the students to understand and implement the concepts of Project Management.

Outline of the Course:

Week No.	Content
1-2	Developing the Project Plan Identify Project Scope & Objectives, Access the Project Information, Identify project Resources and Tasks.
3-5	Building the Schedule Schedule the Tasks, Allocate Resources, Planning Resource and Task Budget, Working with the Critical Path Methods (CPM)and Critical Tasks, Controlling Schedule Changes with Task Types
6-7	Checking and Adjusting Project Plan Bringing in the Project Finish Date, Reducing Project Costs, Balancing Resource Workloads, Changing Project Scope, Program Evaluation and Review Technique (PERT).
8	Tracking the Progress of Project Development Setting a Baseline of the Project, Updating Task Progress, Updating Actual Costs.
9	Responding to Changes in Your Project Monitoring and correcting the Schedule, Monitoring and Adjusting Costs, Monitoring and Adjusting Resource Workload.
10	Reporting and Analyzing Project Information Using Views to Report Project Information, Generating Text and Visual Reports, Revising a Built-In Report, Building a Custom Report, Saving Project Data Fields, Sharing Project Reports

11	Managing Multiple Projects Structuring Master Projects with Subprojects, Opening Multiple Project Files as a Set, Sharing Resources by Using a Resource Pool, Exchanging Information Between Project Plans, Copying and Moving Information Between Projects, Linking Information Between Project Plans.
12	Integrating Project 2010 with Other Programs Integrating Project 2010 with Visio, Integrating Project 2010 with Excel.

Total Hours (Lab): 36

Total: 36

Core Books:

1. Carl Chatfield, Timothy Johnson: Microsoft Project 2016 Step by Step, Microsoft Press, 2016.

Reference Books:

1. Cynthia Snyder: Project 2016 For Dummies, John Wiley & Sons, 2016.

Web References:

1. <http://www.kasal.biz/MP.2016.Step.by.Step.pdf>[For e-book Content]
2. <http://www.sjfc.edu/dotAsset/137092.pdf> [Basics of MS Project]
3. https://s3.amazonaws.com/sample-courseware-int/sample_project-2016-foundation-manual.pdf [For Sample Project]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : To form the baseline plan of project and schedule it.

C02 : To adjust, change and track the progress of project plan.

C03 : To generate various standard and visual reports and analyze them.

C04 : To shareresources by using a Resource Pool and exchangeinformation between Project Plans

C05 : To integrate project with other programs.

Course Outcomes Mapping:

Week No.	Content	Course Outcomes				
		C01	C02	C03	C04	C05
1 -2	Developing the Project Plan	✓				
3 - 5	Building the Schedule	✓				

6 - 7	Checking and Adjusting Project Plan		✓			
8	Tracking the Progress of Project Development		✓			
9	Responding to Changes in Your Project		✓			
10	Reporting and Analyzing Project Information			✓		
11	Managing Multiple Projects				✓	
12	Integrating Project 2010 with Other Programs					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	3	2	3	3	3	2	-	2	2	3	2
CO2	2	2	3	2	3	3	3	2	-	2	2	3	2
CO3	2	2	3	2	3	3	3	2	-	2	2	3	2
CO4	2	3	3	2	3	3	3	2	-	2	2	3	2
CO5	2	2	3	2	3	3	3	2	-	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA321: Basics of Mobile Applications
(100 Marks)

Contact Hours : 3

Pre-requisite: Knowledge of Core Java programming language.

Methodology & Pedagogy During Practical sessions, students will be required to develop Mobile Application using JAVA programming language in Android. Student shall also develop applications dealing with data storage, documents sharing among applications and application based on Google maps.

Outline of the Course:

Week No.	Practical	Description
1	Overall Android architecture and features. Different Android versions: Froyo, Gingerbread, Ice Cream Sandwich, Jellybean, and Kitkat. Android development tools like SDK and Android Studio introduction.	Introduction to Android Framework
2	Introduction of Android Manifest file and R.java file. Android application architecture overview. Introduction of Gradle. Different screen size AVD creation and Sample App. First android application	Getting started with Android development
3	Activity management and Inter Activity communication in Android application	Development of core component Activity of Android
4	Fragment: Life Cycle, Architecture Interaction between fragments,	Understanding and development of partitioning in Activity using Fragments
5	Views and Layout management in Android Studio	Different Layout implementation and arrangement of Views in Layout
6 – 7	Compound Views, Styles and Themes implementation	Designing of Android Activity using different Styles, Themes and Views

8 – 9	Dialogs: AlertDialog, DatePickerDialog TimePickerDialog, ProgressDialog Menu: Option Menu, and Context Menu Animation basics: Frame by Frame, and Tweened animation	Implementation of different types of Menus and Dialogues in Android. Basic Android animation techniques demonstration
10 – 11	Data Persistence using User Preferences Persisting Data to Files: Internal Storage, External Storage Database management through SQLite	Implementation of RDBMS and Session management in Android
12	Introduction to cross platform application development framework like PhoneGap and Xamarin	Basics of cross platform application development with given frameworks

Total Hours (Lab): 36

Total: 36

Core Books:

1. Wei-Meng Lee: Beginning Android 4 Application Development, Wiley India Pvt Ltd.
2. Mark L. Murphy: The Busy Coder's Guide to Android Development.

Reference Books:

1. Uwe Hansmann, LotharMerk, Martin S. Nicklons and Thomas Stober: Principles of Mobile Computing, Springer, New York, 2003.
2. Jonathan Simon: Head First Android Development, O'REILLY publication.
3. Mark L. Murphy: Beginning Android 2, APRESS publication.

Web References:

1. <https://developer.android.com/index.html>[Google APIs for Android]
2. <https://developers.google.com/training/android/>[Android Training Course]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be exposed to technology and business trends impacting mobile applications
- C02 : Be competent with the characterization and architecture of mobile applications
- C03 : Be competent with understanding enterprise scale requirements of mobile applications
- C04 : Be competent with designing and developing mobile applications using one application development framework.

Course Outcomes Mapping:

Week No.	Content	Course Outcomes			
		C01	C02	C03	C04
1	Introduction to Android Framework	✓	✓		
2	Getting started with Android development		✓		
3	Development of core component Activity of Android			✓	✓
4	Understanding and development of partitioning in Activity using Fragments	✓		✓	✓
5	Different Layout implementation and arrangement of Views in Layout		✓		✓
6-7	Designing of Android Activity using different Styles, Themes and Views				✓
8-9	Implementation of different types of Menus and Dialogues in Android. Basic Android animation techniques demonstration			✓	✓
10-11	Implementation of RDBMS and Session management in Android			✓	✓
12	Basics of cross platform application development with given frameworks			✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C01	3	3	-	-	-	1	1	-	-		-	-	-
C02	3	3	3	3	-	2	2	-	-	-	-	3	3
C03	3	3	3	2	-	2	1	-	-	2		3	3
C04	3	3	3	3	2	2	3	-	-	2	2	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA322: Computer Graphics-II (Marks 100)

Contact Hours : 3

Pre- requisite: CA216: Computer Graphics - I.

Methodology & Pedagogy: During practical sessions, the course will cover the usage of adobe Photoshop – a graphics editor tool.

Outline of the Course:

Week No.	Content
1 & 2	Getting Started with Photoshop and Working with images. Introductions to images, Difference between vector graphics and raster graphics, Introduction Photoshop Interface, Exploring the Toolbox, Options Bar, Panels & Menus, Creating & Viewing a New Document ,Importing & placing images, Image size and resolution , Zooming & Panning an Image, Working with Multiple Images, Rulers, Guides & Grids, Undoing Steps with History, Saving the file in different File Formats.
3	Getting Started with Layers Understanding the Background Layer, Creating, Selecting, Linking & Deleting Layers, Locking & Merging Layers, Copying Layers, Using Perspective & Layer Styles, Filling & Grouping Layers, Introduction to Blending Modes, Blending Modes, Opacity & Fill, Creating & Modifying Text
4	Working with Basic Selections Introduction Selections Tool & Transformation, Selecting images, Combining Selections, Resizing & Cropping images ,Transforming Objects
5	Drawing, Painting and text Using the Brush Tool, Working with Colors & Swatches, Creating & Using Gradients, Creating & Working with Brushes, Using the Pencil & Eraser Tools, Painting with Selections, Type tool, History brush tool, Art History brush tool, Creating Patterns
6	Photo Retouching Retouching the Image, Healing Brush Tool, Clone Stamp Tool, Pattern Stamp Tool, Eraser Tool
7 & 8	Introduction to Color Correction & Masking Adjusting Levels & Adjustment Layers, Vector Masking, clipping Masking and Test Masking
9	Working with The Pen Tool & shape Tool Understanding Paths & the Pen Tool, Creating Straight & Curved Paths, Creating a Clipping Path, Built in shapes, Custom shapes, User defined Shape

10	Creating Special Effects Getting Started with Photoshop Filters, Smart Filters, Creating Text Effects, Applying Gradients to Text
11	Animation in Photoshop
12	Exporting Your Work

Total Hours (Lab): 36

Total: 36

Core Books:

1. Lesa Snider, Photoshop CS5 - The missing manual, O'reilly Media, First Edition, May 2010.
2. Lisa DaNaeDayley and Brad Dayley, Photoshop CS5 - Adobe Photoshop CS5 Bible, Wiley India Pvt. Ltd., June 2010.

Reference Books:

1. Michael Toot, Sherry Kinkoph, Master Adobe Photoshop, Illustrator, Premiere and after Effects Visually, First Edition 2002
2. Douglas K. Van Duyne, James A. Landay, Jason I. Hong, The Design of sites – Patterns for creating Winning web sites, Prentice Hall, Second Edition

Web References:

1. <https://www.youtube.com/watch?v=IjM9p0ISGZo> [Photoshop]
2. <http://mashable.com/2010/08/12/12-beginner-tutorials-for-getting-started-withphotoshop/> - [Overview of tools]
3. <http://www.photoshopessentials.com/photo-effects/> -[Photo effects]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Able to get familiar with the Photoshop tool, workspace and it's working.
- C02 : Able to work with the selection, resize, crop, navigate, color modes, text for editing and creating new images as well as shape.
- C03 : Able to understand and implement retouching and repairing techniques to correct images
- C04 : Use basic Photoshop skills and concepts to develop effective graphics for both web and print media.
- C05 : Able to create simple animation.
- C06 : Able to design HTML template in Photoshop and convert the template in HTML,CSS

Course Outcomes Mapping:

Week No.	Content	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1-3	Getting Started with Photoshop and Working with images, Getting Started with Layers	✓					
4-6	Working with Basic Selections, Drawing, Painting and text, Photo Retouching		✓	✓			
7-9	Introduction to Color Correction & Masking, Working with The Pen Tool & shape Tool		✓		✓	✓	
10-12	Creating Special Effects, Animation in Photoshop		✓		✓		✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	2	2	3	-	-	2	-	-	-	-	-	2	1
CO2	2	2	3	3	2	2	-	-	-	1	-	2	1
CO3	2	2	3	2	-	3	-	-	2	2	1	2	1
CO4	2	2	-	3	3	1	-	-	2	2	2	2	1
CO5	2	2	-	3	3	1	-	-	2	2	2	2	1
CO6	2	2	-	3	3	1	-	-	2	2	2	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS132.02 C: CONTRIBUTORY PERSONALITY DEVELOPMENT

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	02
Marks	--	100	--	100	

Pre-requisite courses:

- Communication and Soft Skills

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Concept of Personality	06
2.	Soft Skills and Personality Development	08
3.	Developing Contributory Personality	06
4.	Life skills and Personality Development	06
5.	Contemporary Issues in CPD	04
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	Concept of Personality	06 Hours	20%
	Meaning of Personality, Types of Personality, Factors contributing to Personality, Personality Traits, Personality Profiling		
2.	Soft Skills and Personality Development	08 Hours	26%
	Positive Thinking and Mind Set, Leadership, Assertiveness and Negotiation Skills, Self-Management, Interpersonal Skills, Being a Team Player		
3.	Developing Contributory Personality	06 Hours	20%

	Concept of Contributory Personality, Characteristics of a Contributor, The Contributor's Vision of Success & Career, The Scope of Contribution in a field, Embarking on the Journey to Contributor ship, Developing Contributor Personality, Reviewing Some Contributors Personalities		
4.	Life skills and Personality Development	06 Hours	20%
	Concept of life skills, Self-awareness, Empathy, Decision Making, Problem Solving		
5.	Contemporary Issues in CPD	04 Hours	14%
	Contemporary Trends and Practices in Contributory Personality Development, Case Study & Presentations		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify one's individual personality strengths and challenges.
CO2	Develop more assertive and optimist attitude towards work and life.
CO3	Develop quintessential soft skills to groom one's personality.
CO4	Identify traits of contributor personality.
CO5	Contribute to self, society, nation, and globe.
CO6	Develop skills of global citizenship to perform societal responsibilities.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	2	2	-	-	3
CO3	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	3	-	2	-	-	-	-
CO6	-	-	-	-	-	-	-	-	2	-	3	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

Recommended Study Material:

❖ **Text book:**

1. Personality Development & Soft Skills, Oxford University Press
2. Soft Skills, Bookboon
3. Personality Development, Swami Vivekananda; Advaita Ashrama

❖ **Reference book:**

1. Contributor Personality Program Workbook (Volume 1,2),
2. Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd

❖ **Web material:**

1. <https://www.coursera.org/learn/wharton-succcess>
2. <https://www.coursera.org/learn/personality-types-at-work>
3. <https://www.coursera.org/learn/self-awareness>

CA323: Minor Project
(300 Marks)

Contact Hours: 15

Guidelines: Minor Project is in house project development. Every student is required to carry out Minor Project work under the supervision of a guide provided by the MCA Programme placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Minor Project work during the semester as per the schedule provided by the MCA Programme placement Coordinator.

Minor Project proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Minor Project to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. MCA programme placement coordinator will provide the format of project report. Student has to submit one copy of Minor Project to the institute. Each Student is required to make a copy of Minor Project in CD and submit along with Minor Project report.

Objective:

- Main objective of Minor project is to understand and gain the knowledge of the principles of software engineering practices, and develops good understanding of SDLC.
- By using standard SDLC steps and concepts of Software Engineering, students must be able to implement their programming knowledge in terms of a working module.
- Students must be in concern of conversation of knowledge into application to solve out a real life problem.

Outcomes:

- Student will understand the implementation of concepts of SDLC and Software Engineering.
- The programming concepts they learn during their academics, it will be converted in to the actual implementations.
- Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.
- Students must understand the difference between a program and professional application/product/software.
- Students will learn different categories of applications like Desktop application, Web applications, etc.

Evaluation: The project report shall normally be written in English in the specified format and shall be characterized by significant contribution to knowledge in the field. The Project report prepared according to approved guidelines and duly signed by the guide and the Head of the Department shall be submitted to the Head of the Institution. The evaluation scheme of Project is as under:

Course	Course Title	Teaching Scheme		Internal	End Semester Examination		Total
		Contact Hours	Credit	Continuous Evaluation	Report	Presentation & Viva	
CA323	Minor Project Work	15	15	100	100	200	400

The internal evaluation of project is done based on progress reports and internal presentations. The final evaluation of the project will be based on the project report submitted and a VivaVoce Examination by a Board of Examiners.

If a candidate fails to submit the project report on or before the specified deadline, he/she is deemed to have failed in the Project Work and shall re-enroll for the same in a subsequent semester. If a candidate fails in the viva-voce examinations of Project work he/she shall resubmit the project report within specified duration decided by university. The resubmitted project will be evaluated during the subsequent academic session. A copy of the approved project report after the successful completion of viva examinations shall be kept in the library of the college / institution.

Web References:

1. <http://techwhirl.com/writing-software-requirements-specifications/>[For effective SRS]
2. http://www.ibm.com/developerworks/websphere/library/techarticles/0306_perks/perks2.html [For best practices of Software Project Development]
3. http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf[Requirement analysis guidelines]
4. <http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf> [Software Design Principles and Guidelines]
5. <http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%20Testing%20->

[2050%20specific%20ways%20to%20improve%20your%20testing.pdf](#)[For Effective Software Testing]

6. <http://www.cs.uics.edu/~jbell/CourseNotes/OO SoftwareEngineering/SE Project Report Template.pdf> [For guidelines to prepare software project report]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO2	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO3	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO4	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO5	3	3	3	2	3	3	3	3	3	2	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”